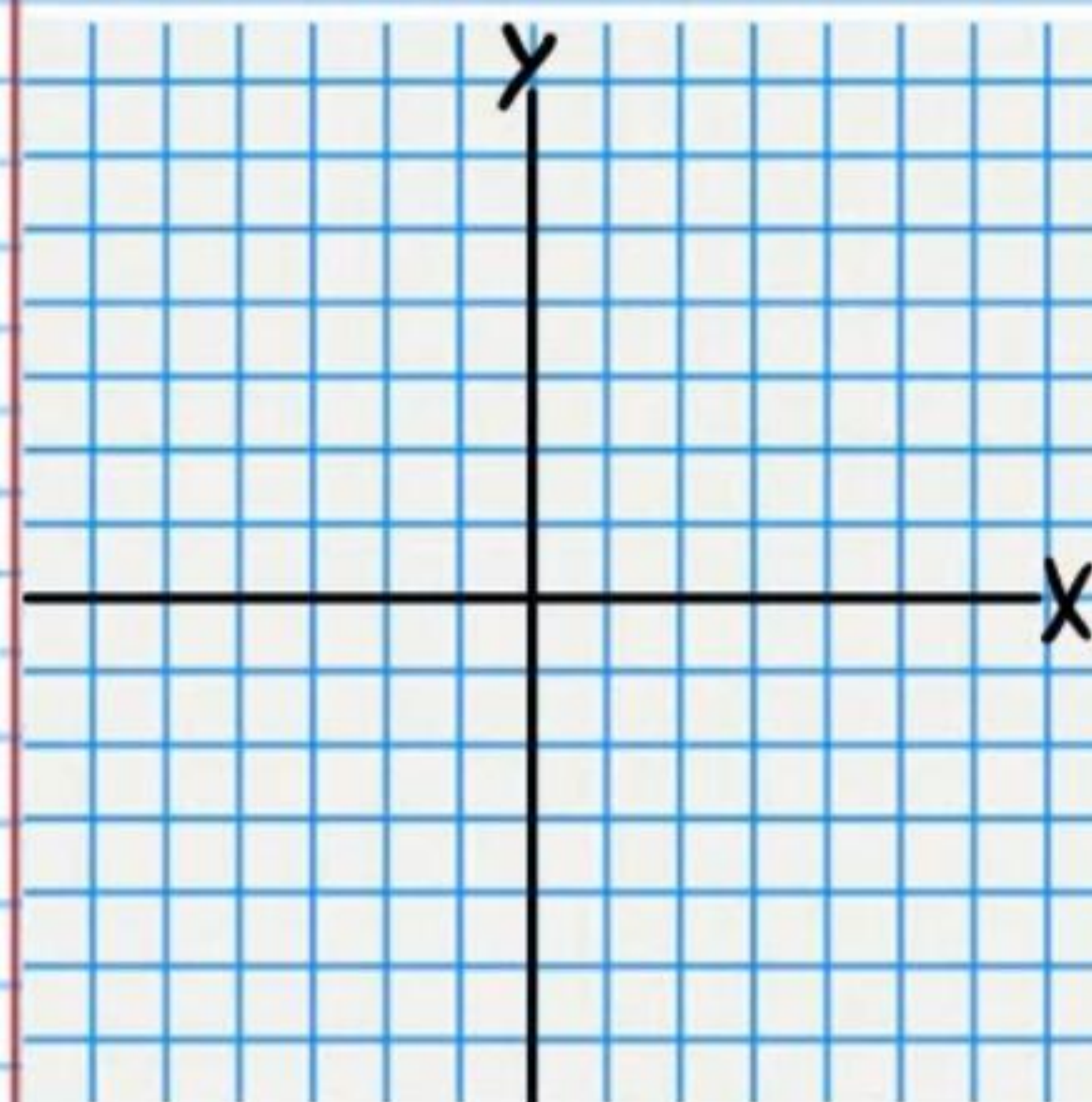


Transformations and the Library of Relations (Part V) Homework

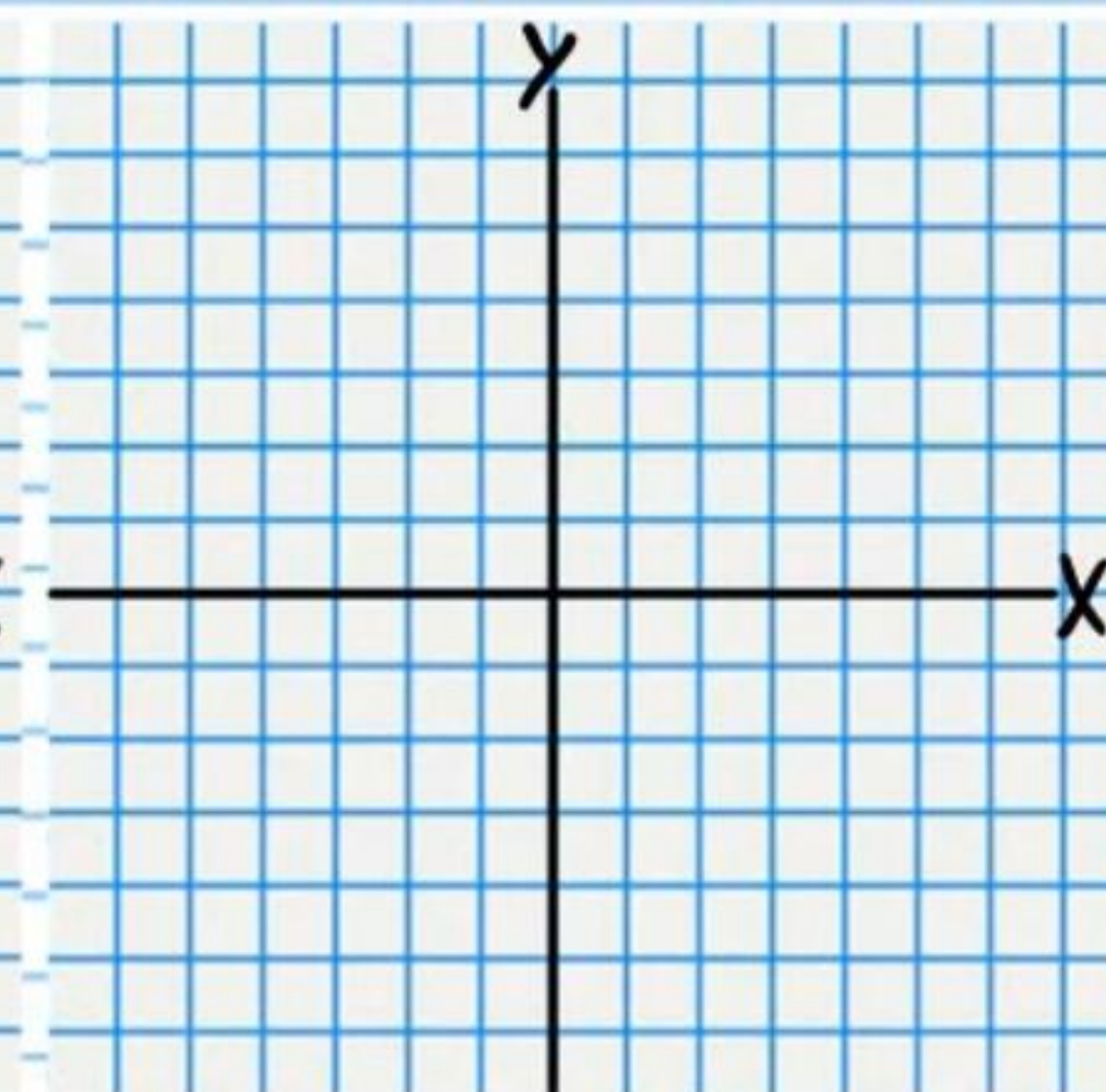
Change the form of each equation below so that it can be graphed with the transformation method. Draw the parent graph. Then draw each transformation, and write the equation for each graph. Draw any asymptotes as dotted lines.

① $y = |x| + 3$

Parent Relation

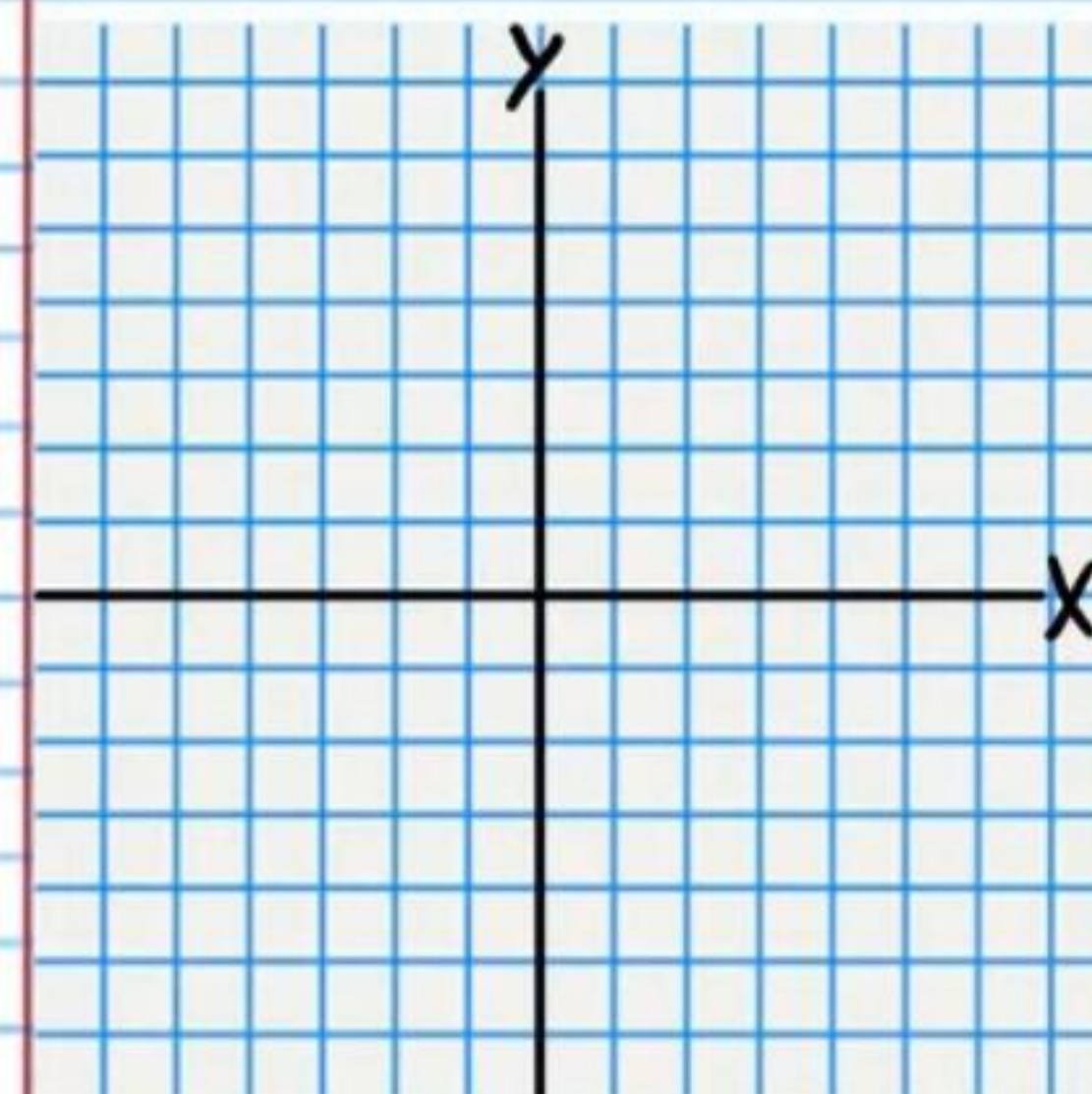


Transformation

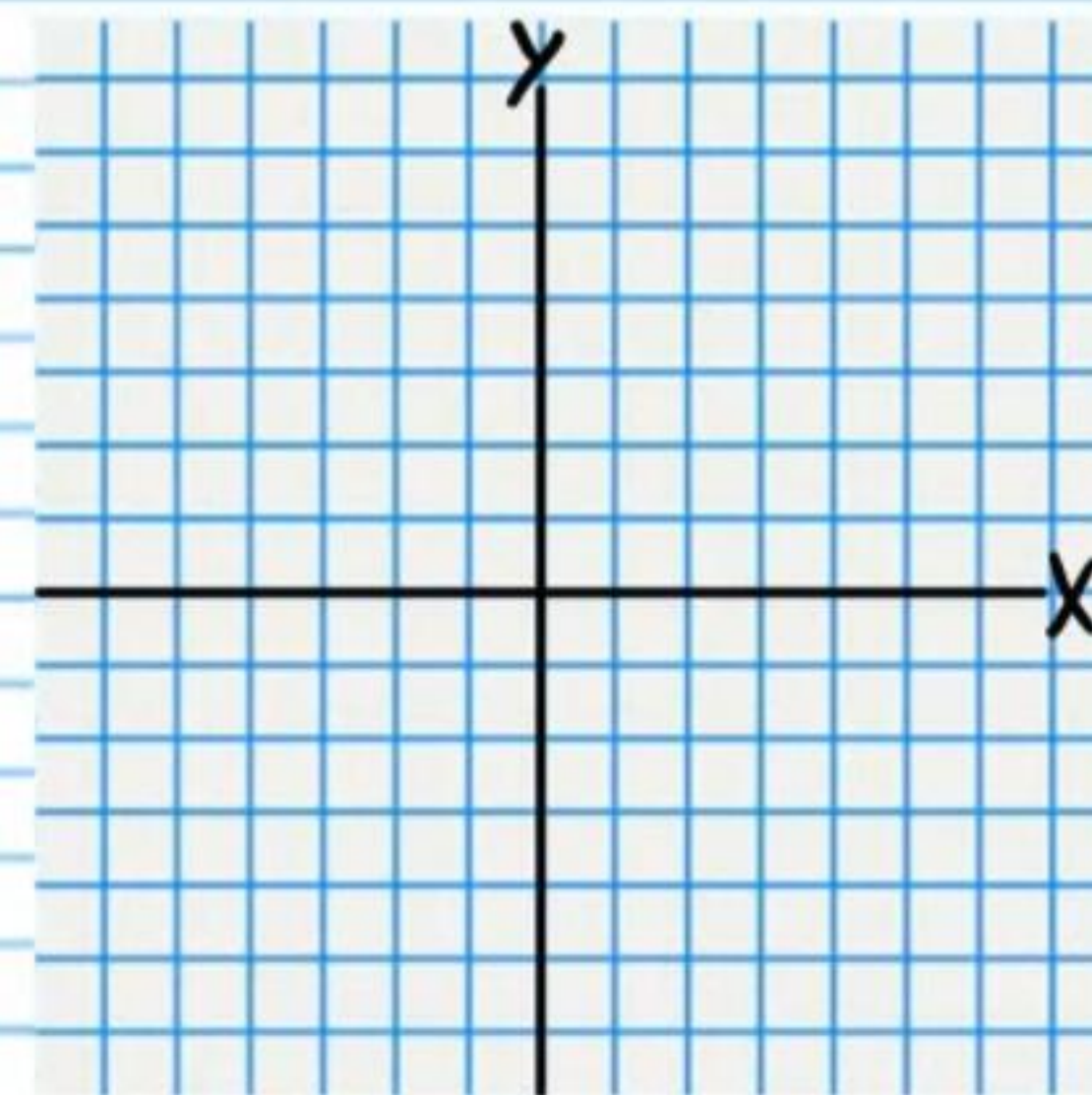


② $y = x^3 - 4$

Parent Relation

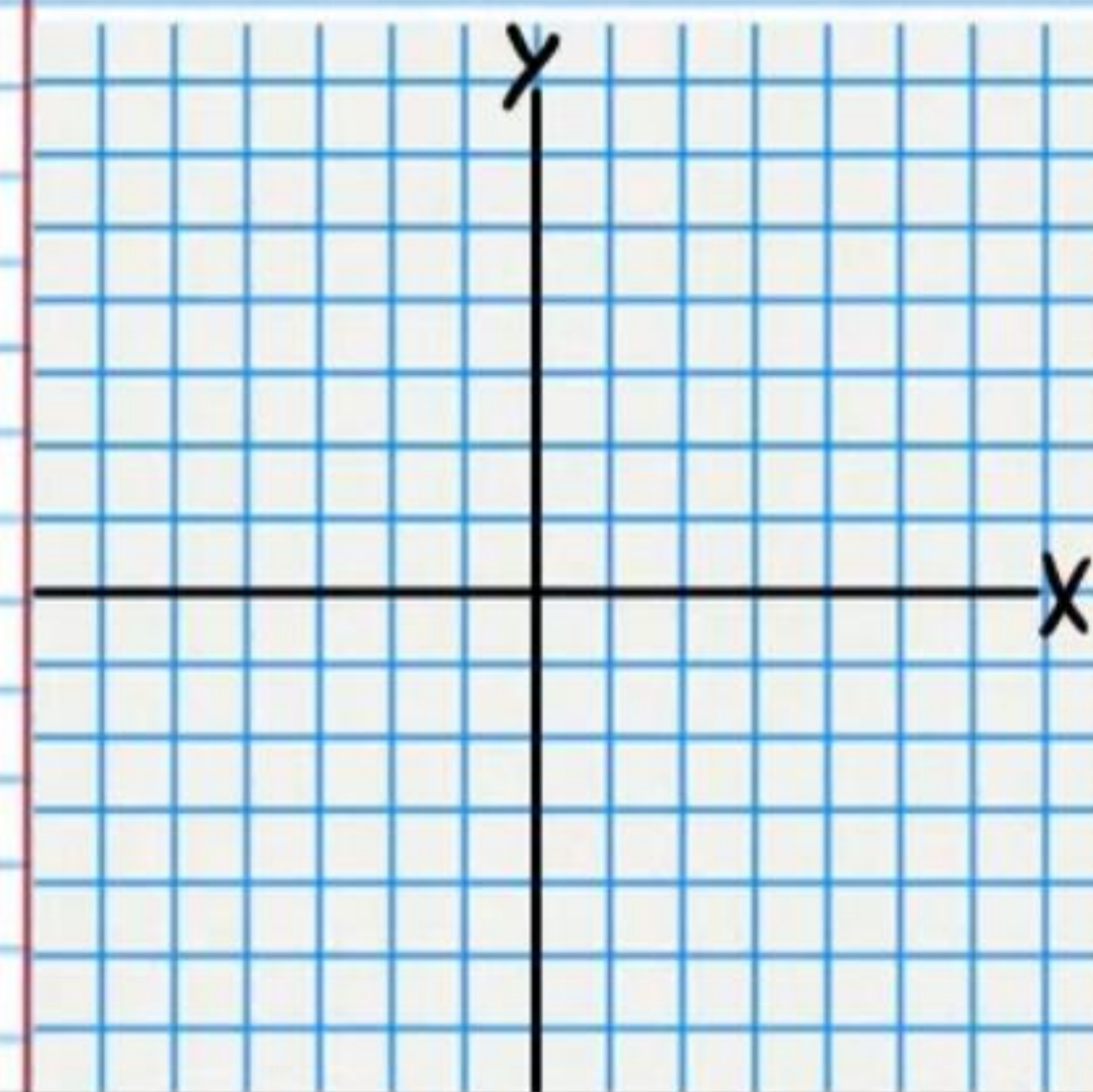


Transformation

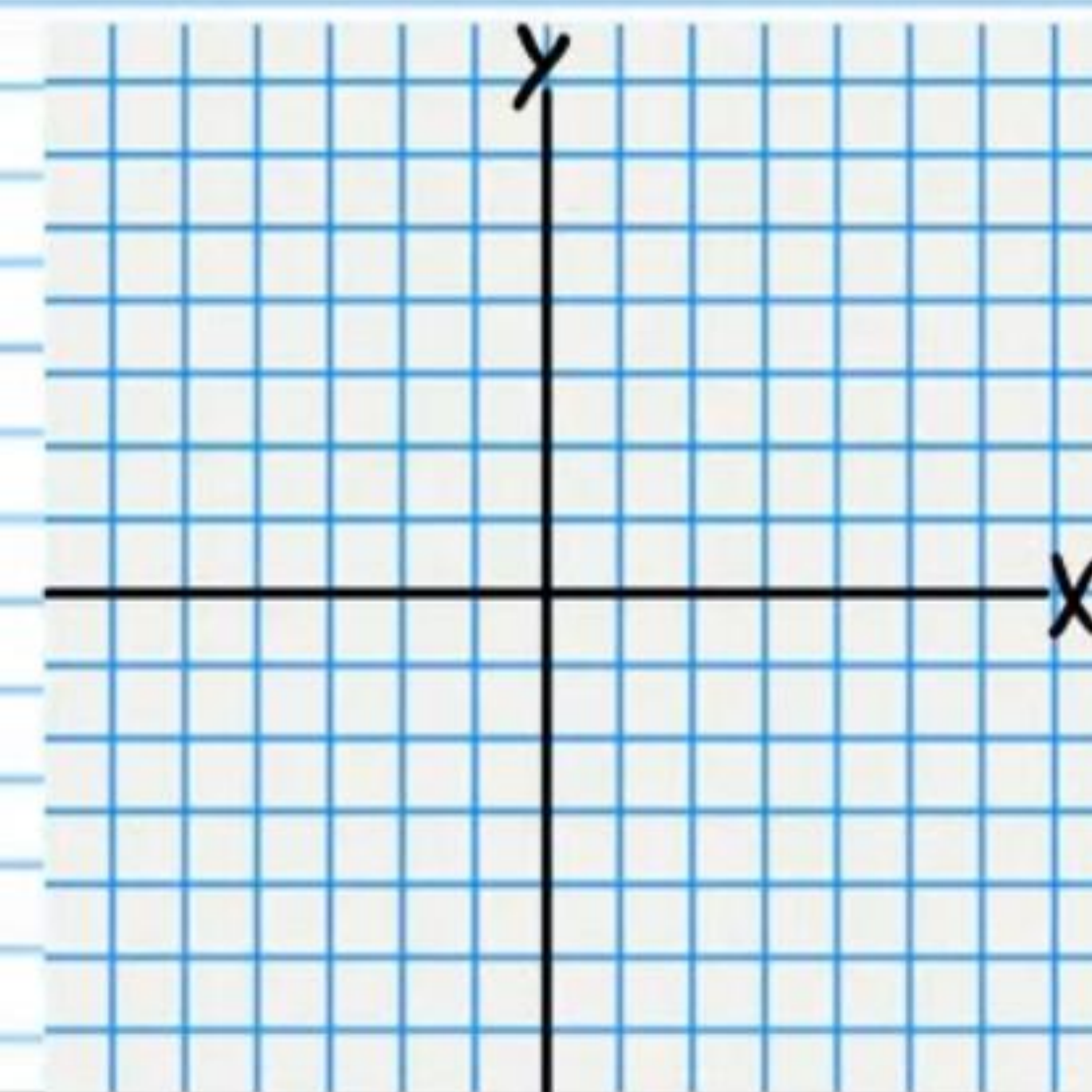


③ $y = \frac{1}{x} + 2$

Parent Relation



Transformation



Logarithm Property III

$$\log_b b = 1$$

Logarithm Property II

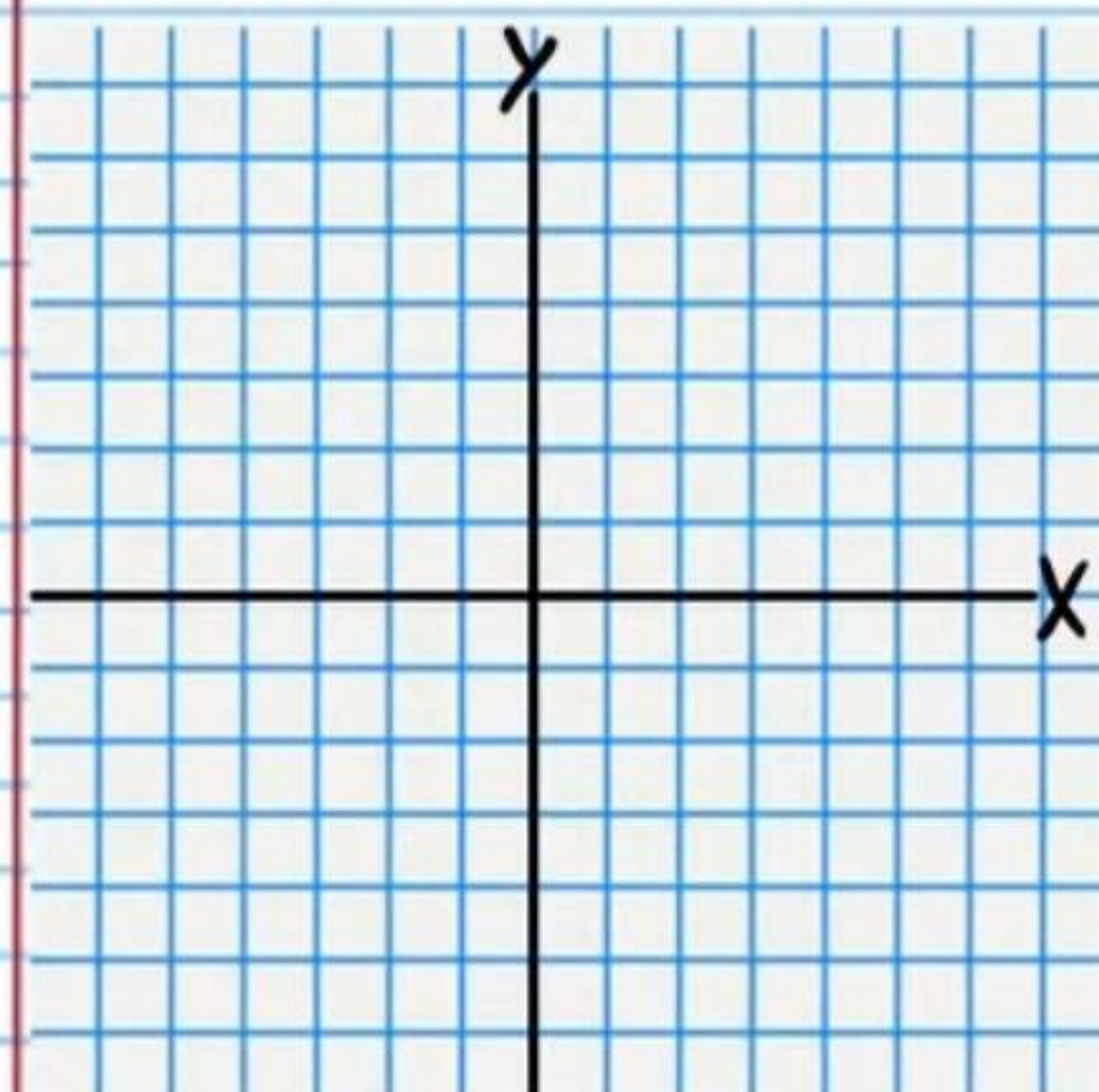
$$\log_b b^x = x$$

Logarithm Property IX

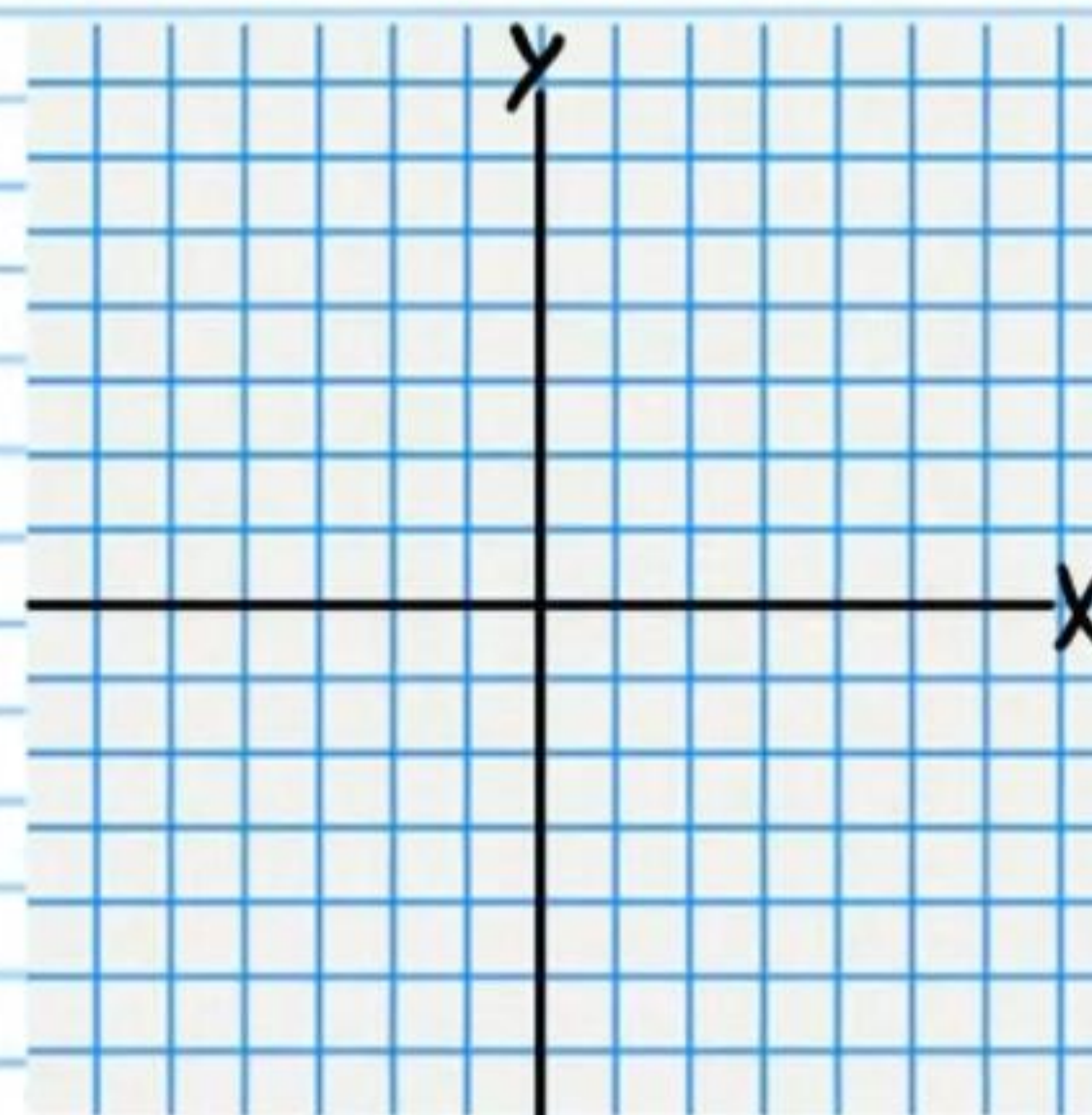
$$\log_b 1 = 0$$

④ $y = \log_3 x + 2$

Parent Relation

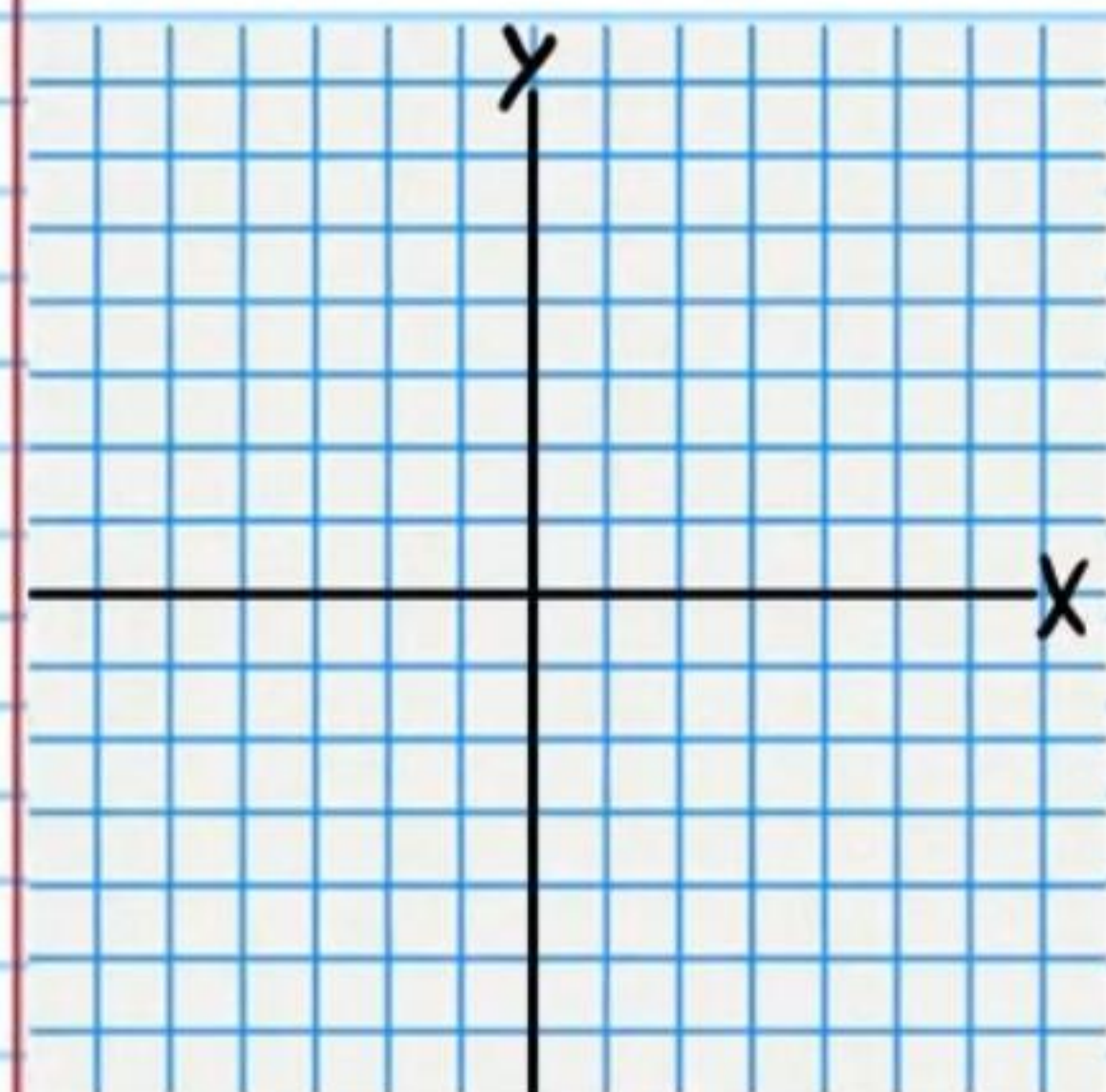


Transformation

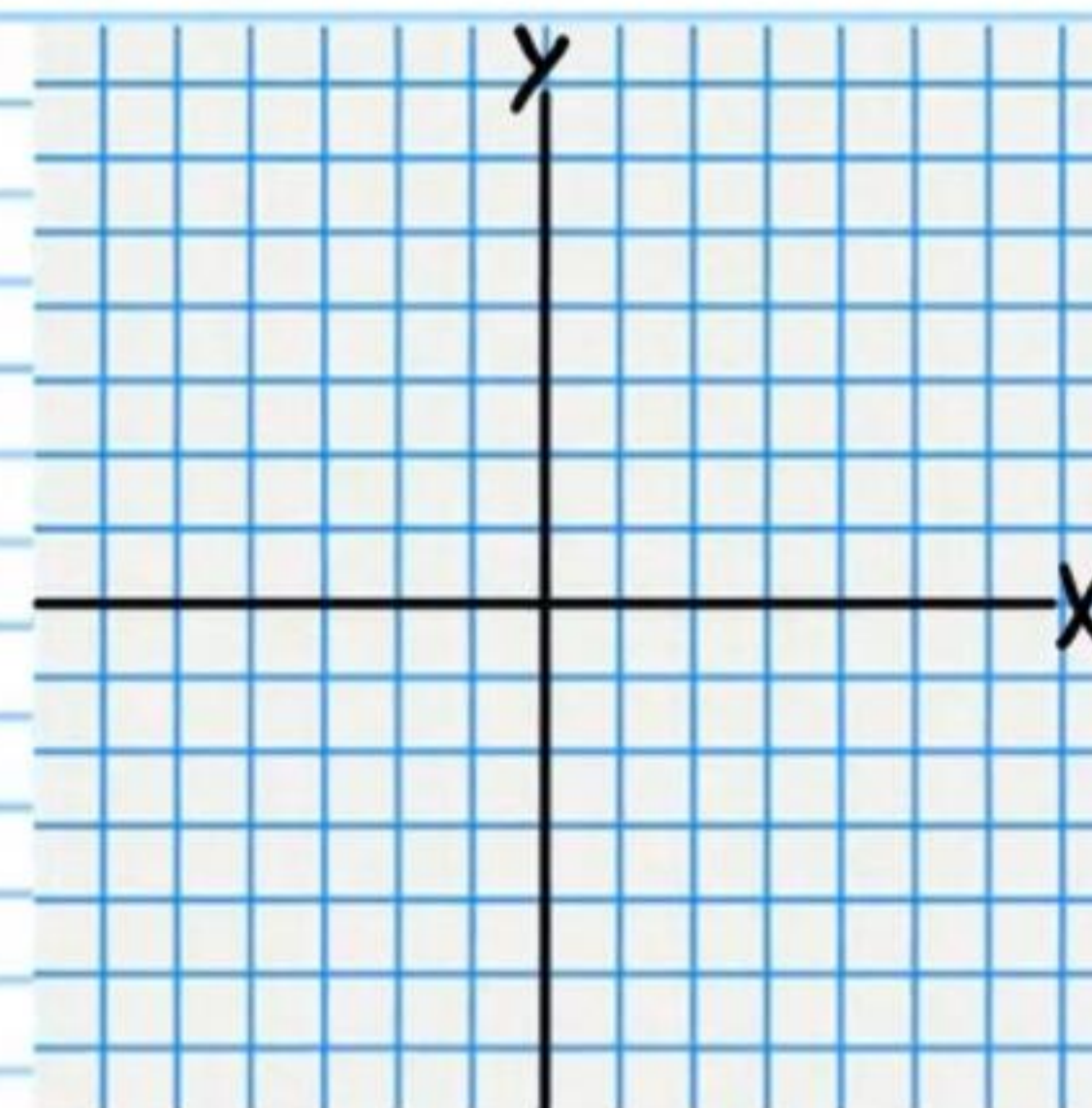


⑤ $y = \frac{x^2}{3}$

Parent Relation

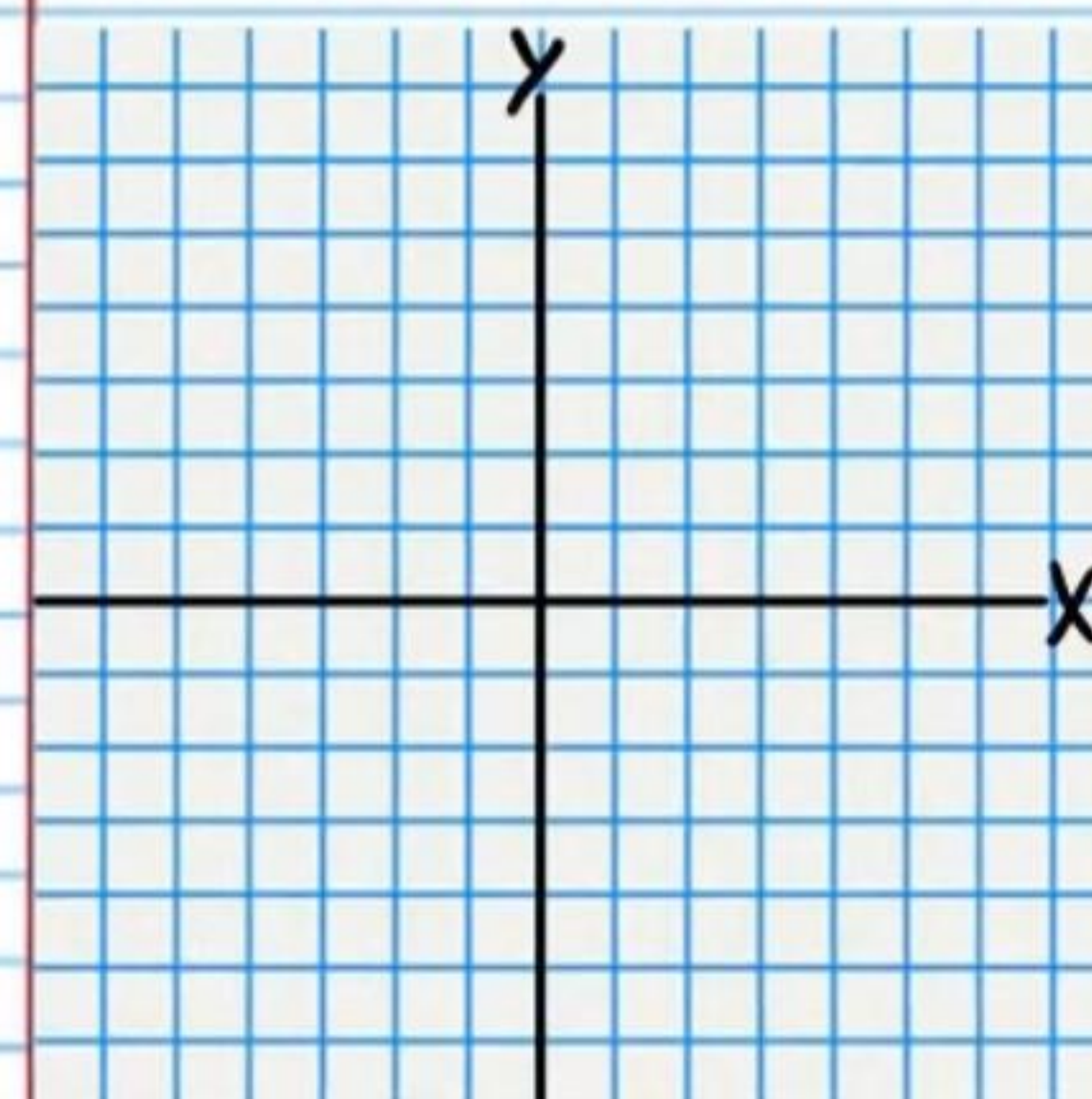


Transformation

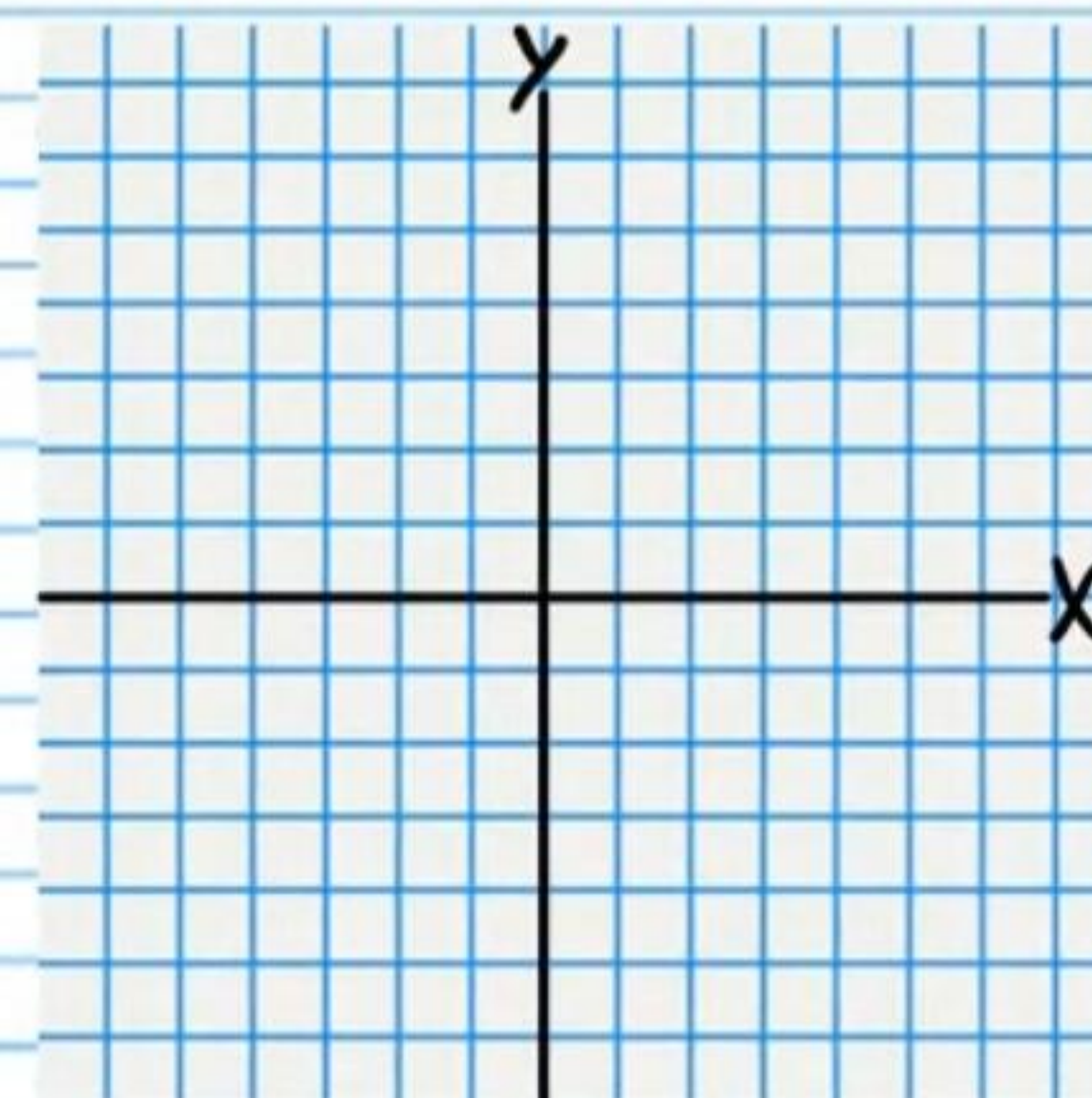


⑥ $y = 2\sqrt{x}$

Parent Relation

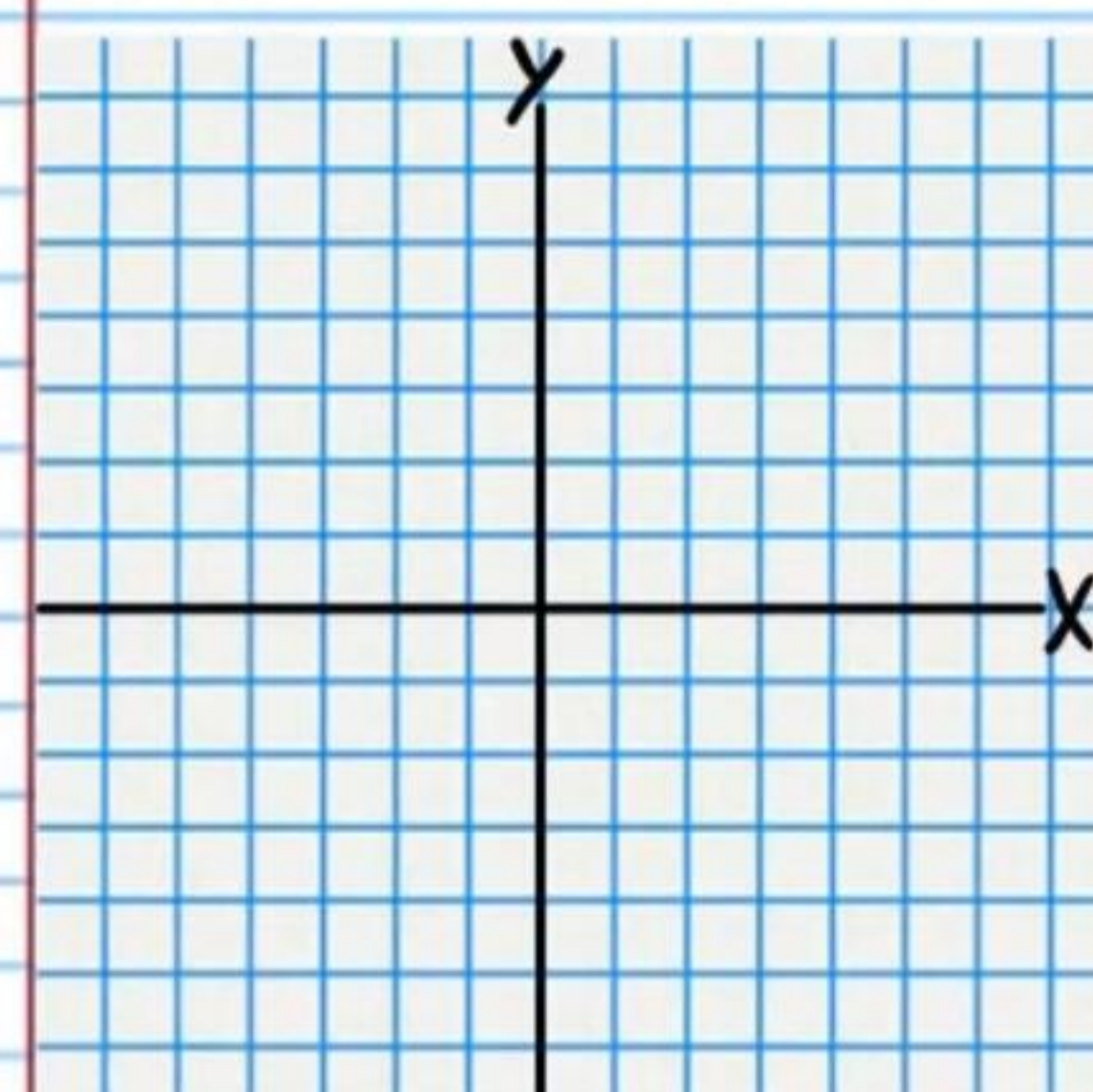


Transformation

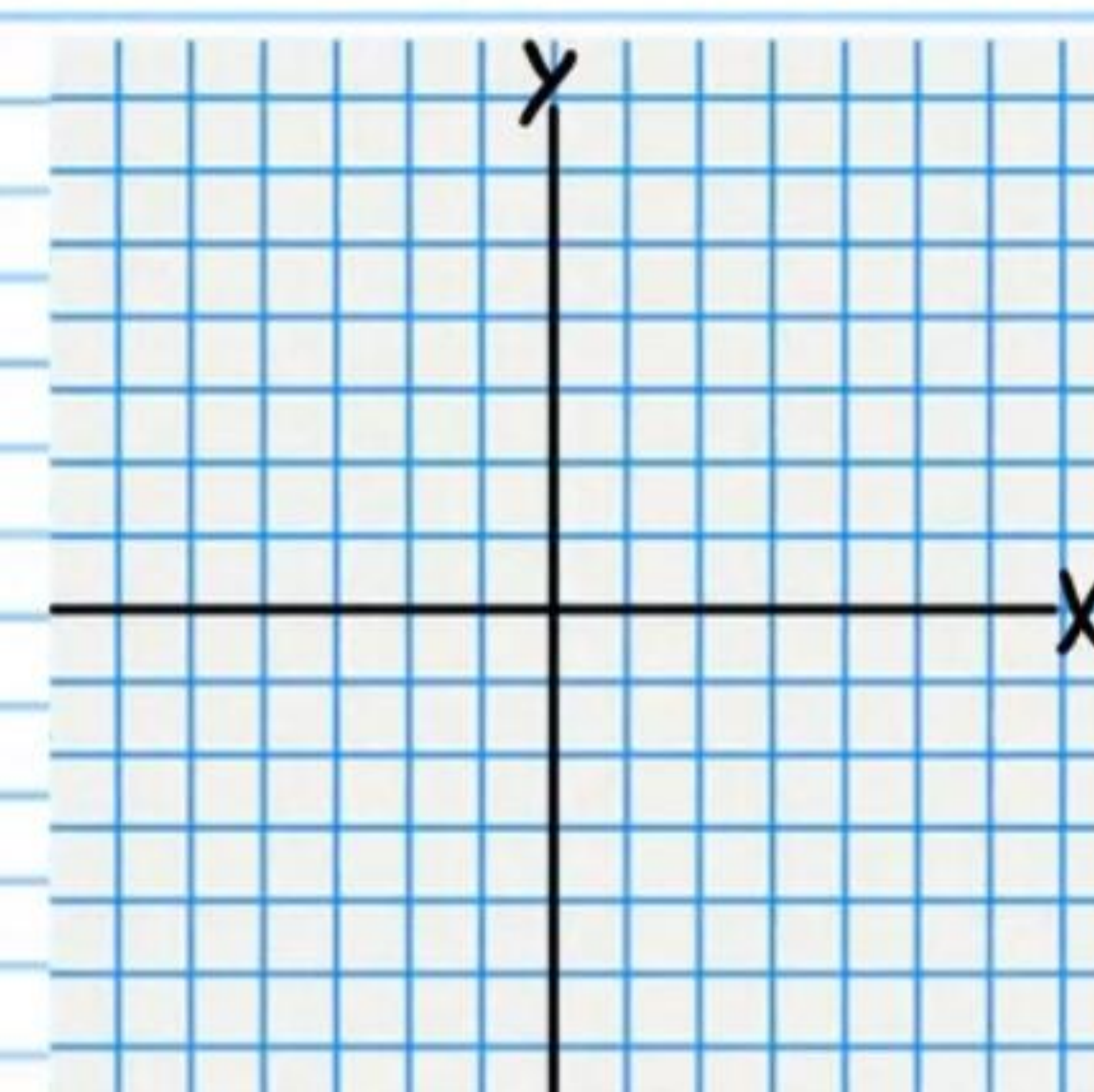


⑦ $y = \frac{3^x}{2}$

Parent Relation



Transformation



Logarithm Property III

$$\log_b b = 1$$

Logarithm Property II

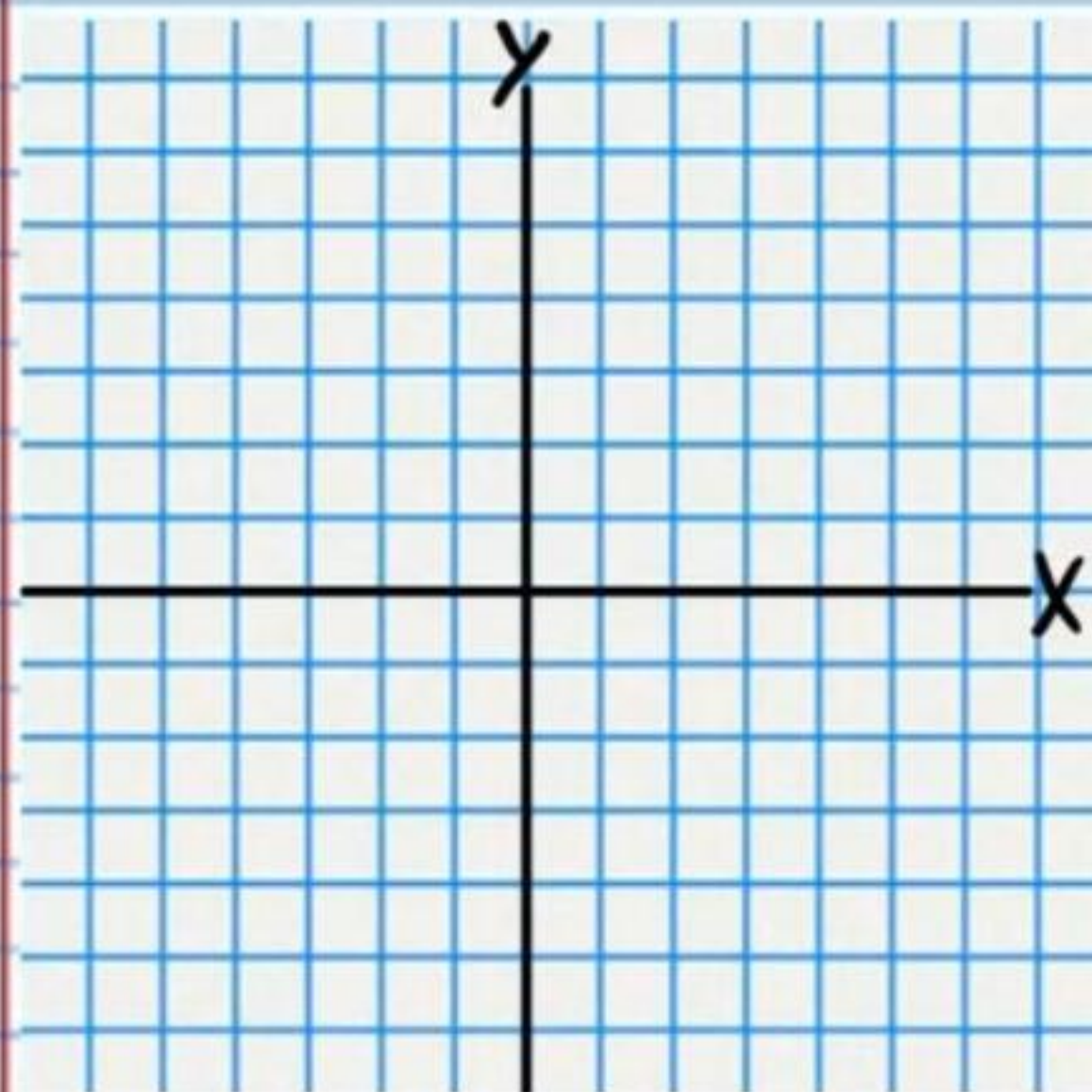
$$\log_b b^x = x$$

Logarithm Property IX

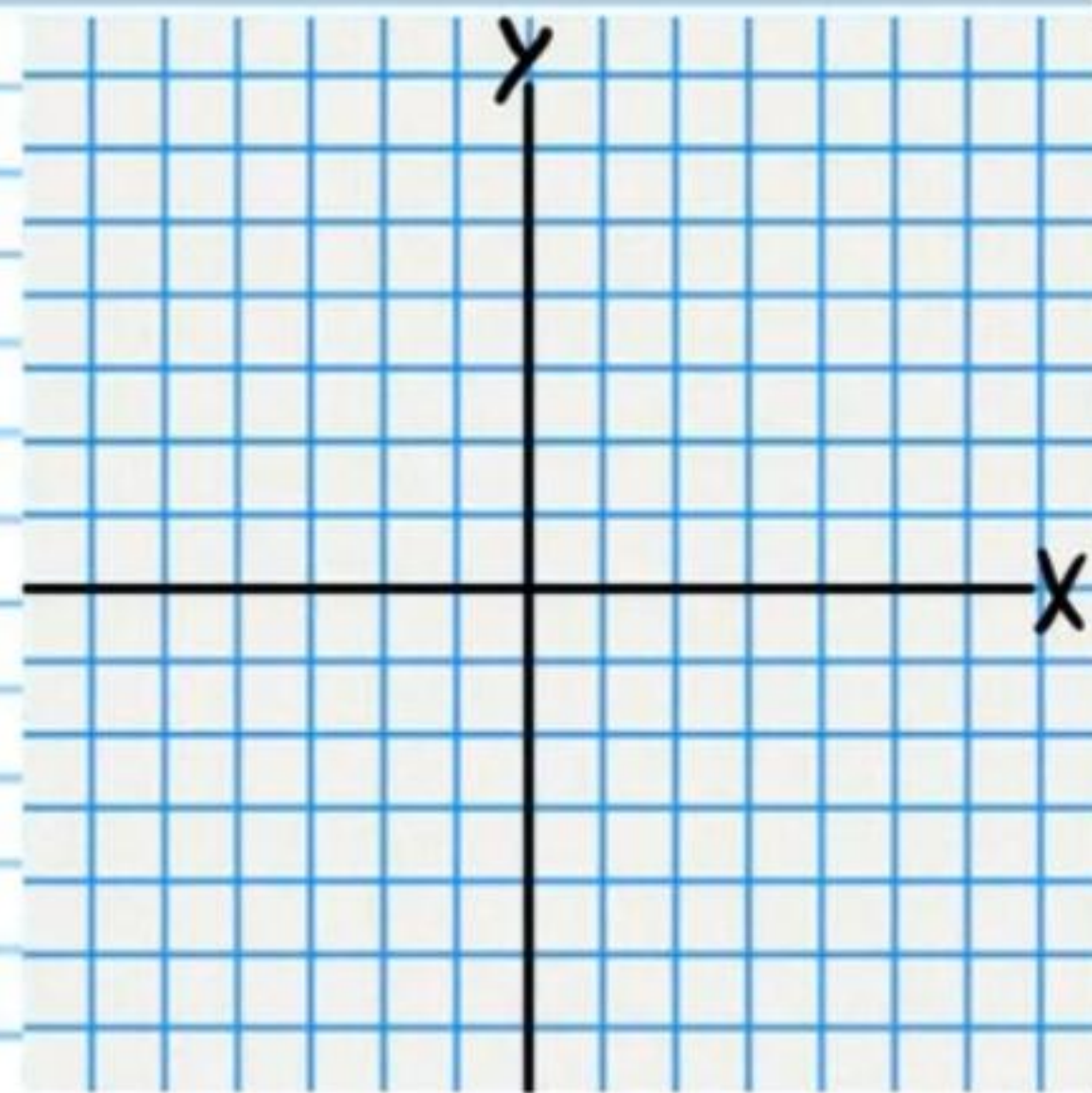
$$\log_b 1 = 0$$

⑧ $y = 3 \log_4 x$

Parent Relation

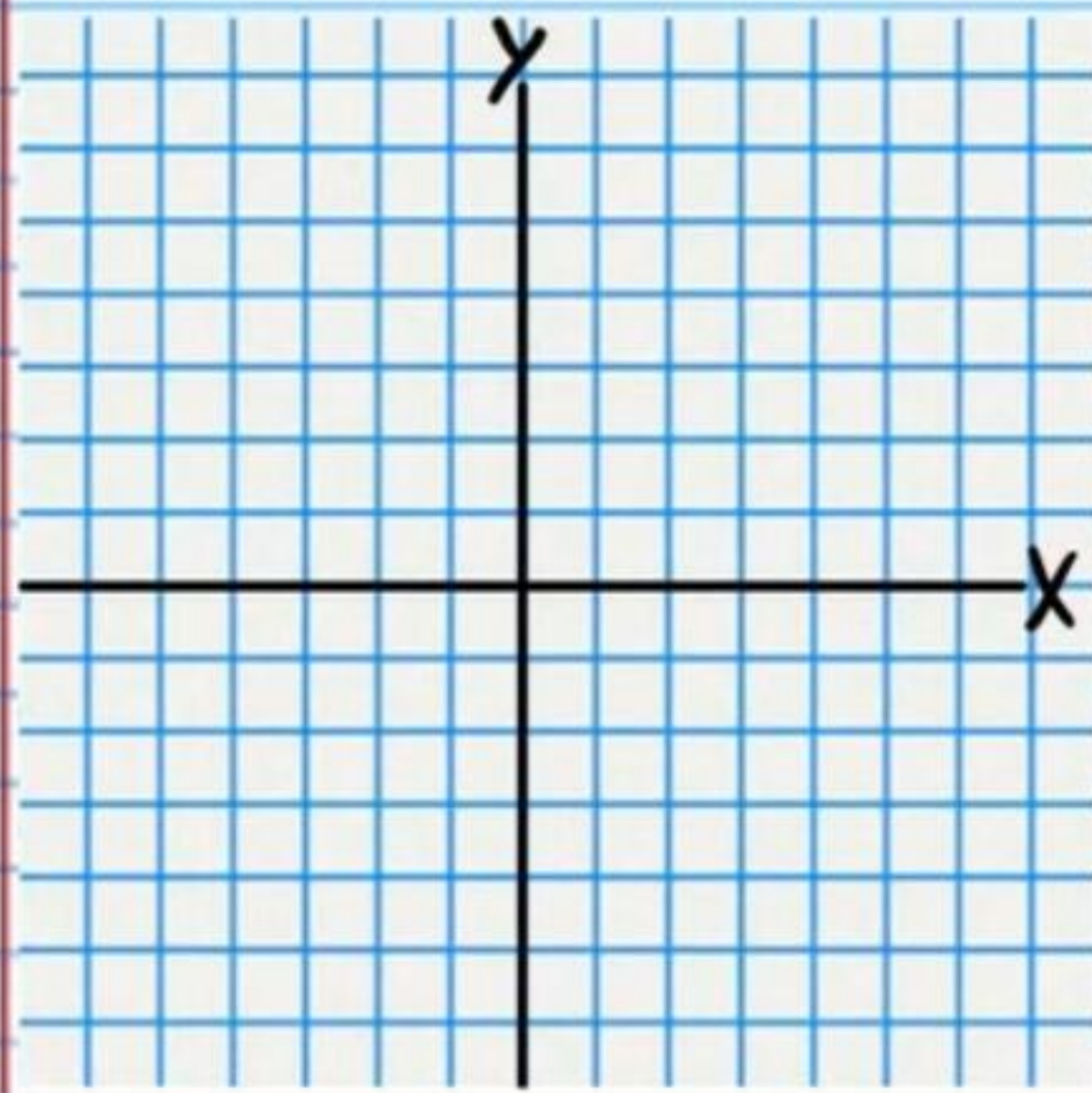


Transformation

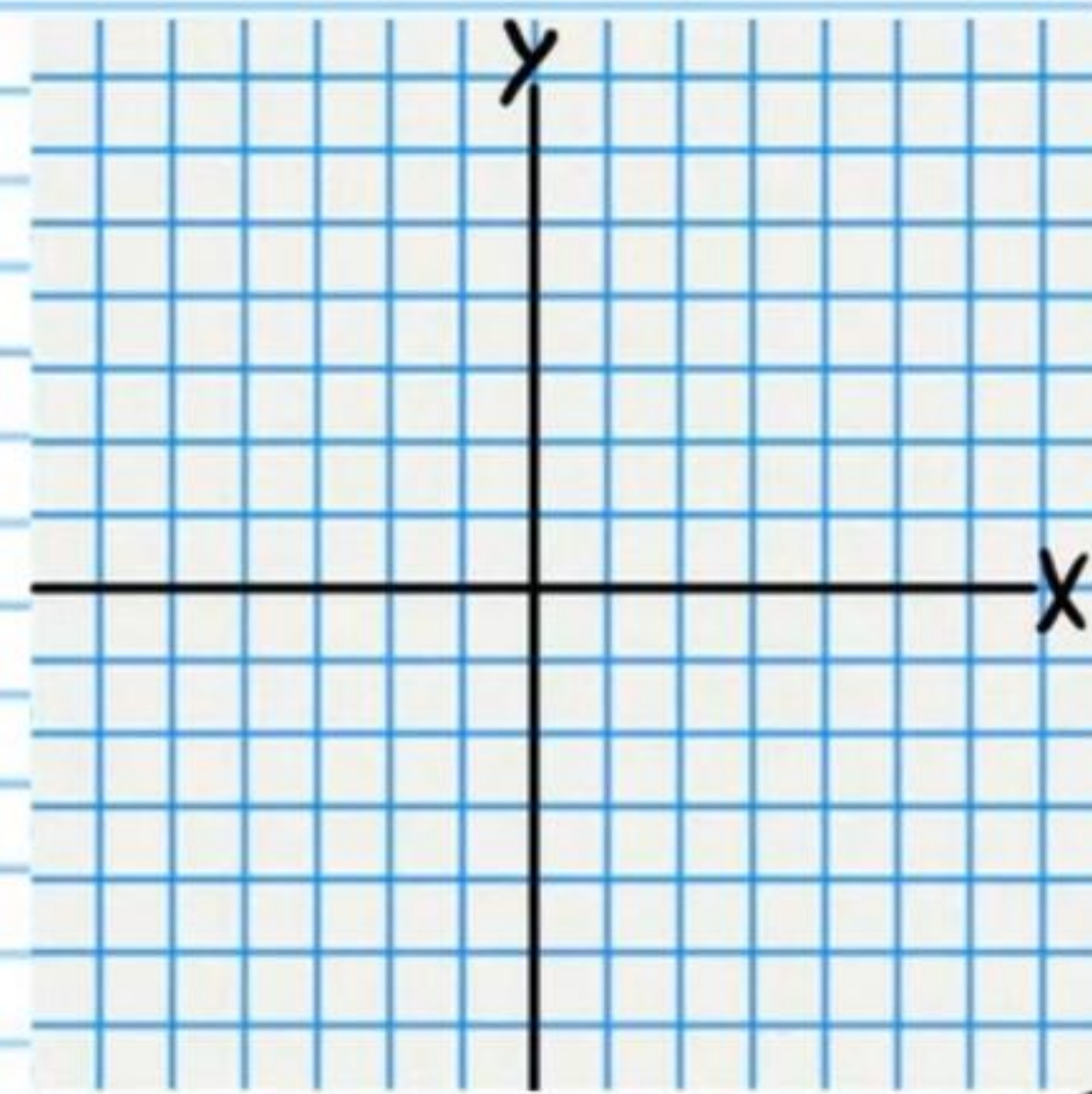


⑨ $y = 2(3^x)$

Parent Relation

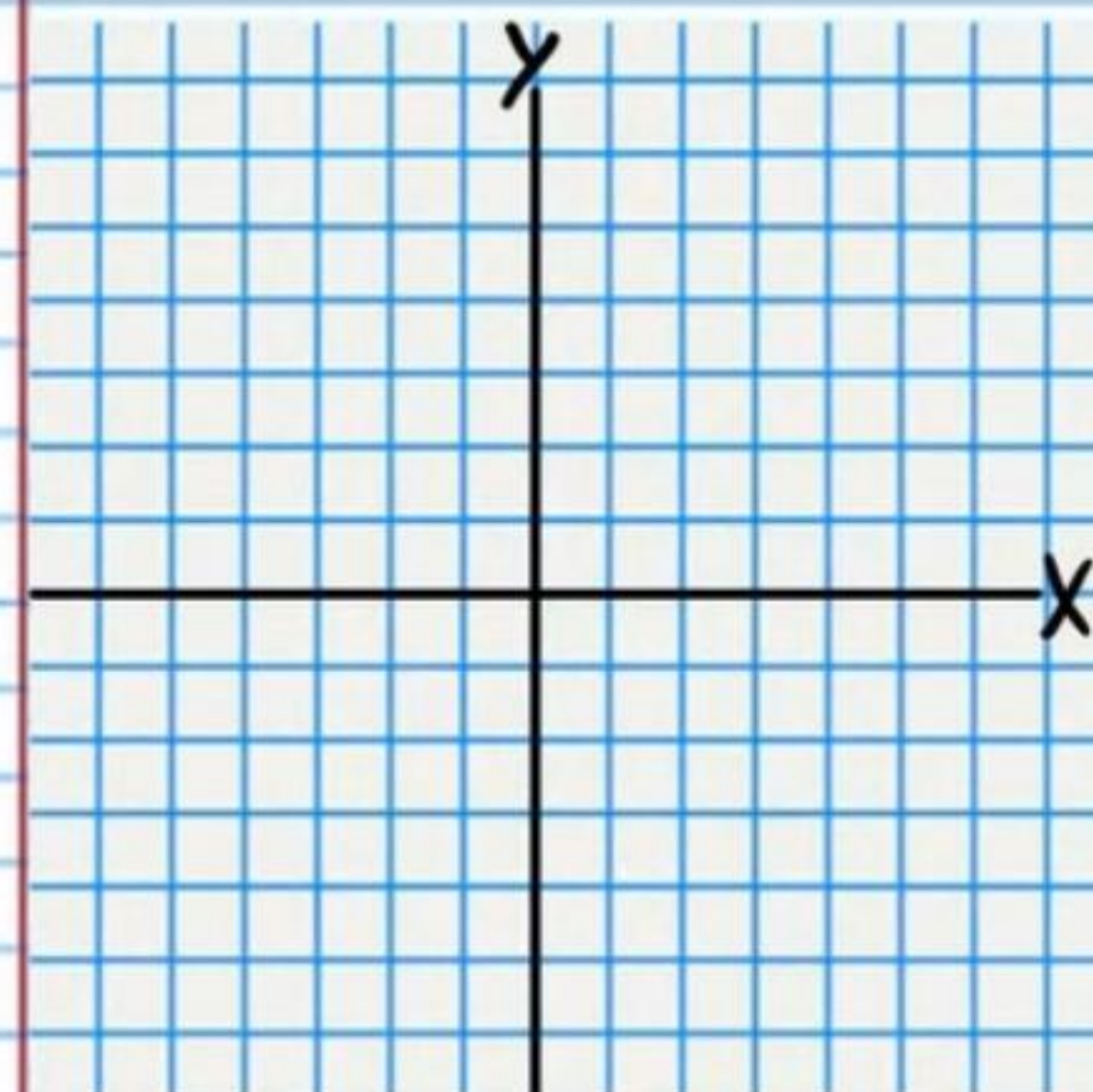


Transformation

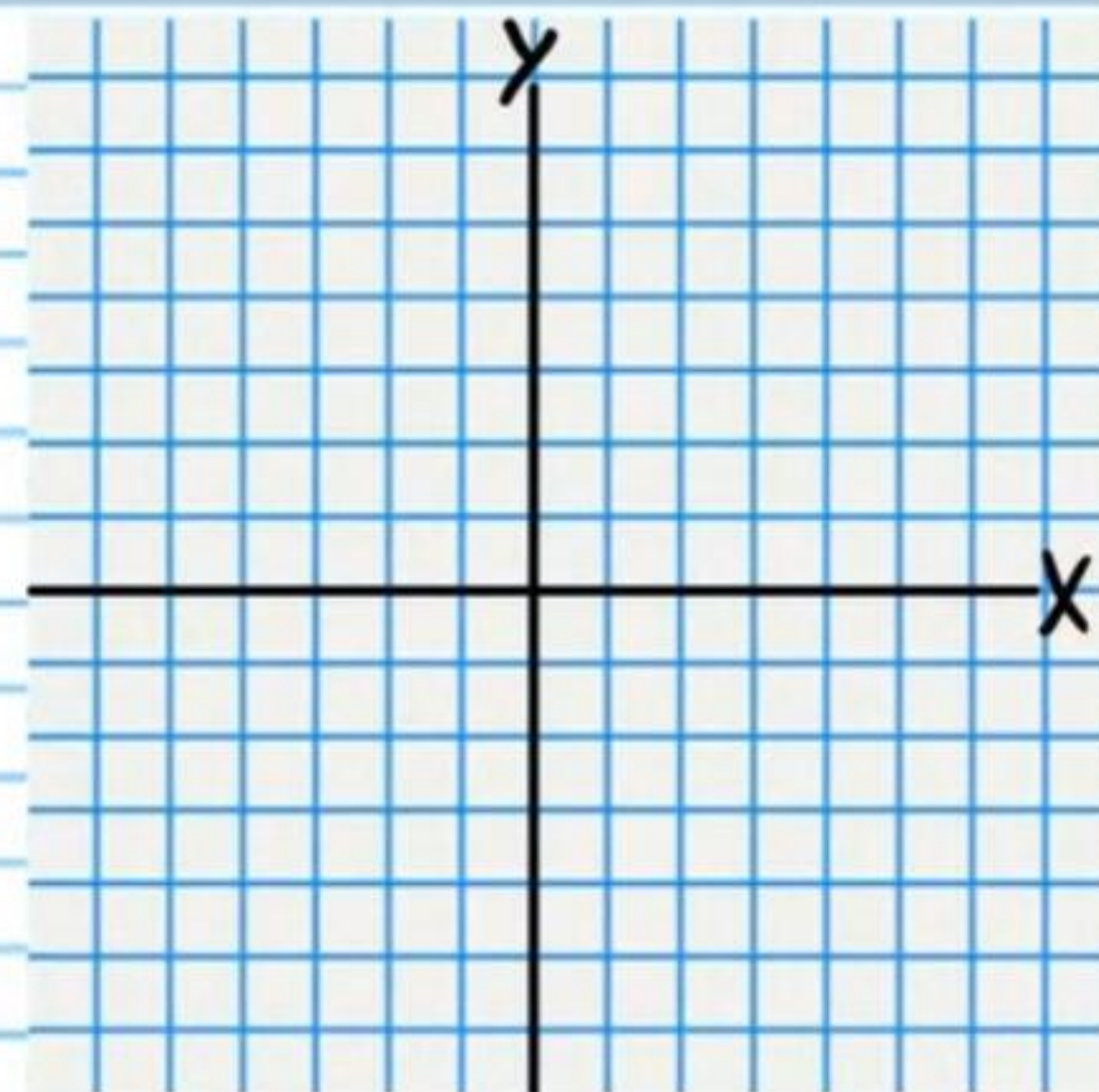


⑩ $y = \frac{4}{x}$

Parent Relation

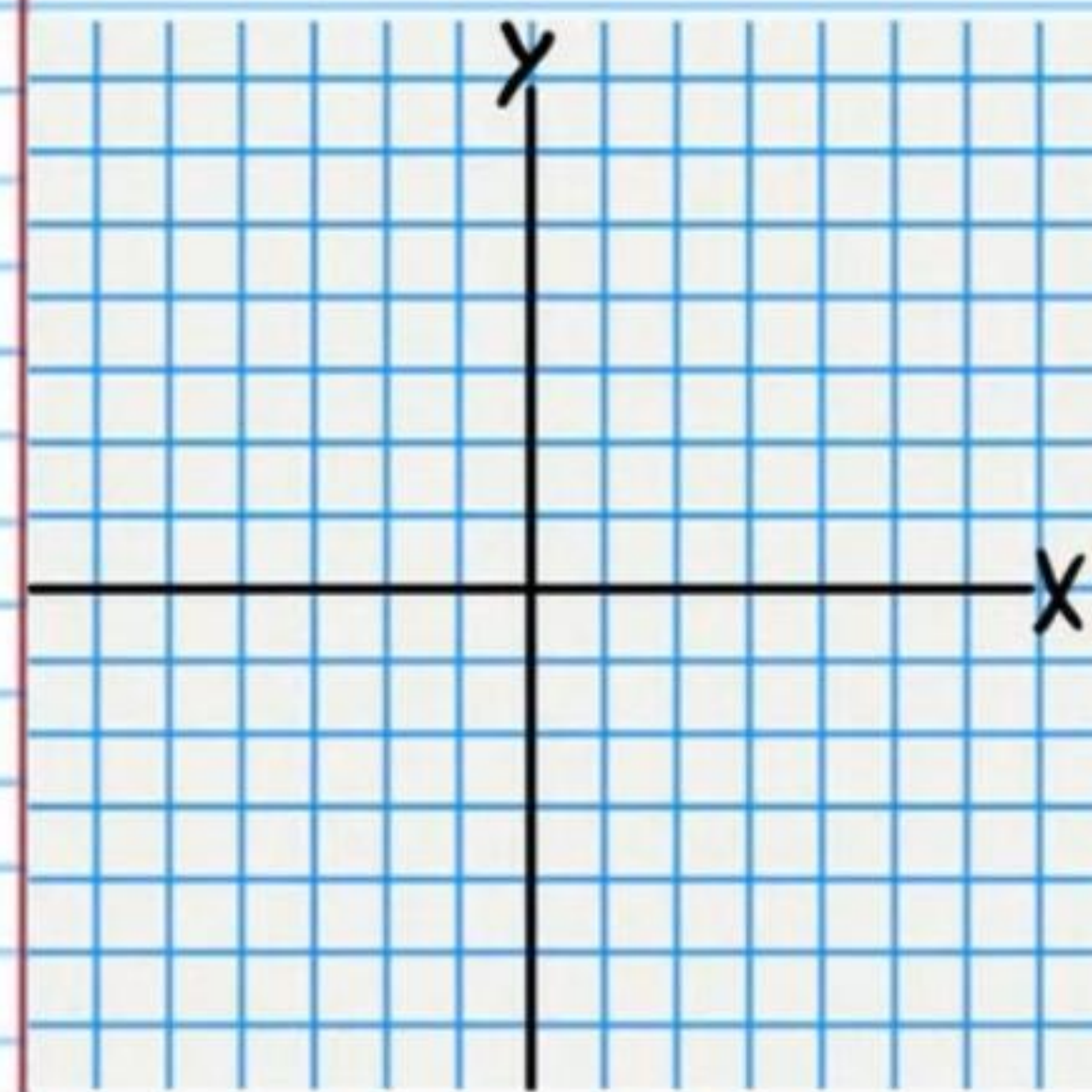


Transformation

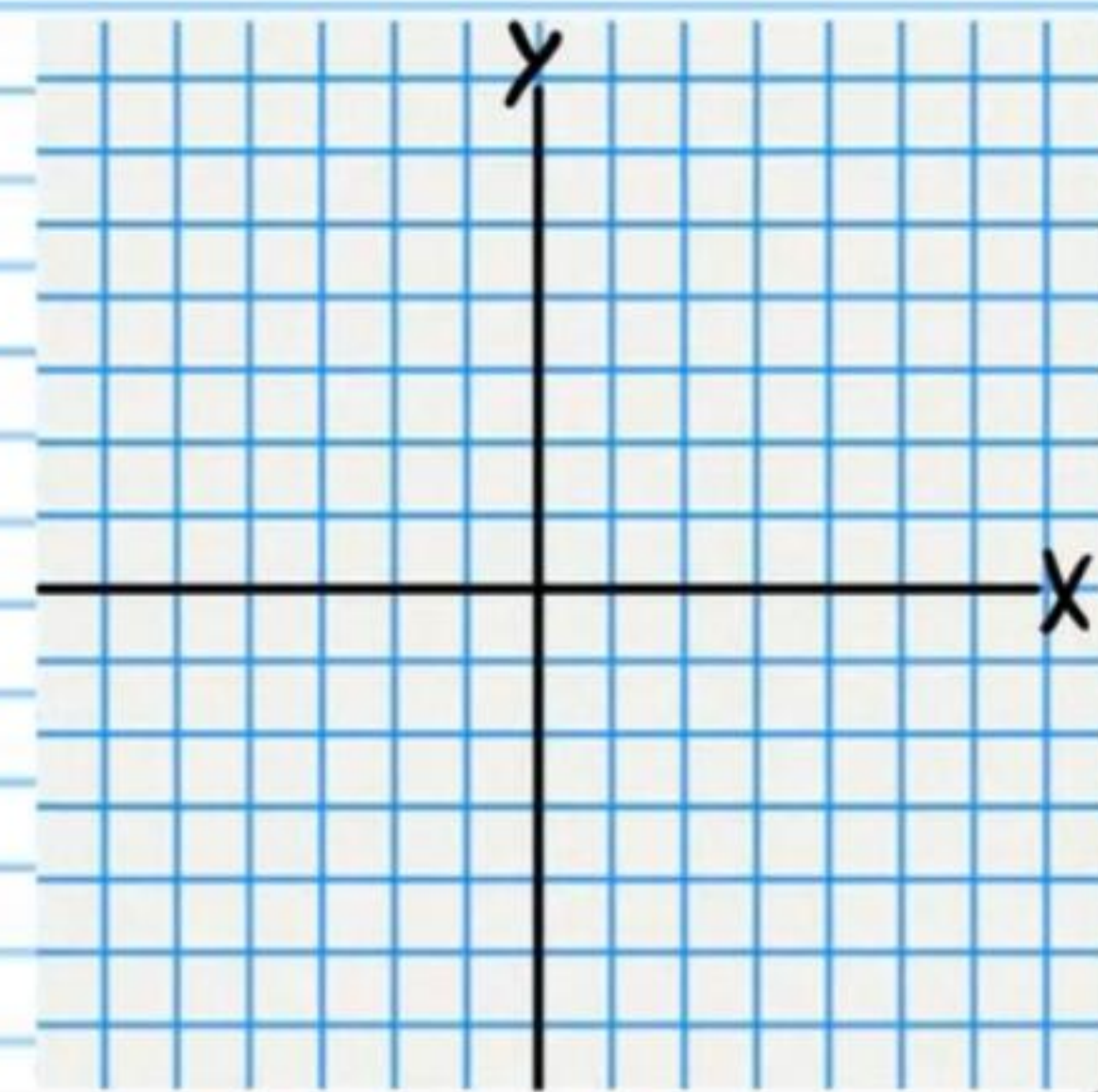


⑪ $y = -|x|$

Parent Relation

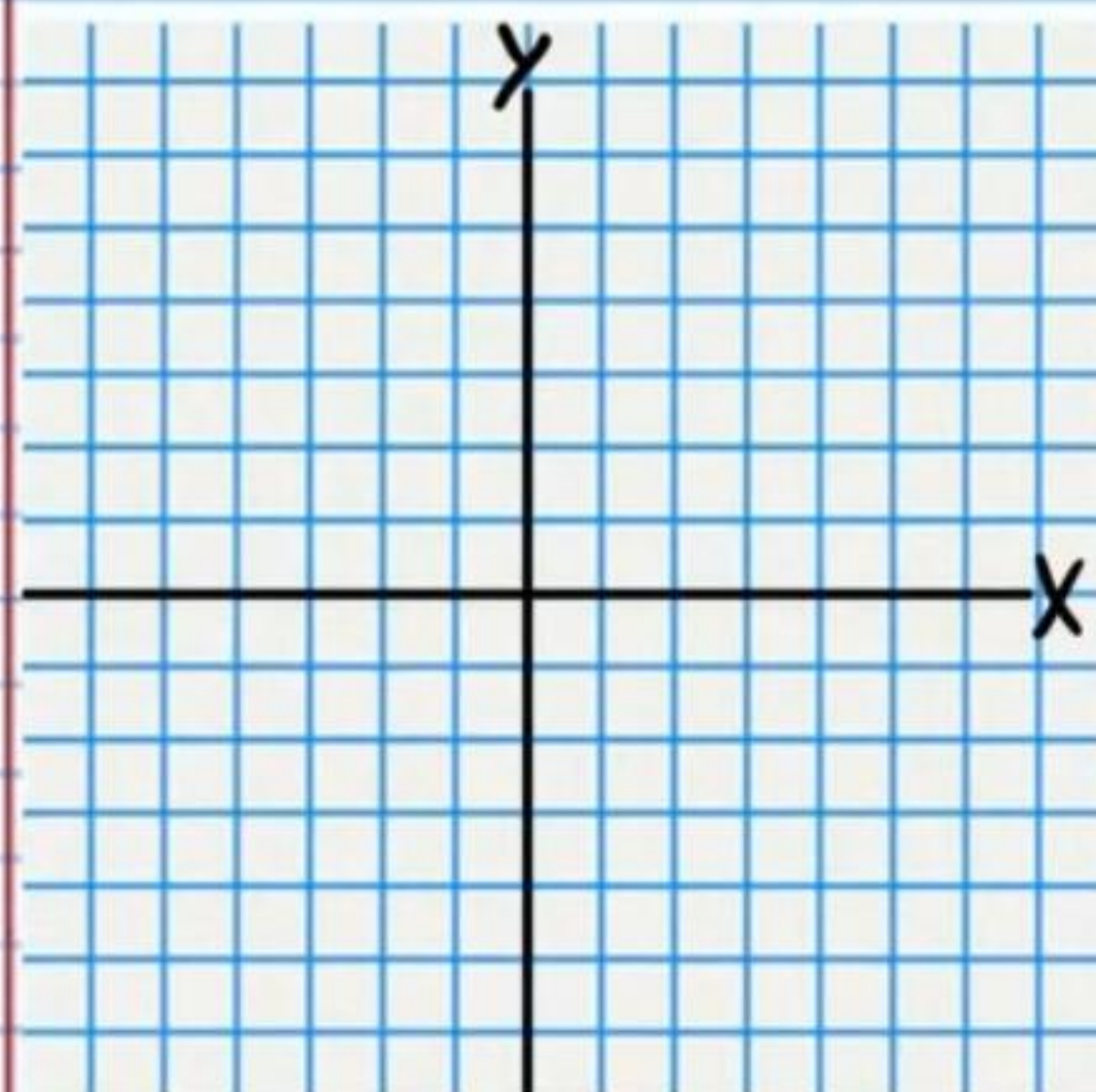


Transformation

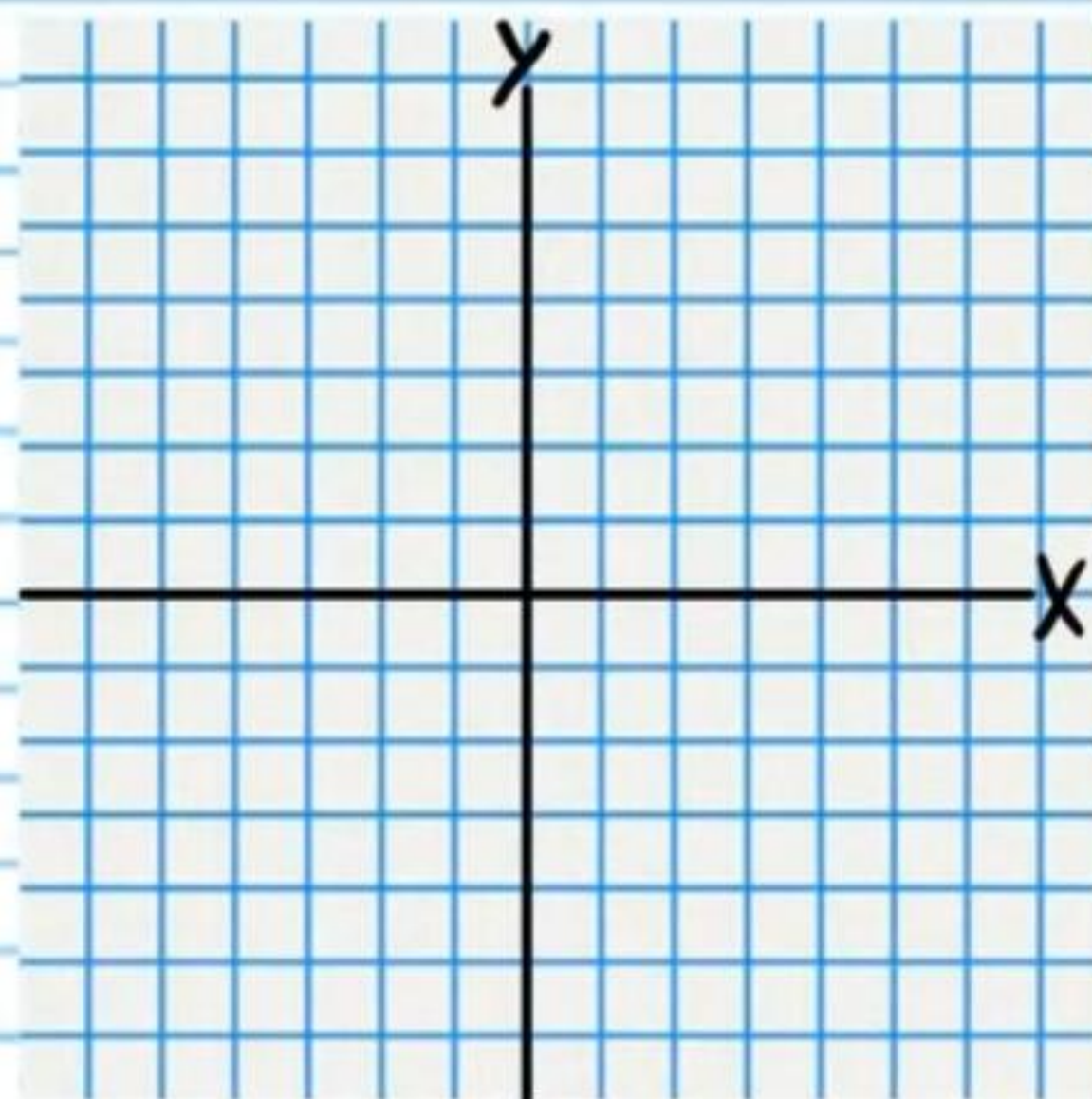


⑫ $y = -x^2$

Parent Relation

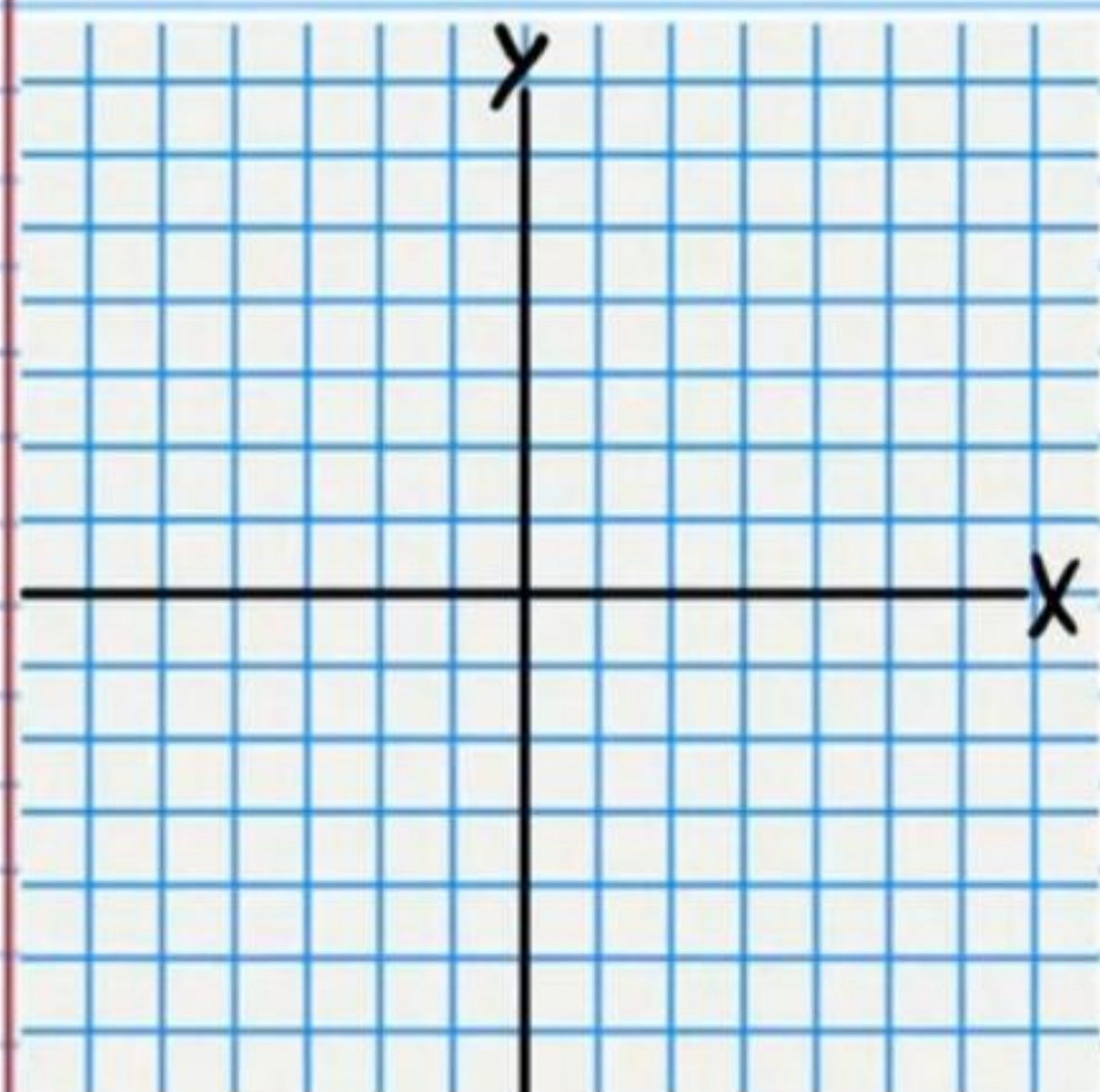


Transformation

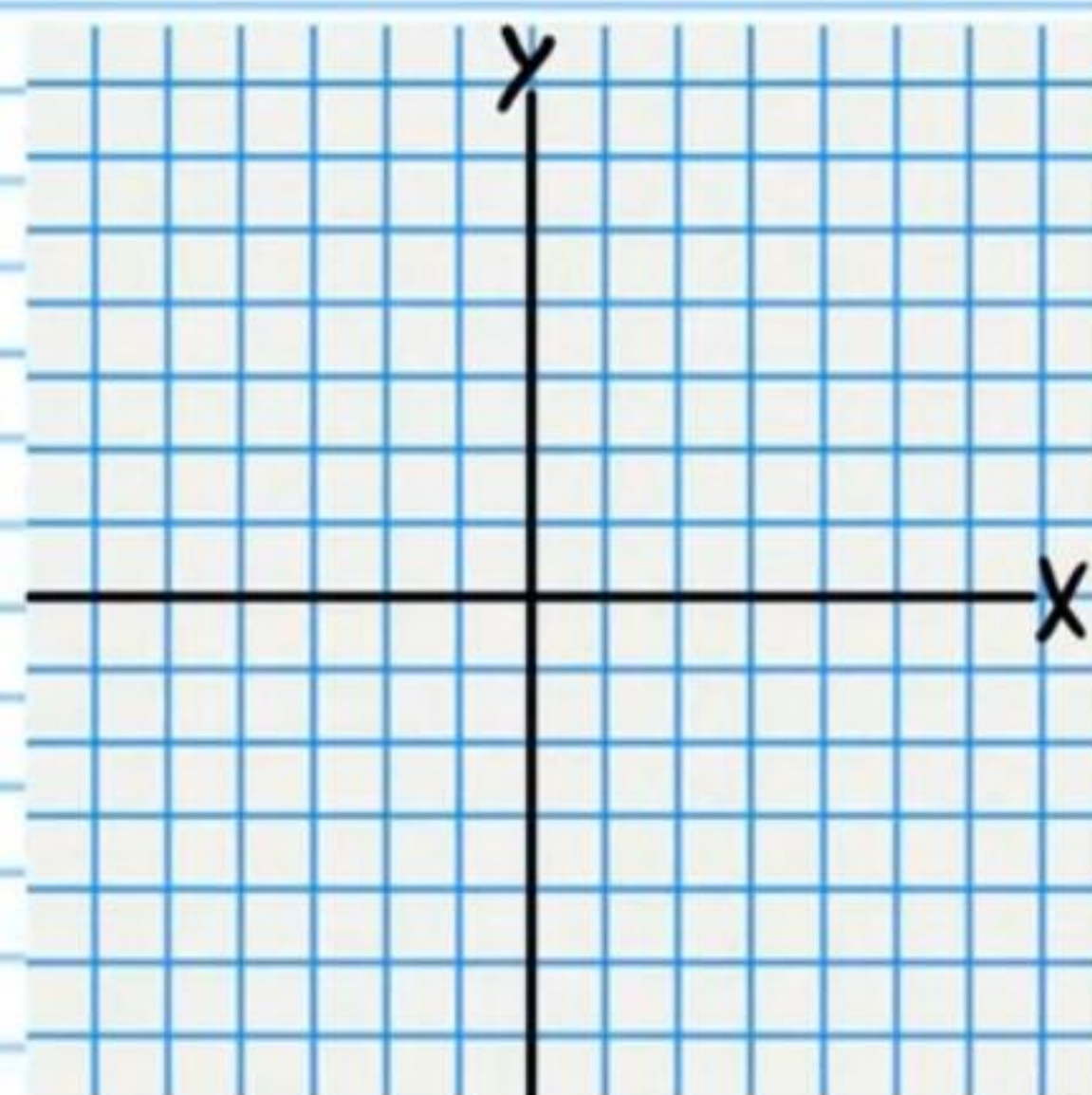


⑬ $y = -\log_2 X$

Parent Relation

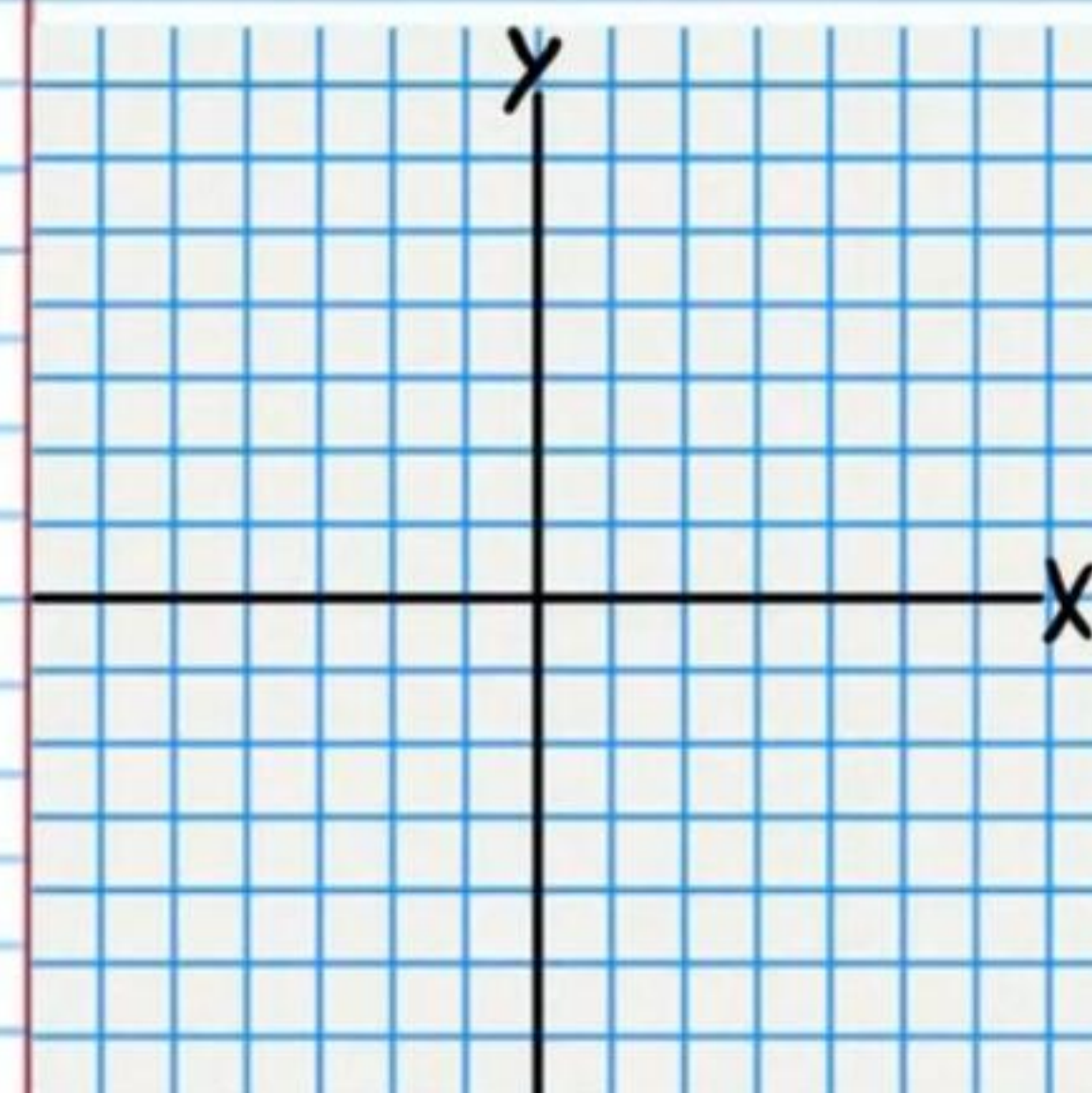


Transformation

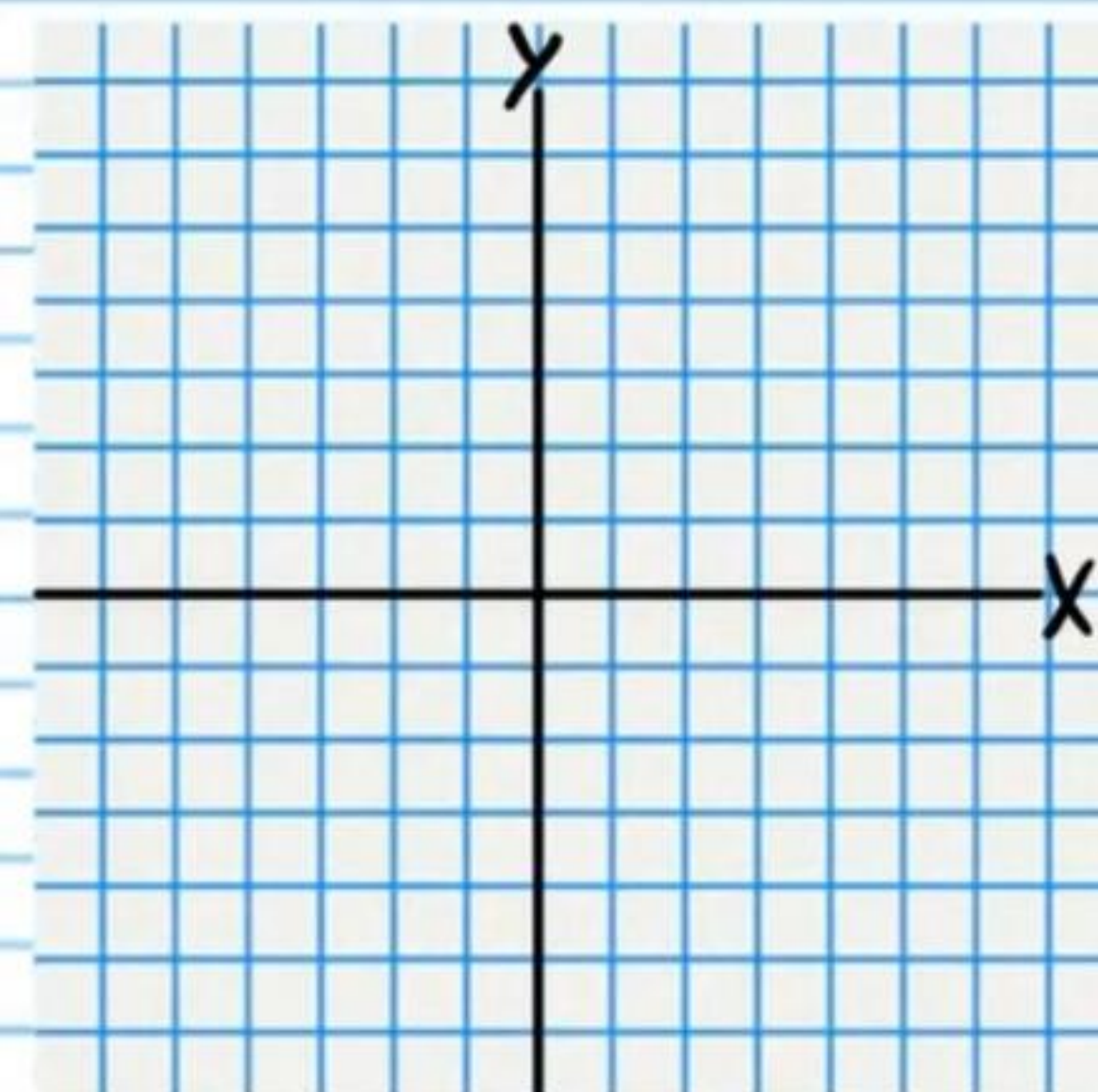


⑭ $y = -2^x$

Parent Relation

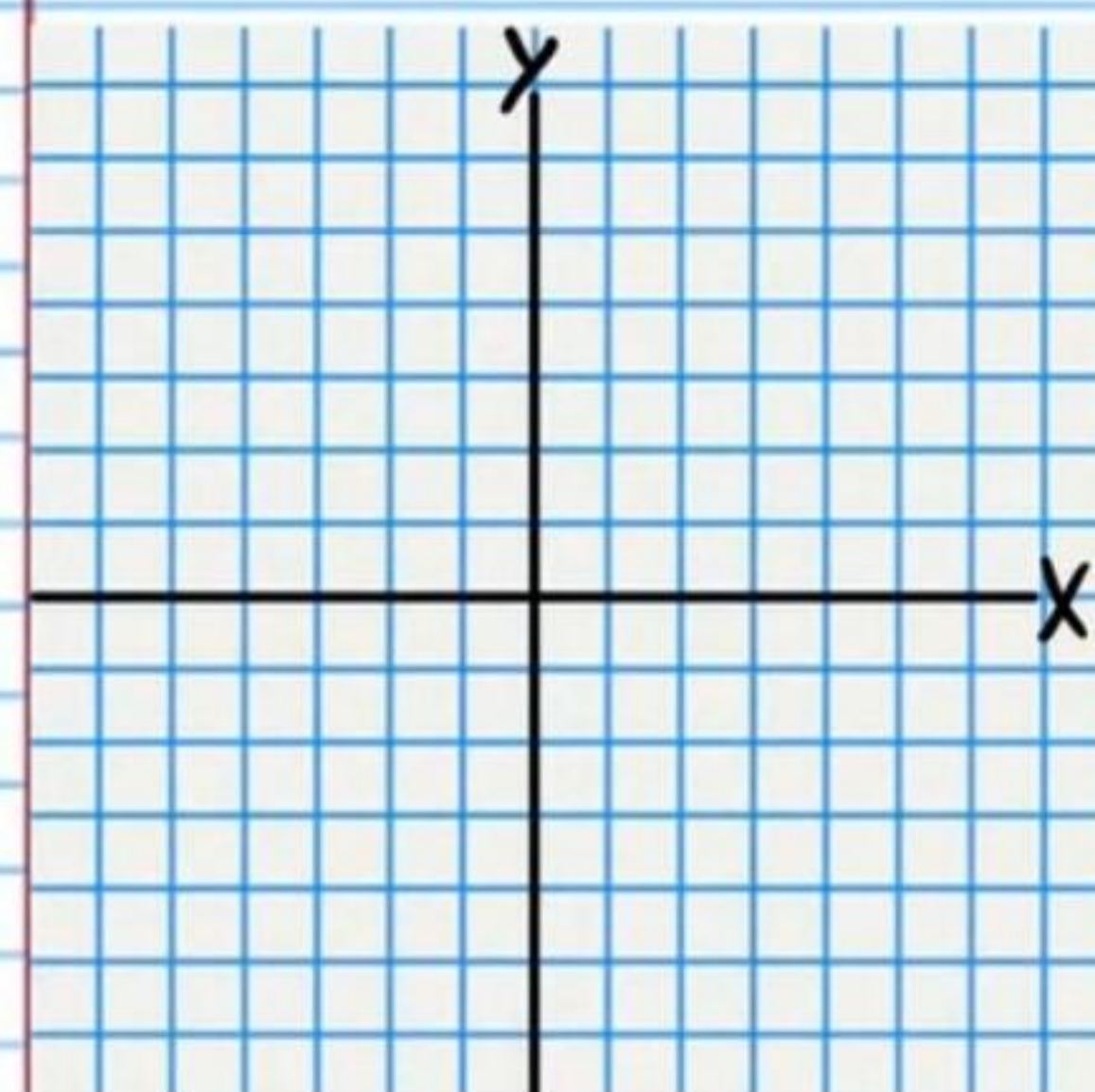


Transformation

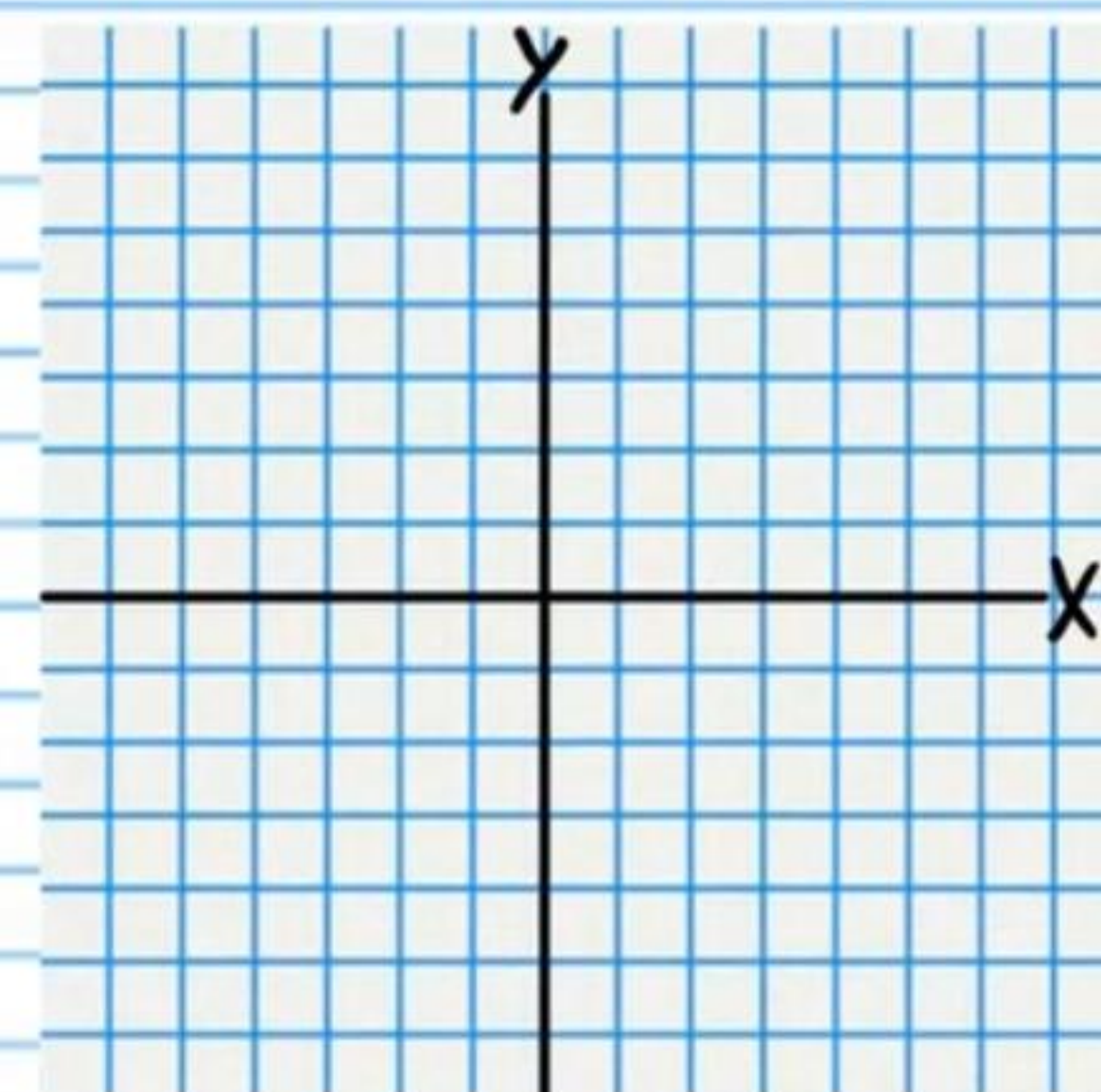


⑮ $y = -\frac{1}{x}$

Parent Relation



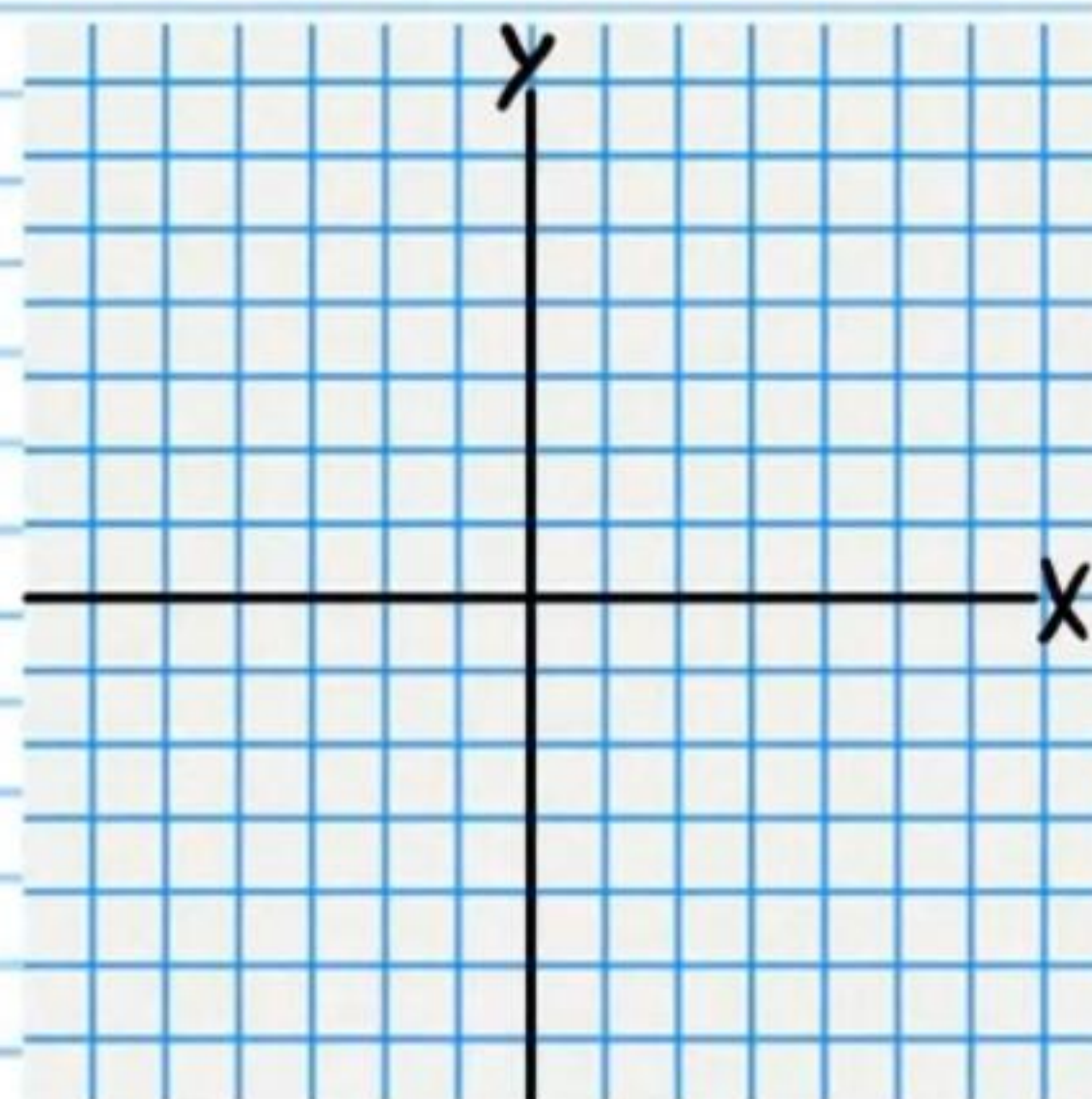
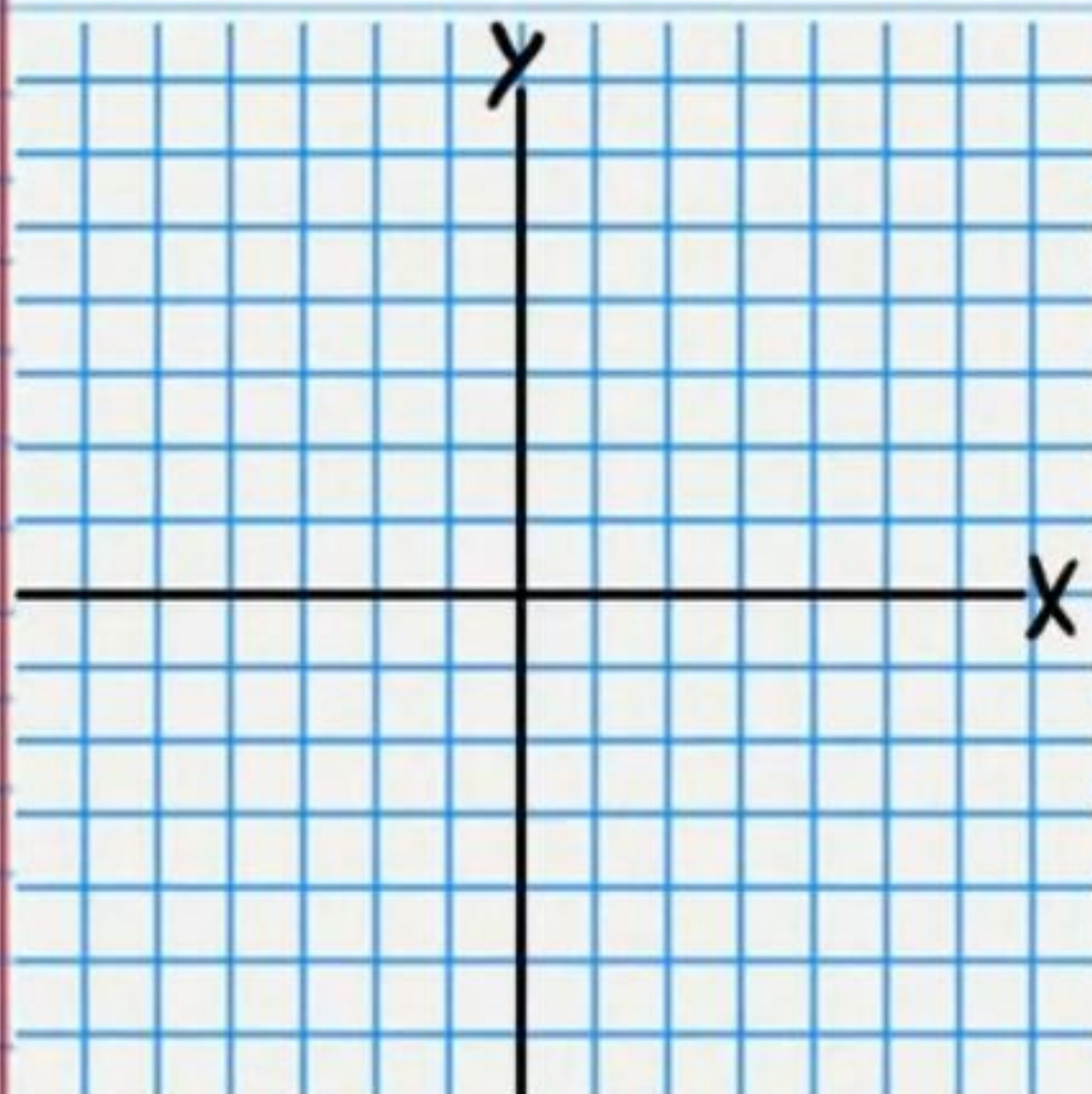
Transformation



⑩ $y = (x-3)^2 + 1$

Parent Relation

Transformation

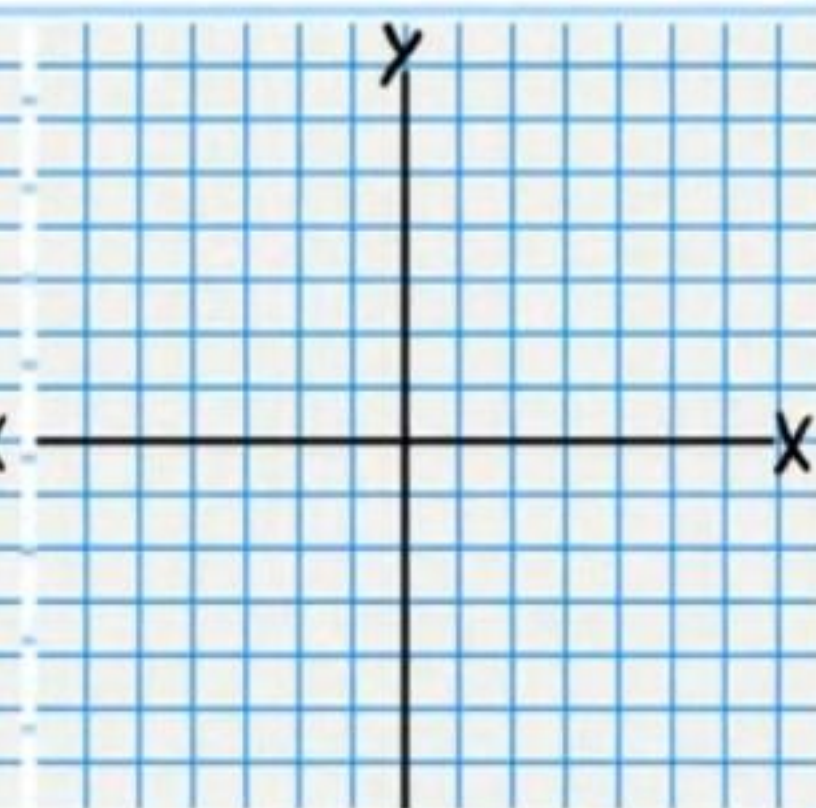
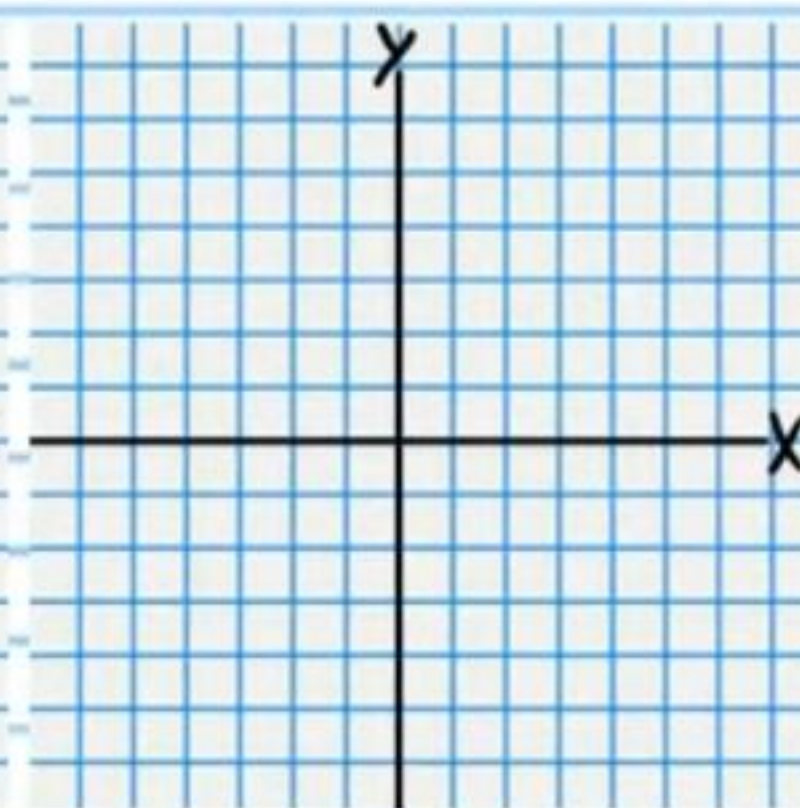
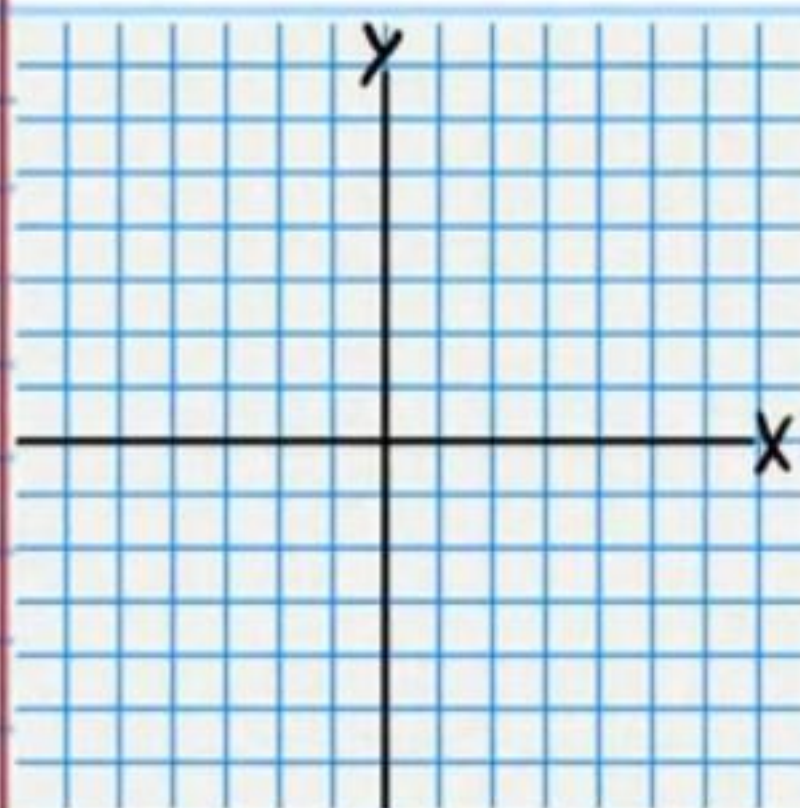


⑪ $y = 3|x+2|$

Parent Relation

Transformation #1

Transformation #2

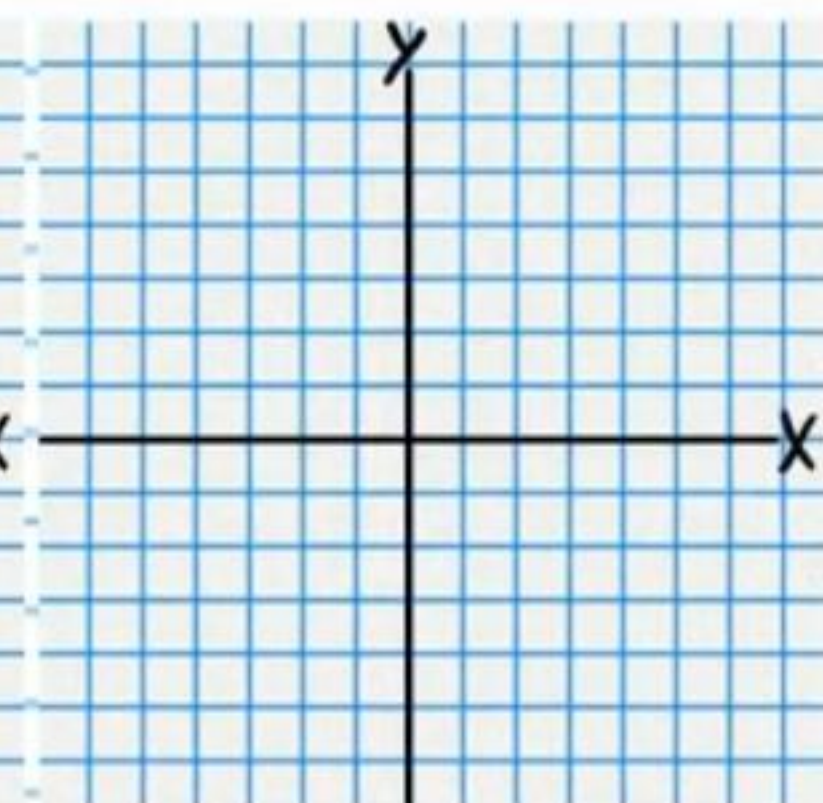
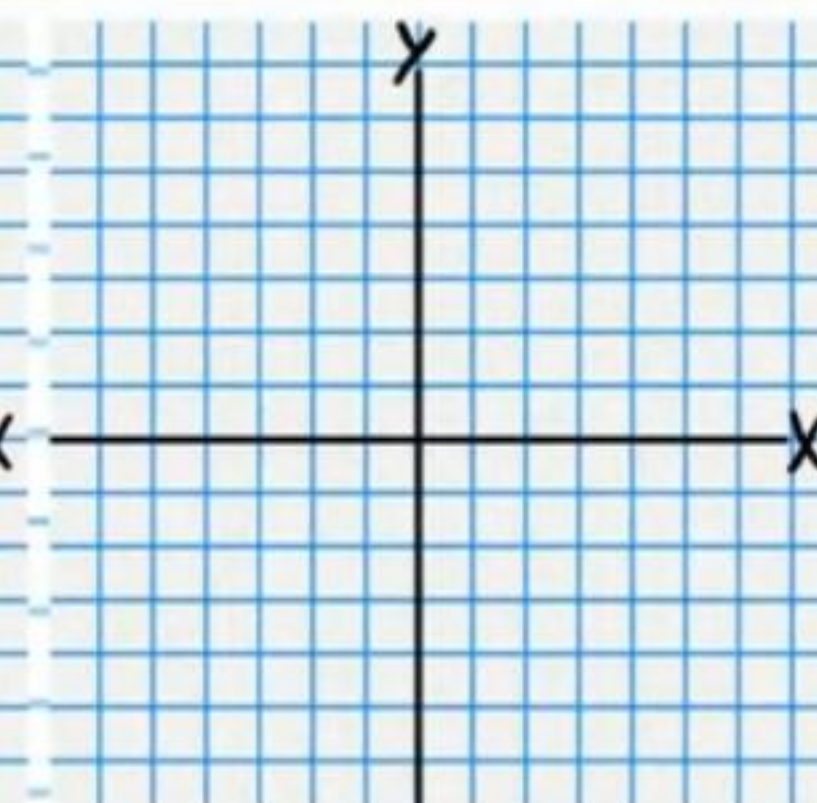
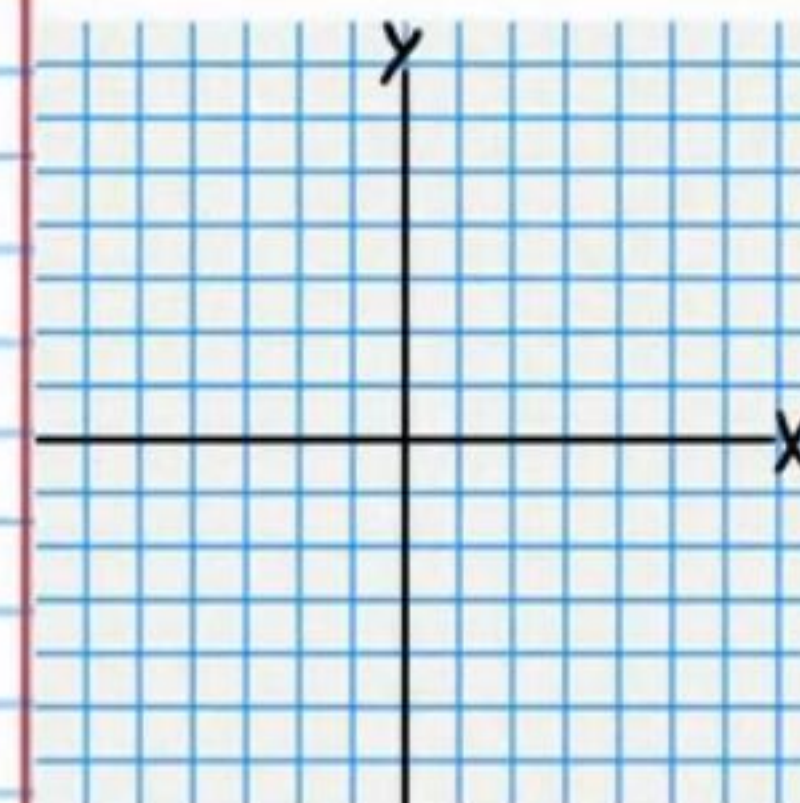


⑫ $y = 2(3^{-x})$

Parent Relation

Transformation #1

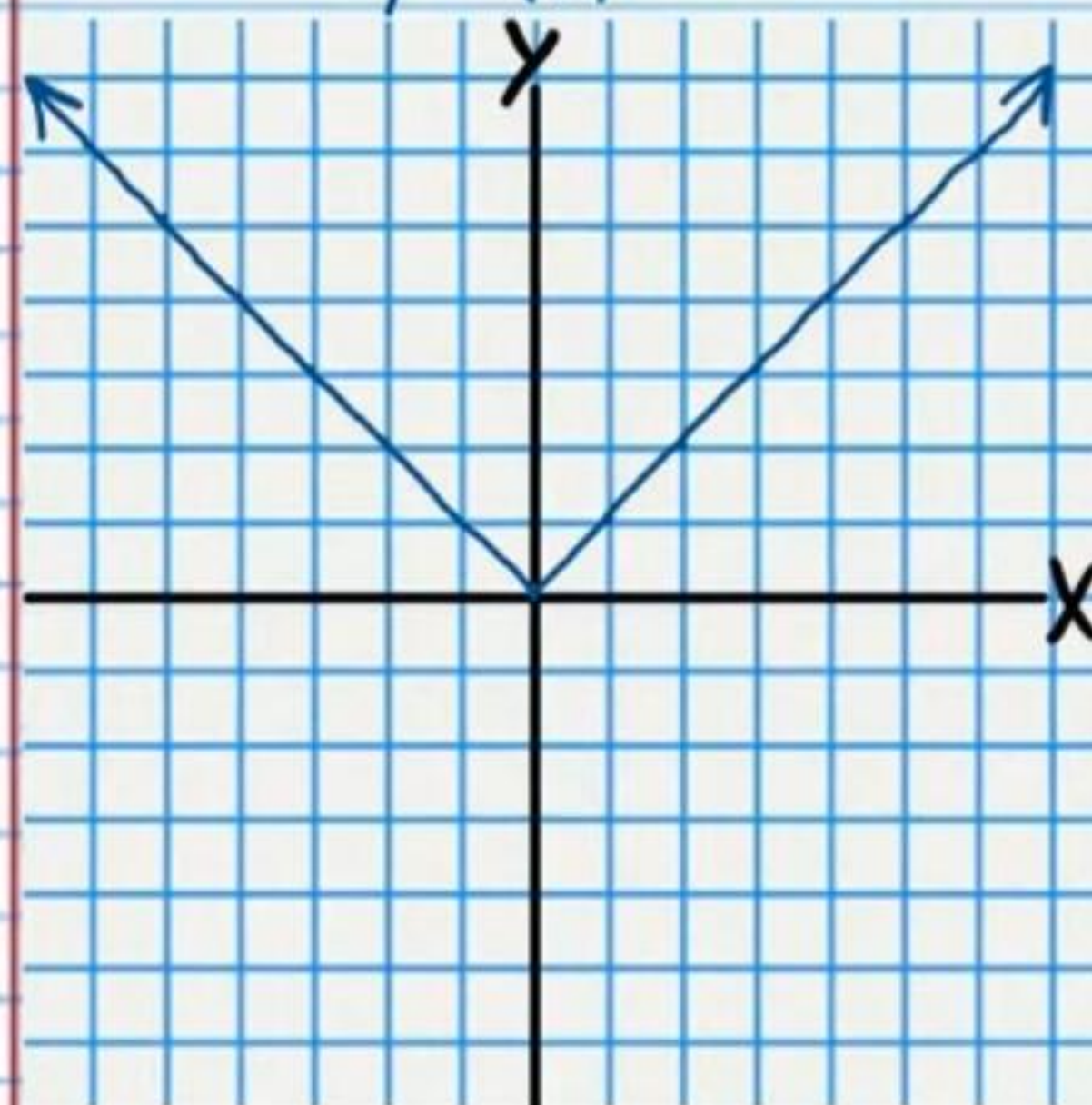
Transformation #2



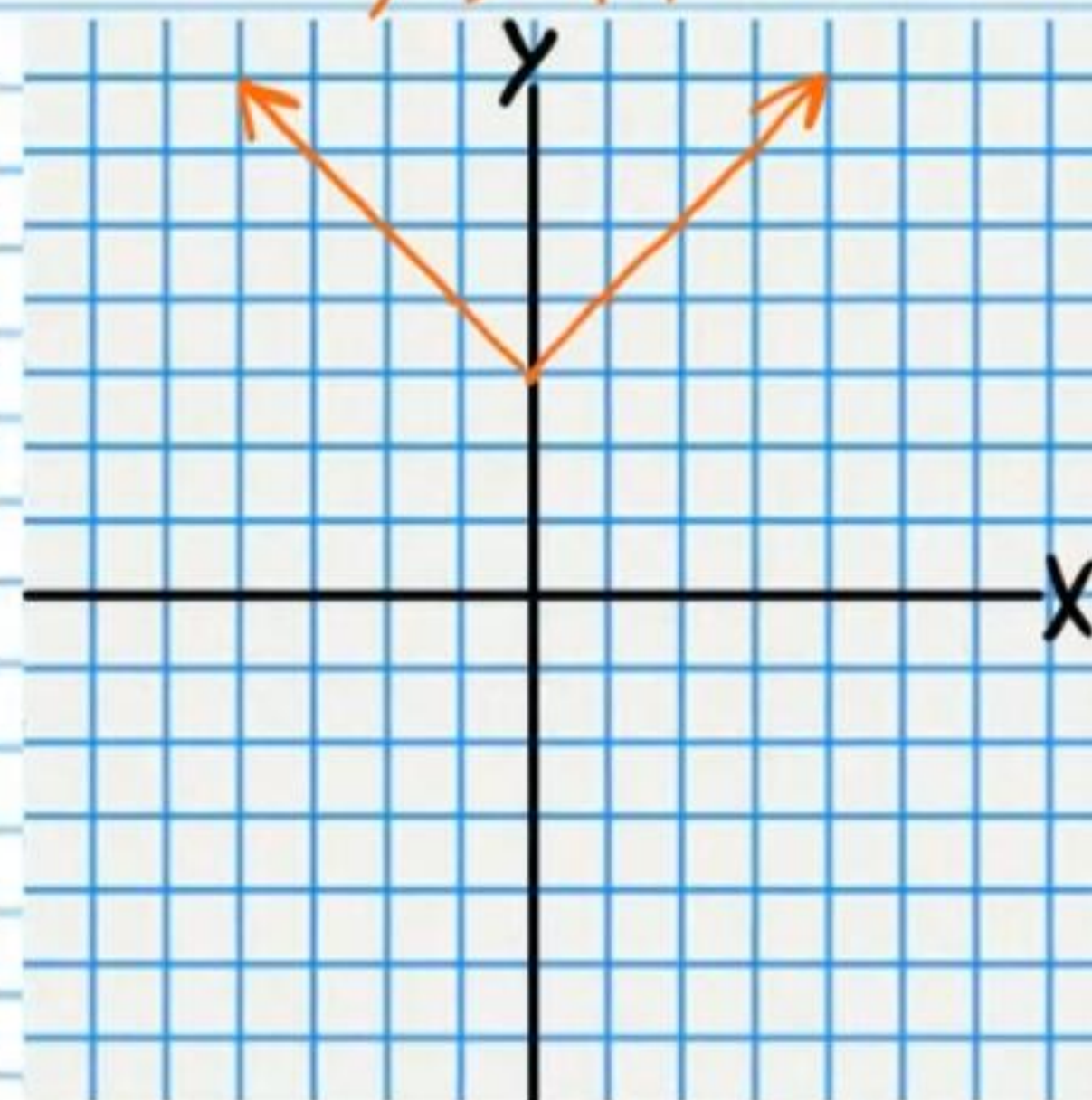
Answers

① $y = |x| + 3$

Parent Relation
 $y = |x|$

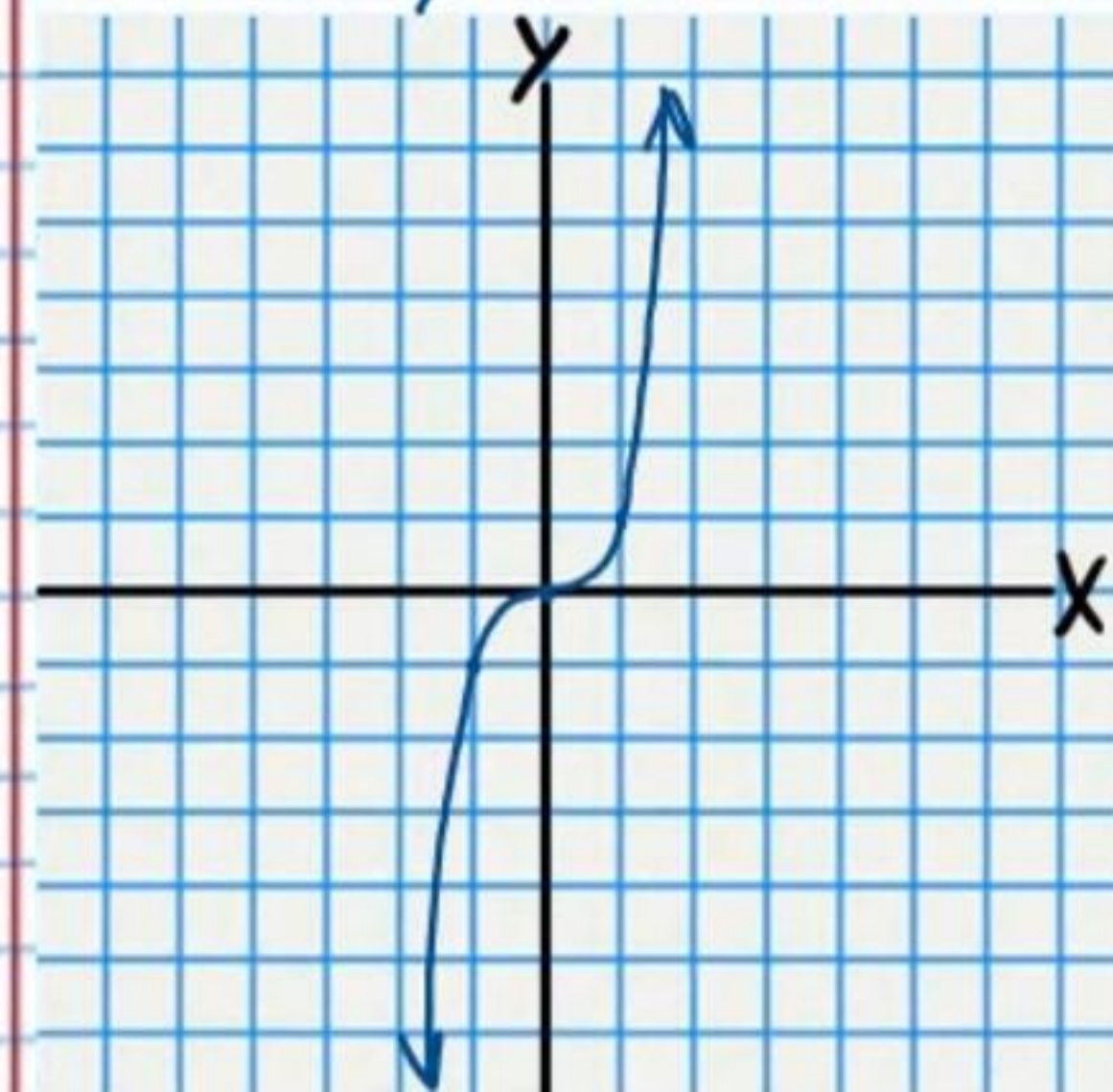


Transformation
 $y - 3 = |x|$

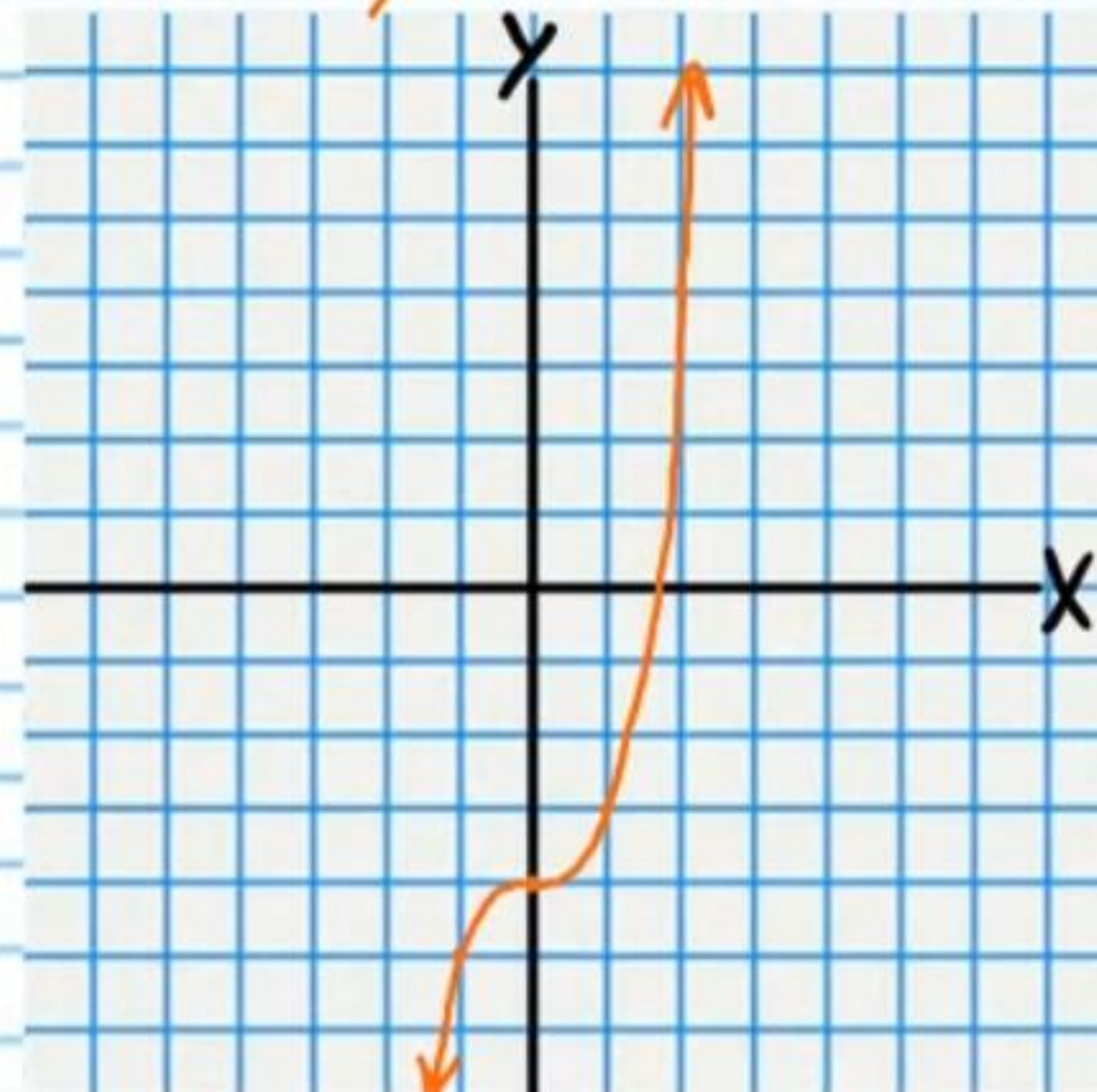


② $y = x^3 - 4$

Parent Relation
 $y = x^3$

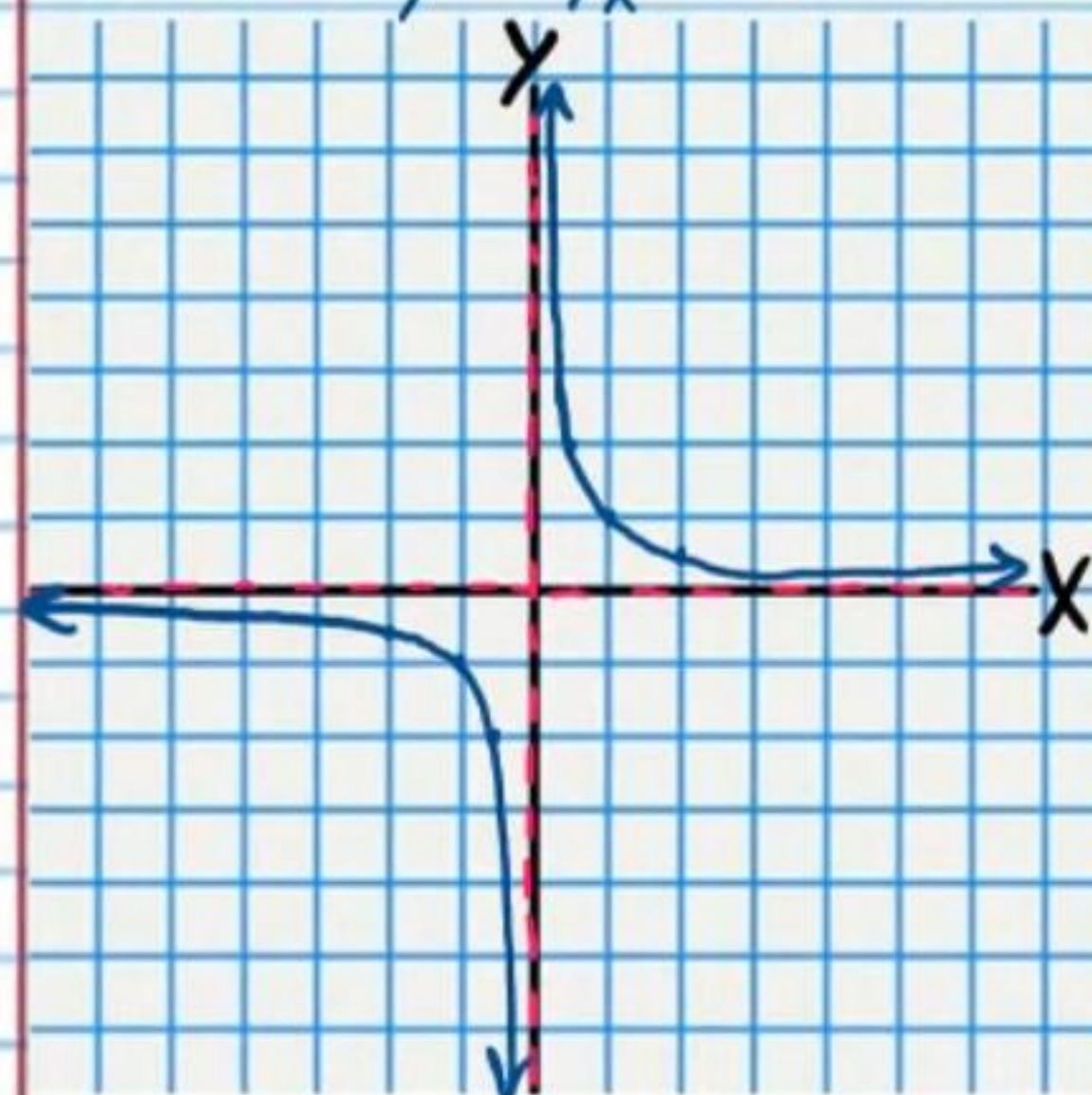


Transformation
 $y + 4 = x^3$

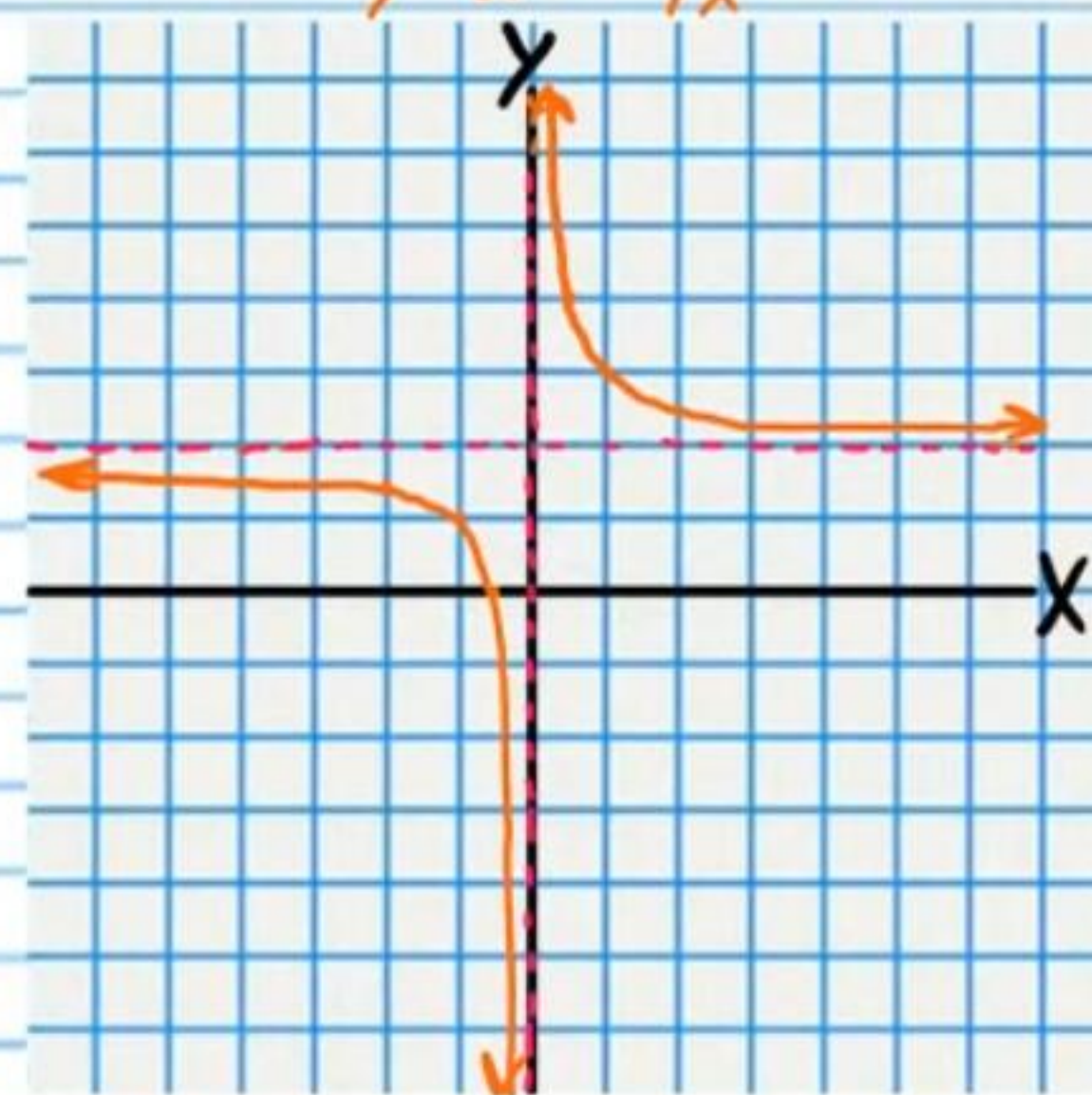


③ $y = \frac{1}{x} + 2$

Parent Relation
 $y = \frac{1}{x}$

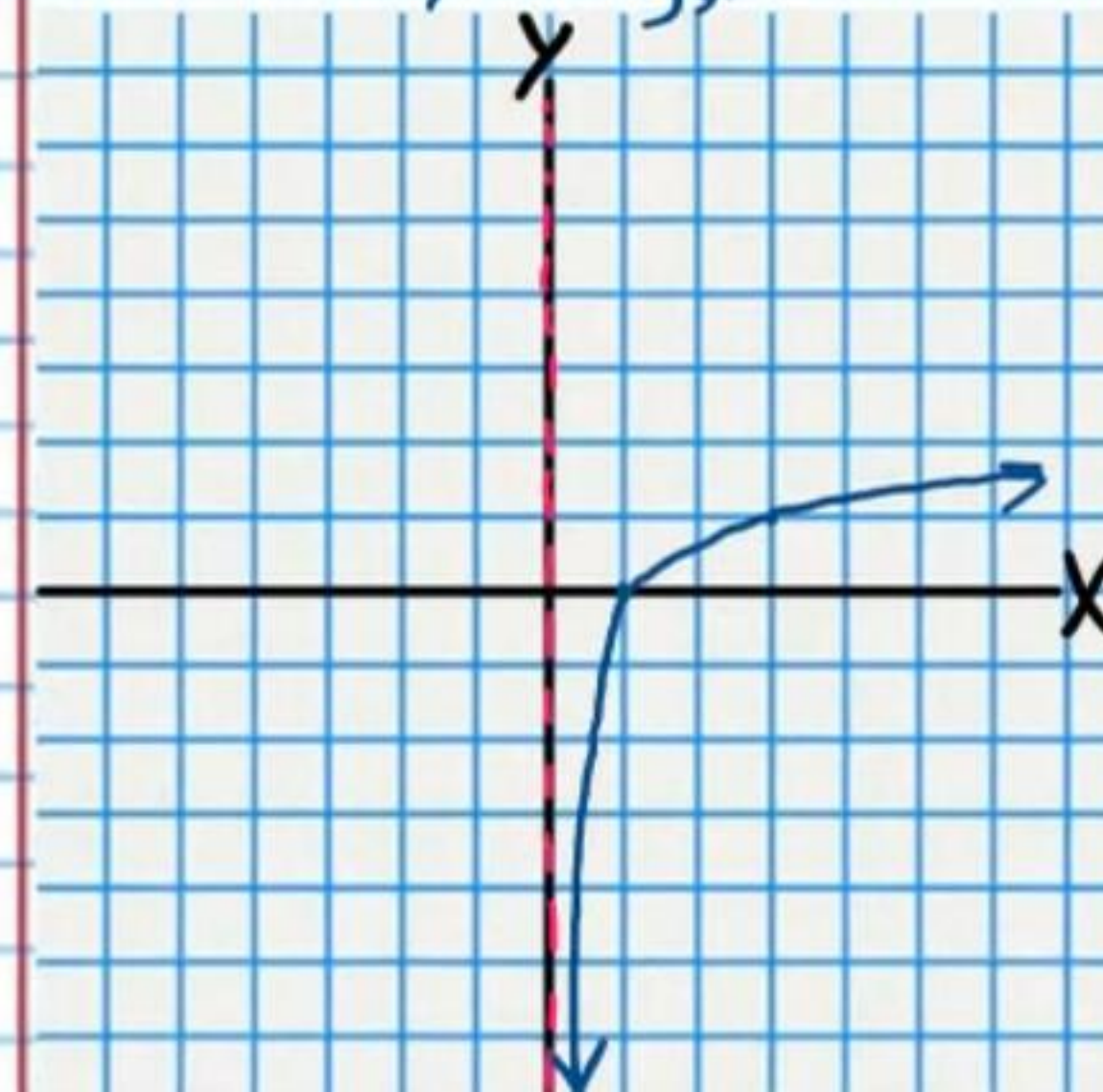


Transformation
 $y - 2 = \frac{1}{x}$

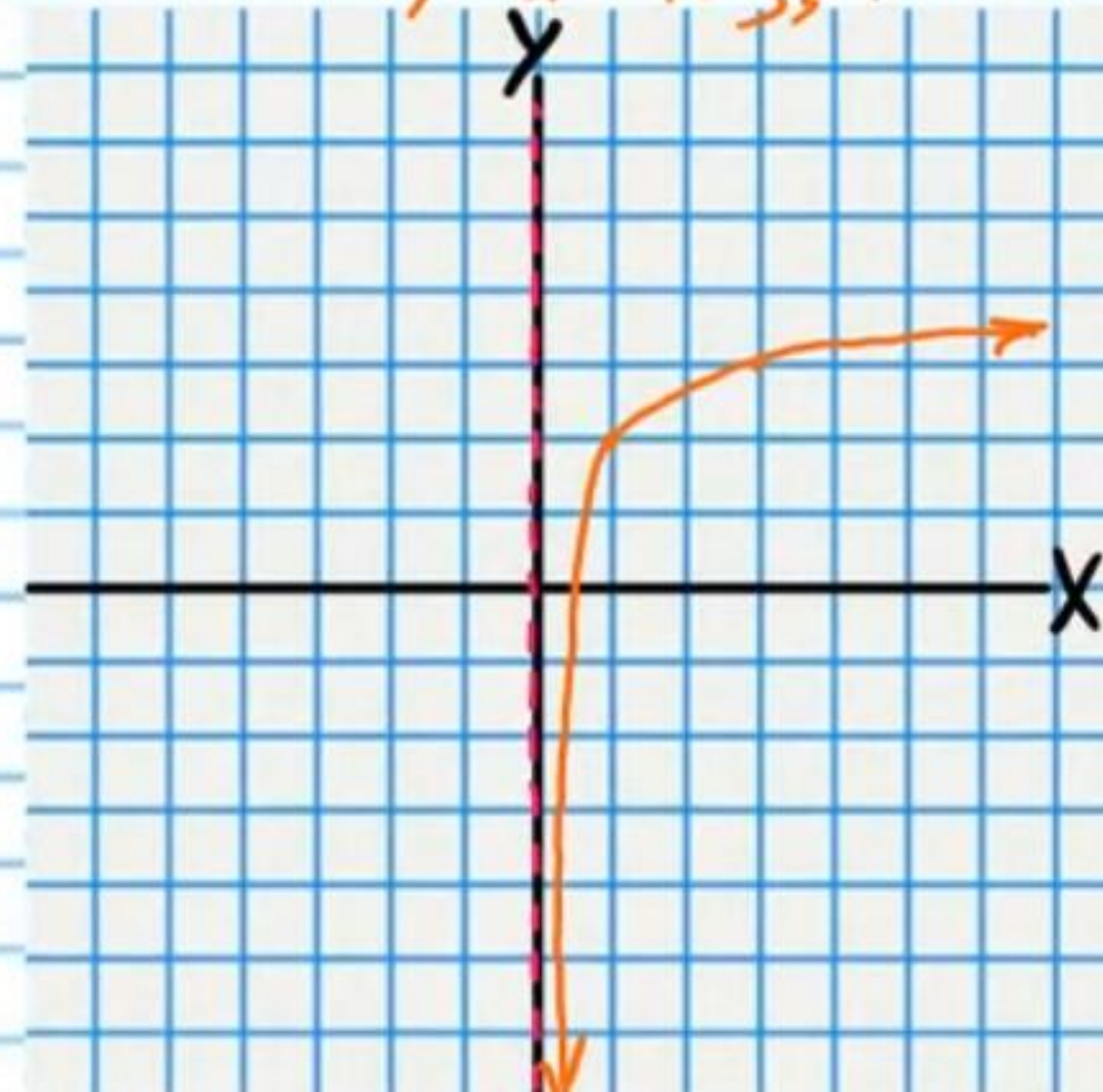


④ $y = \log_3 x + 2$

Parent Relation
 $y = \log_3 x$

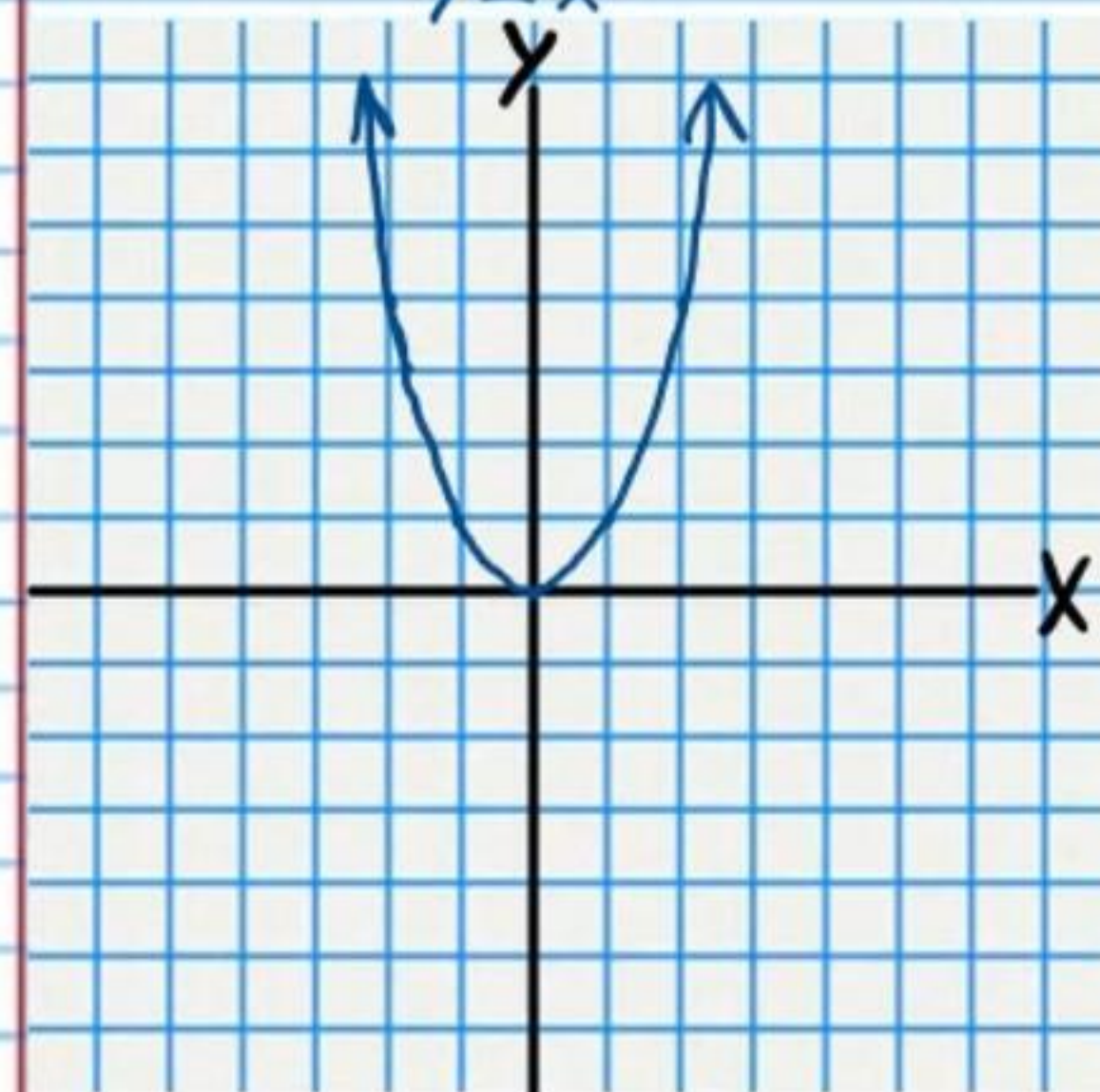


Transformation
 $y - 2 = \log_3 x$

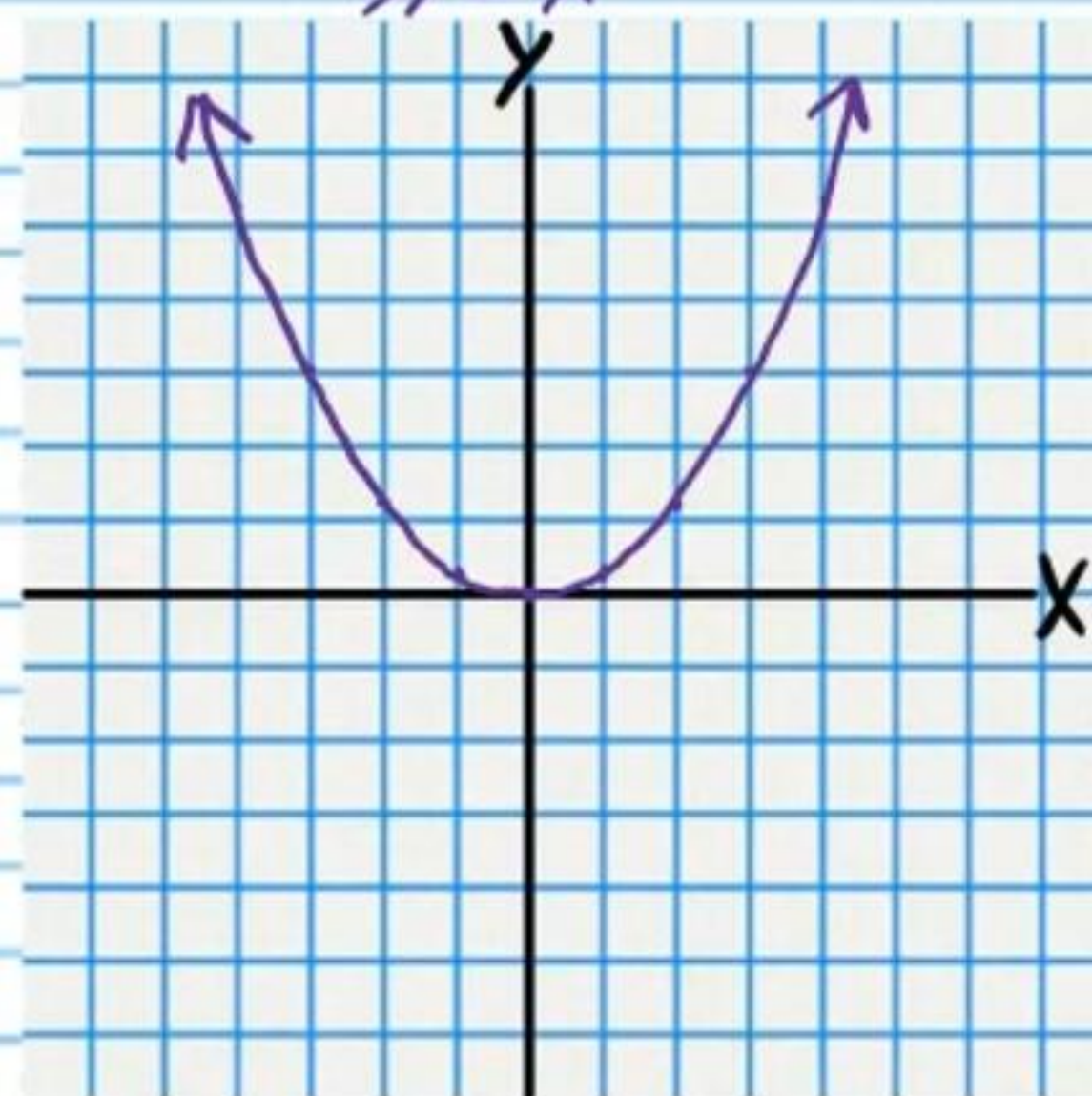


⑤ $y = \frac{x^2}{3}$

Parent Relation
 $y = x^2$

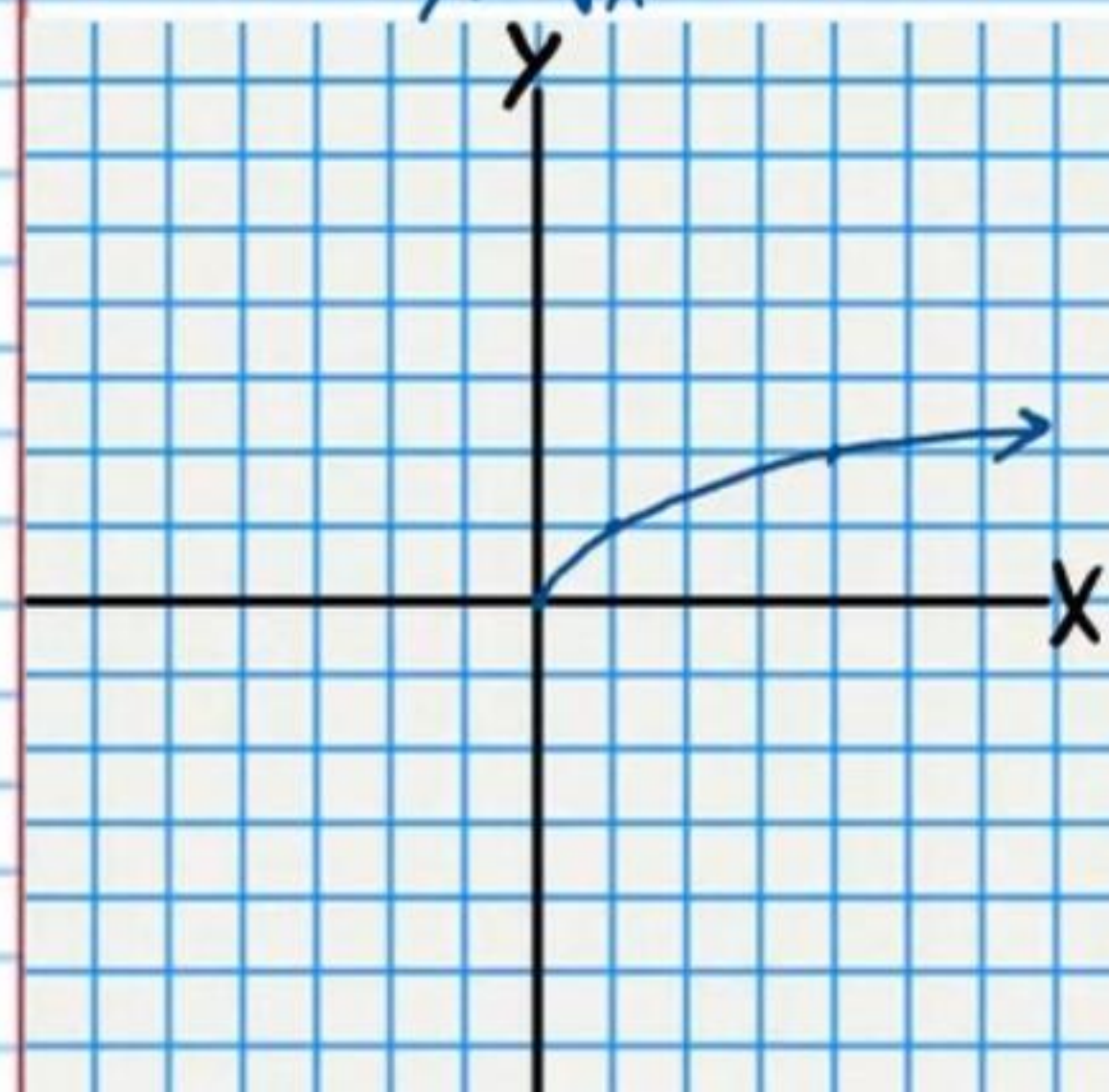


Transformation
 $3y = x^2$

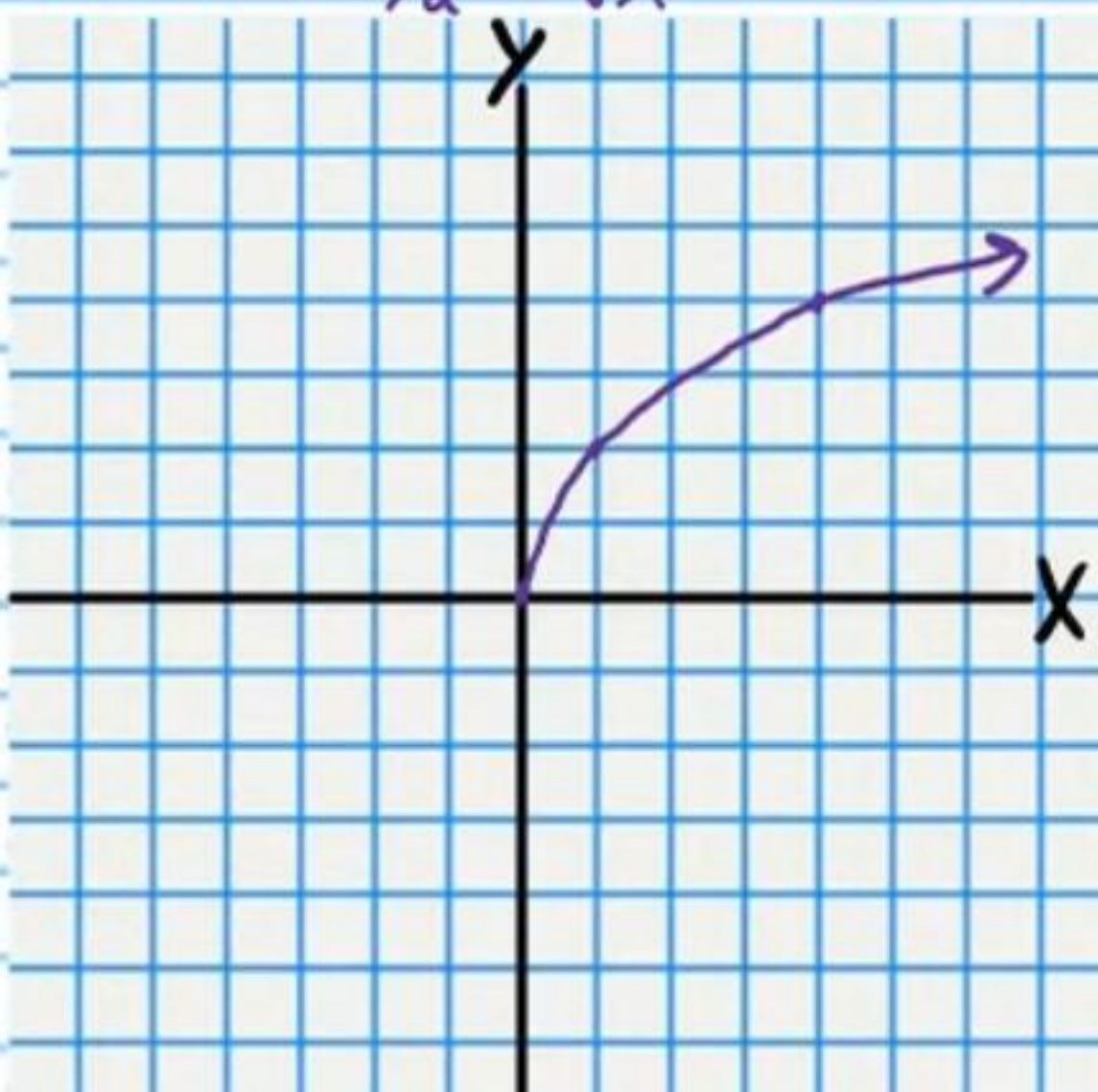


⑥ $y = 2\sqrt{x}$

Parent Relation
 $y = \sqrt{x}$

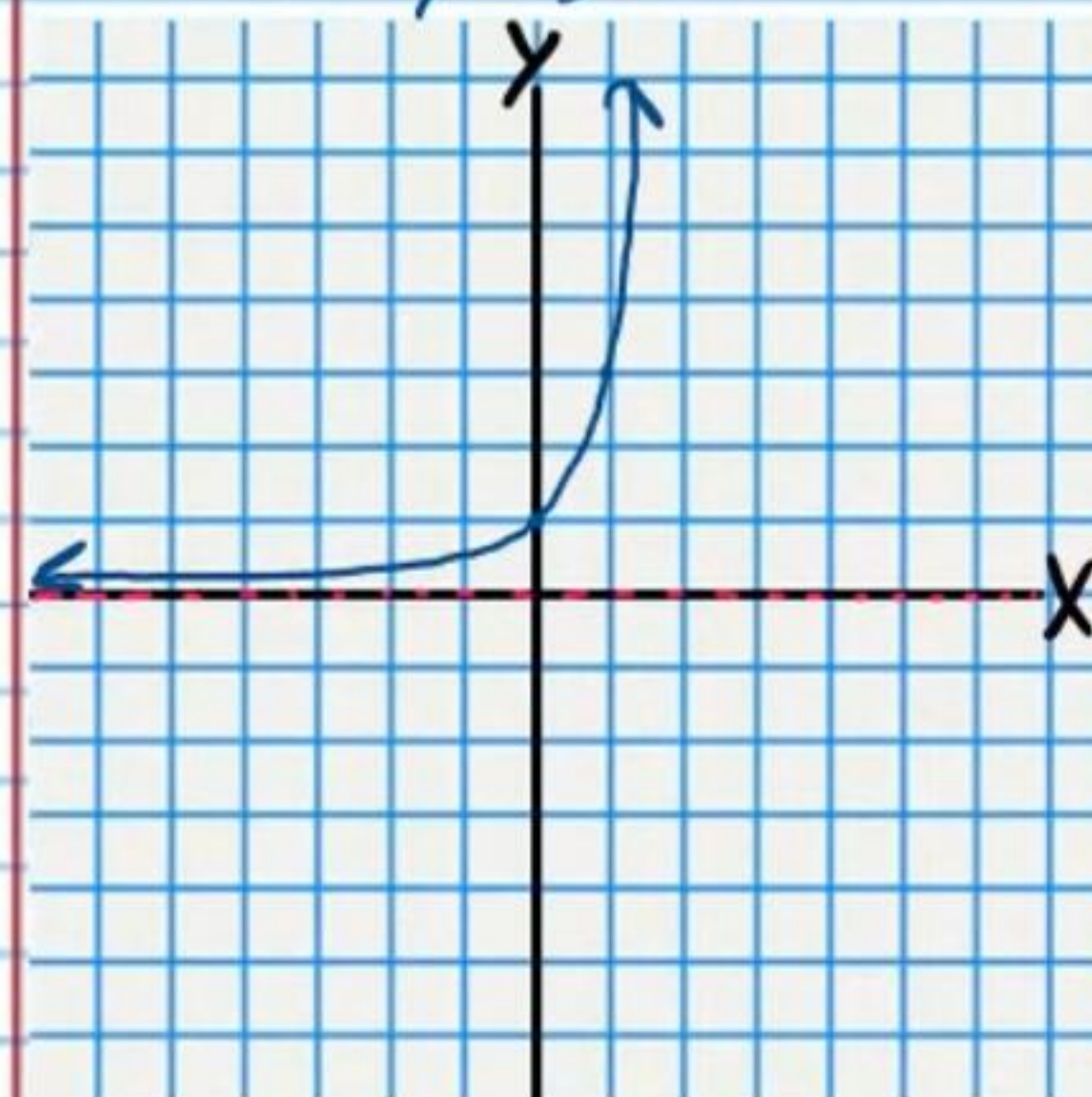


Transformation
 $y/2 = \sqrt{x}$

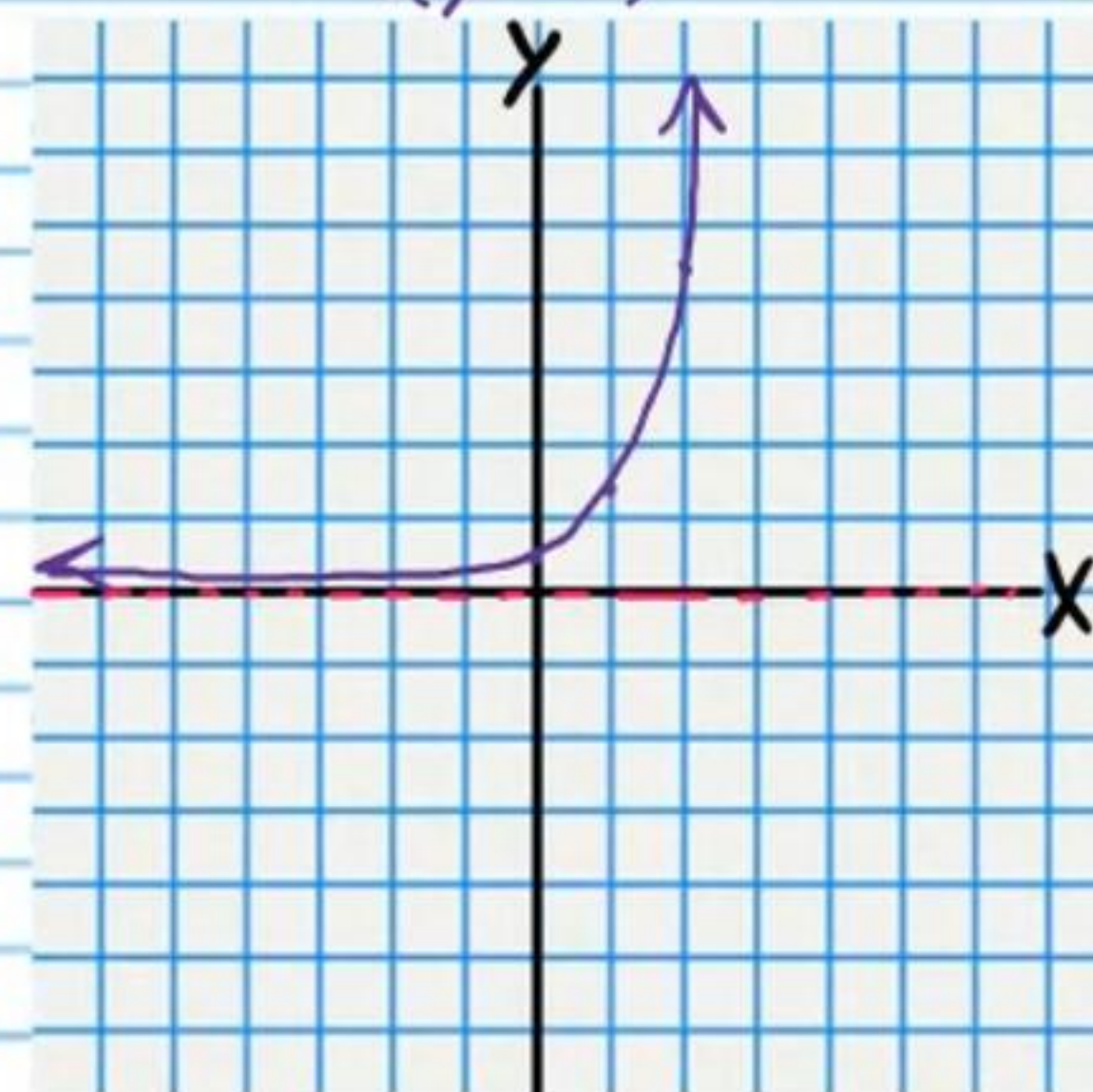


⑦ $y = \frac{3^x}{2}$

Parent Relation
 $y = 3^x$

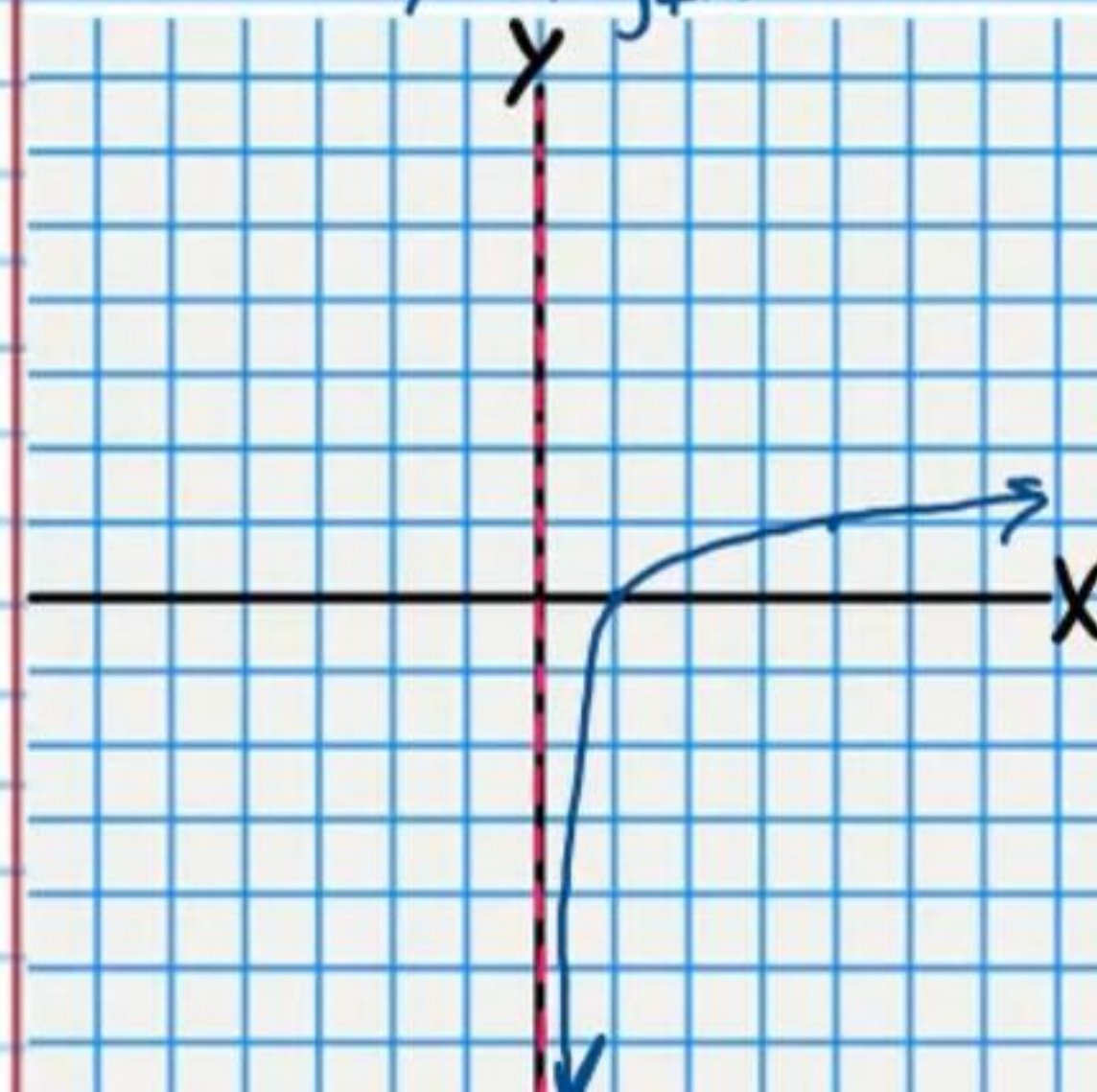


Transformation
 $2y = 3^x$

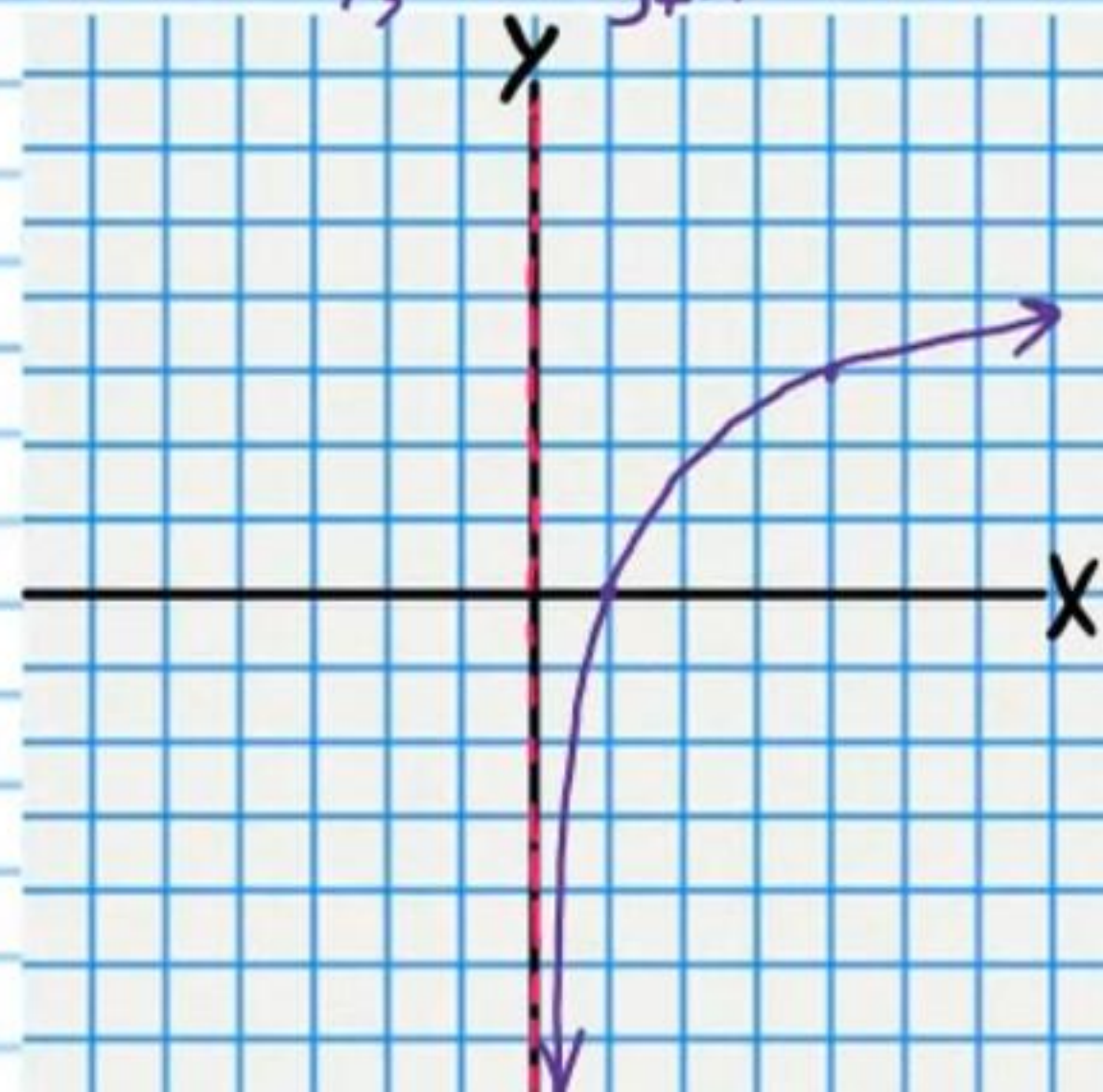


⑧ $y = 3\log_4 x$

Parent Relation
 $y = \log_4 x$



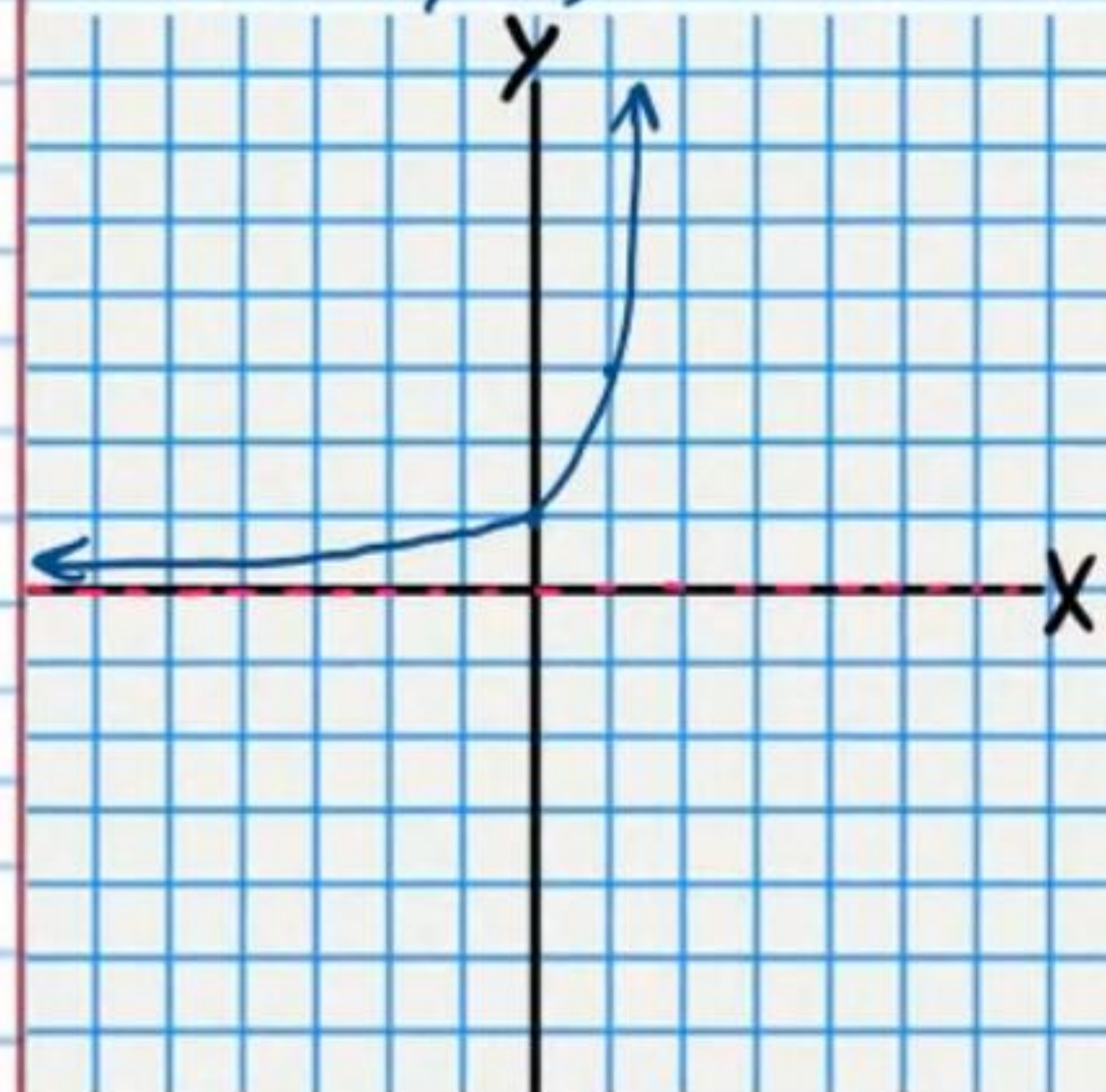
Transformation
 $y/3 = \log_4 x$



⑨ $y = 2(3^x)$

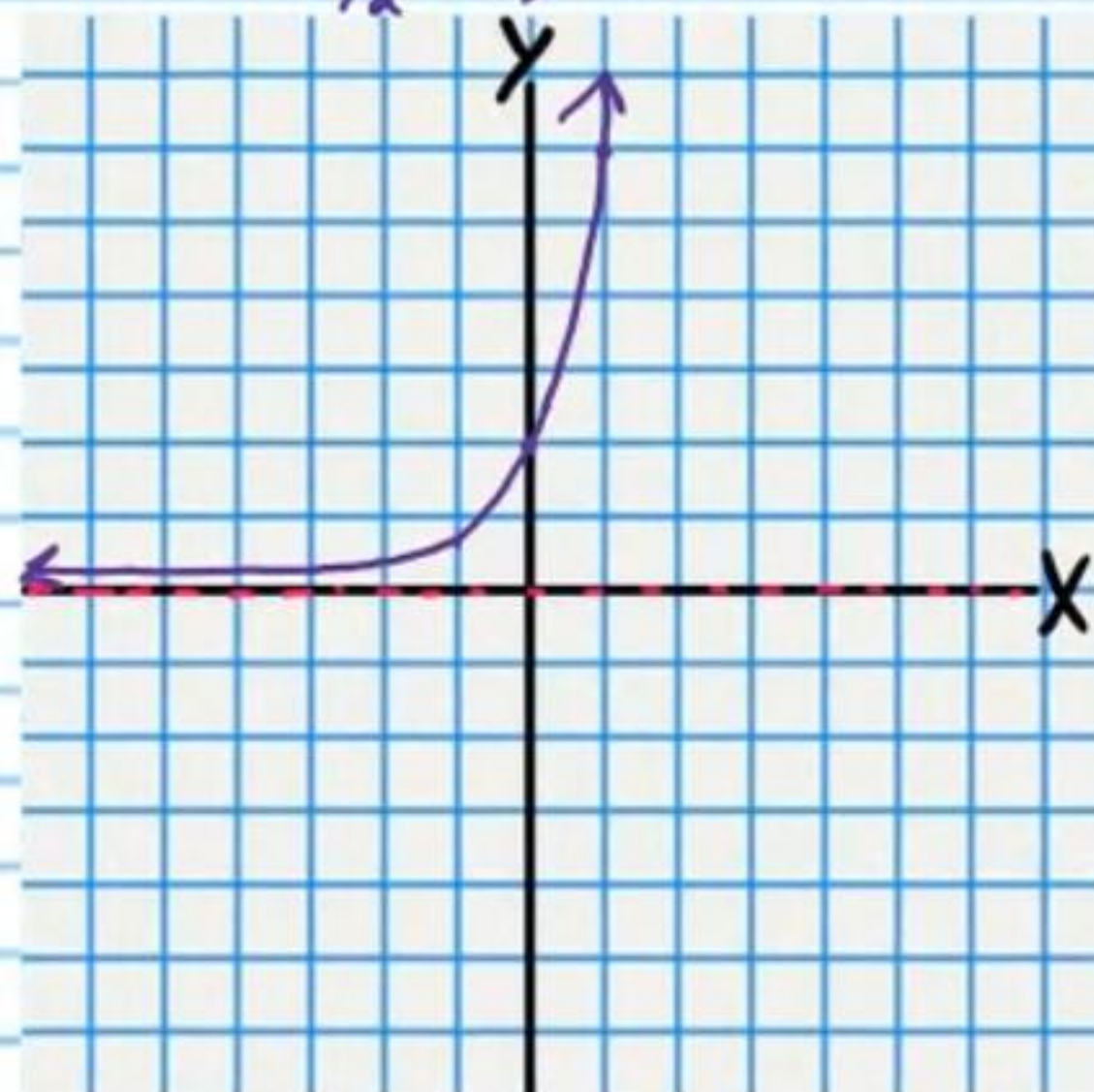
Parent Relation

$y = 3^x$



Transformation

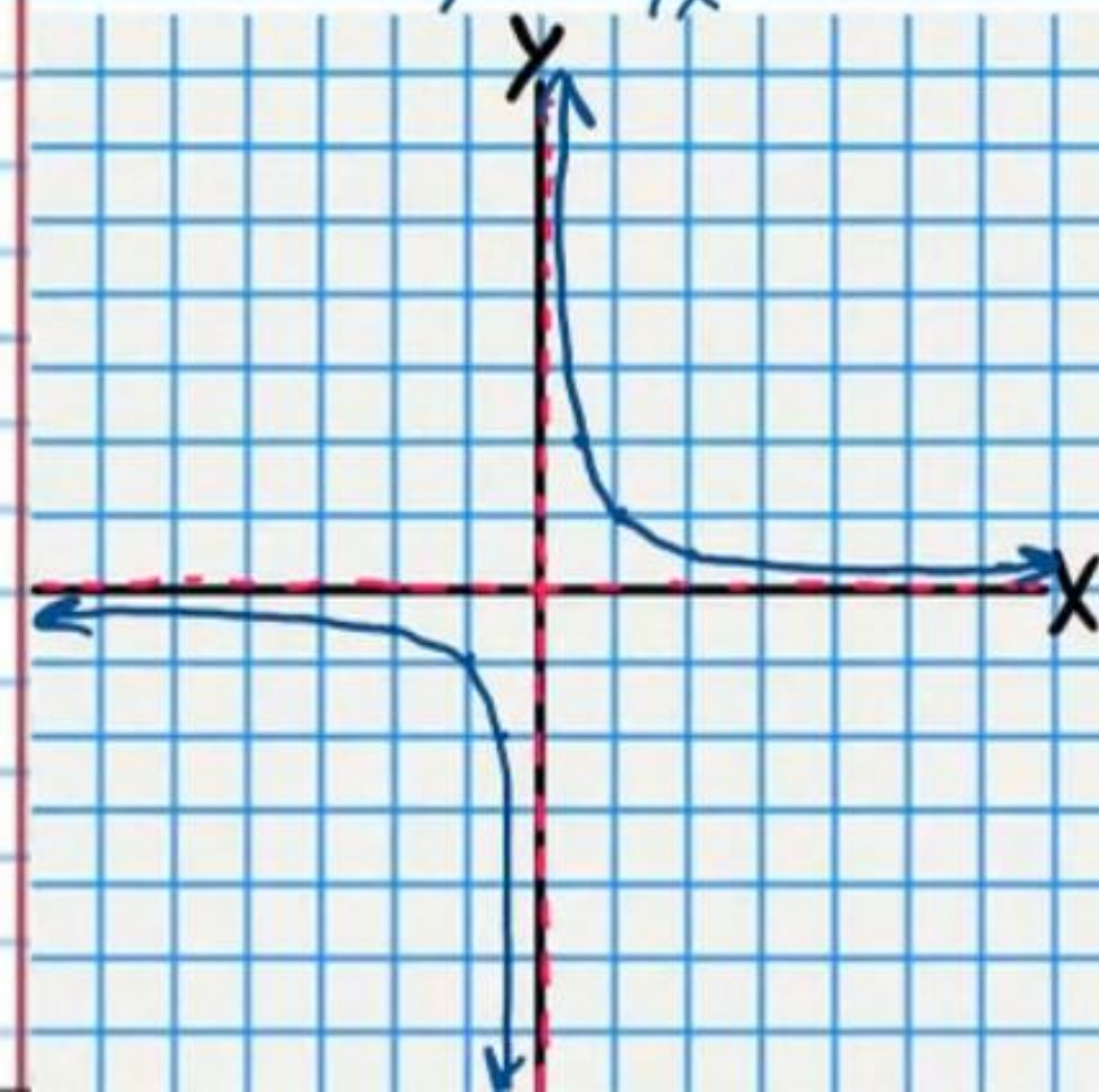
$y/2 = 3^x$



⑩ $y = \frac{4}{x}$

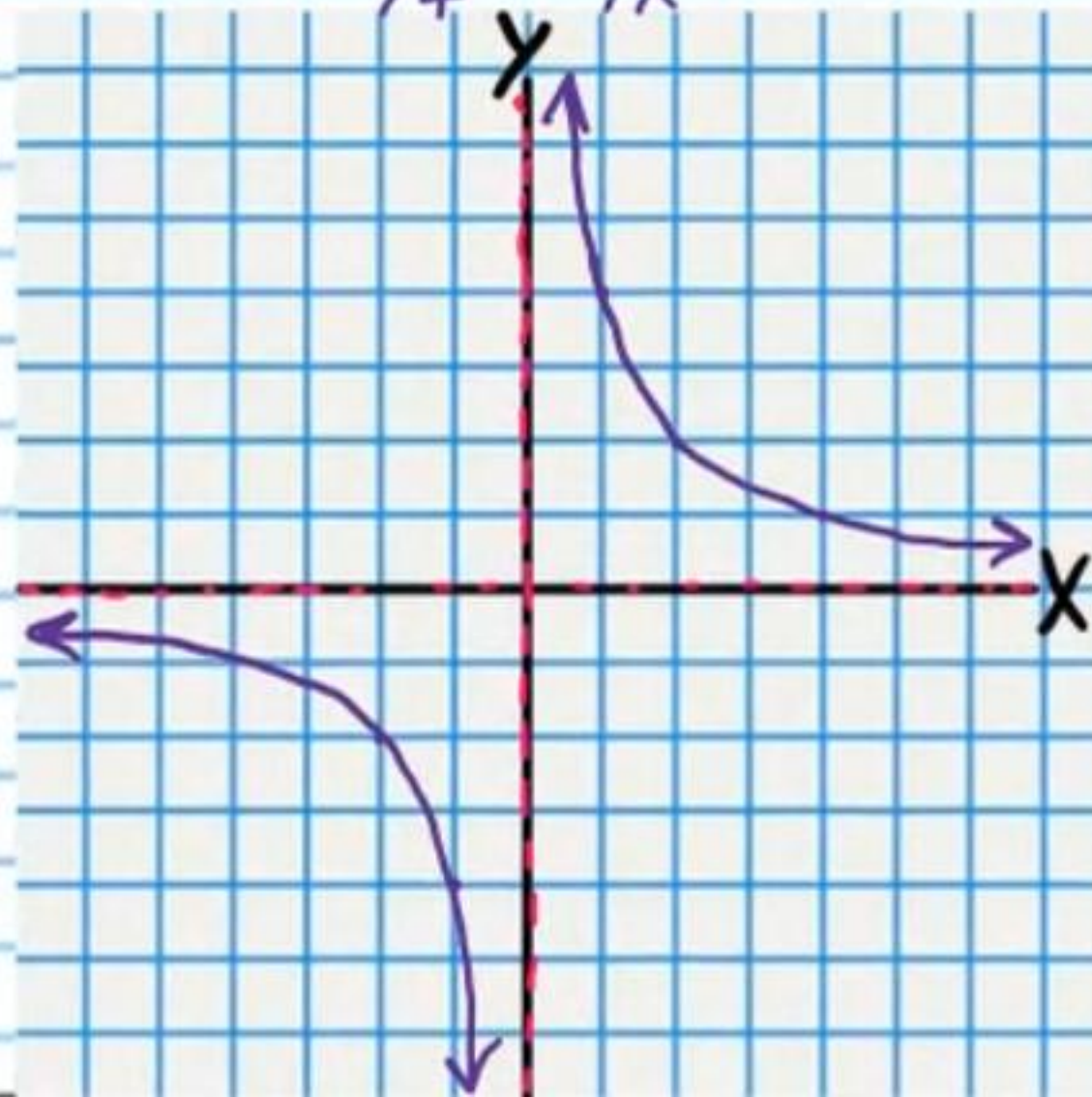
Parent Relation

$y = 1/x$



Transformation

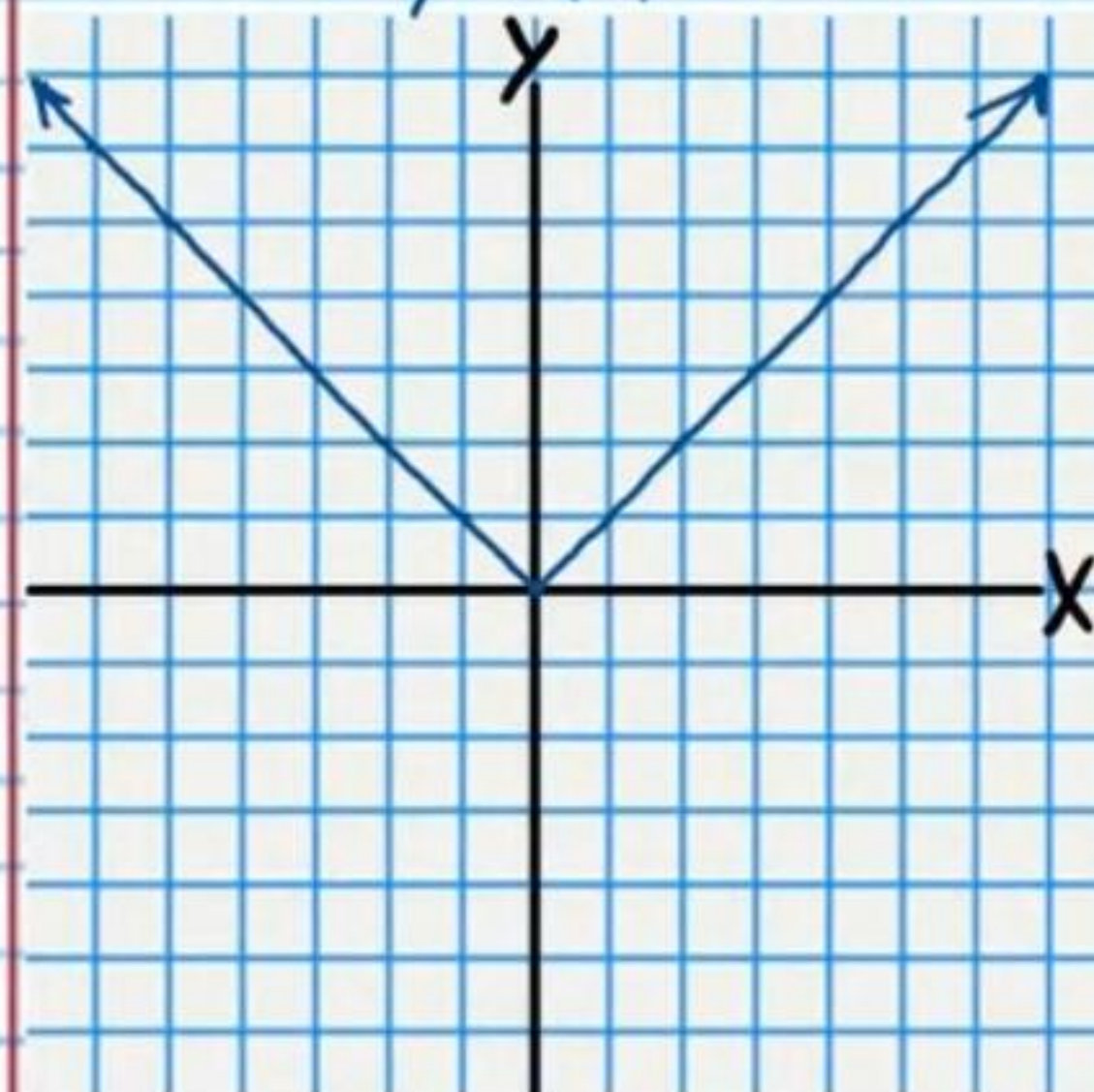
$y/4 = 1/x$



⑪ $y = -|x|$

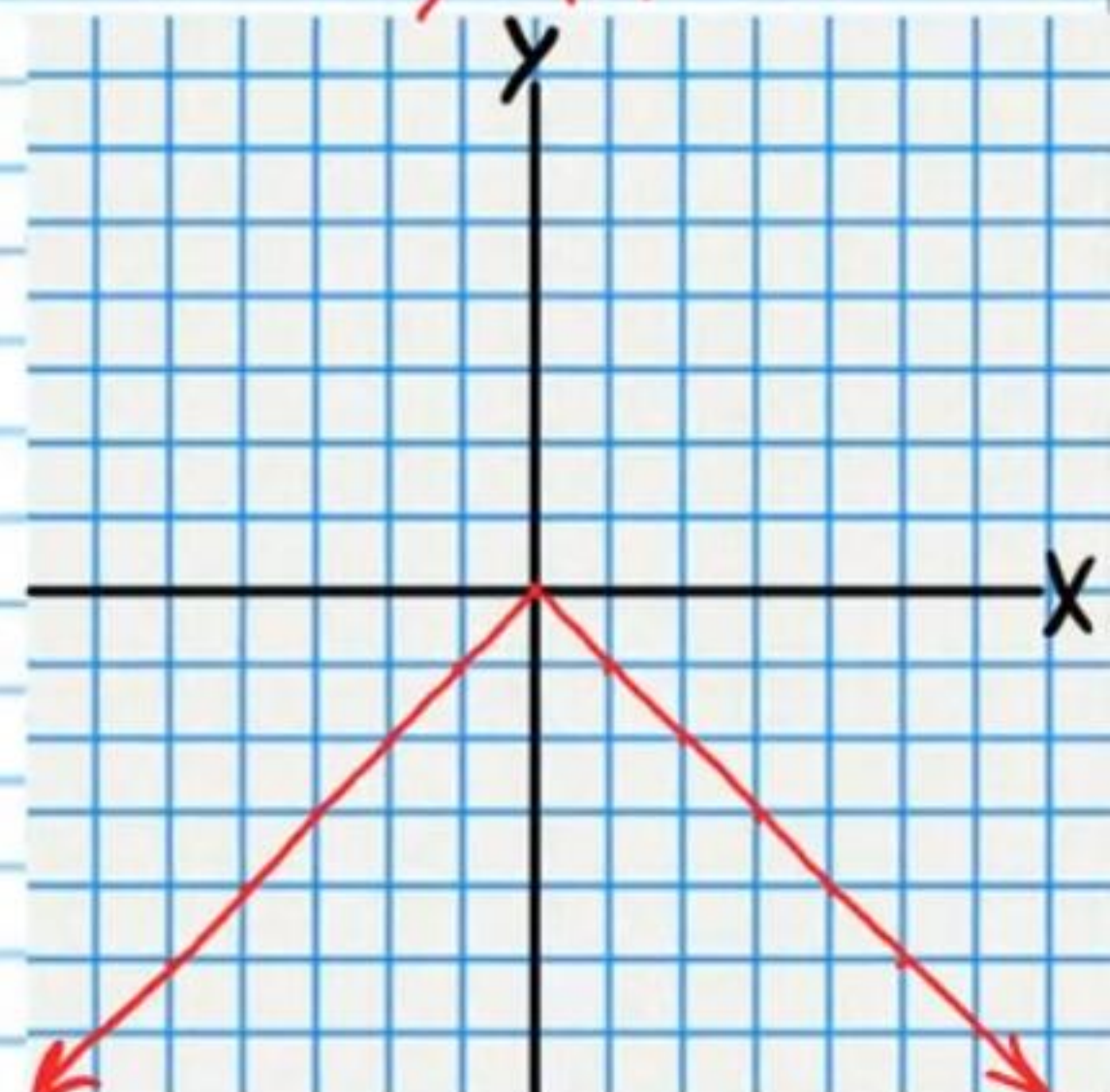
Parent Relation

$y = |x|$



Transformation

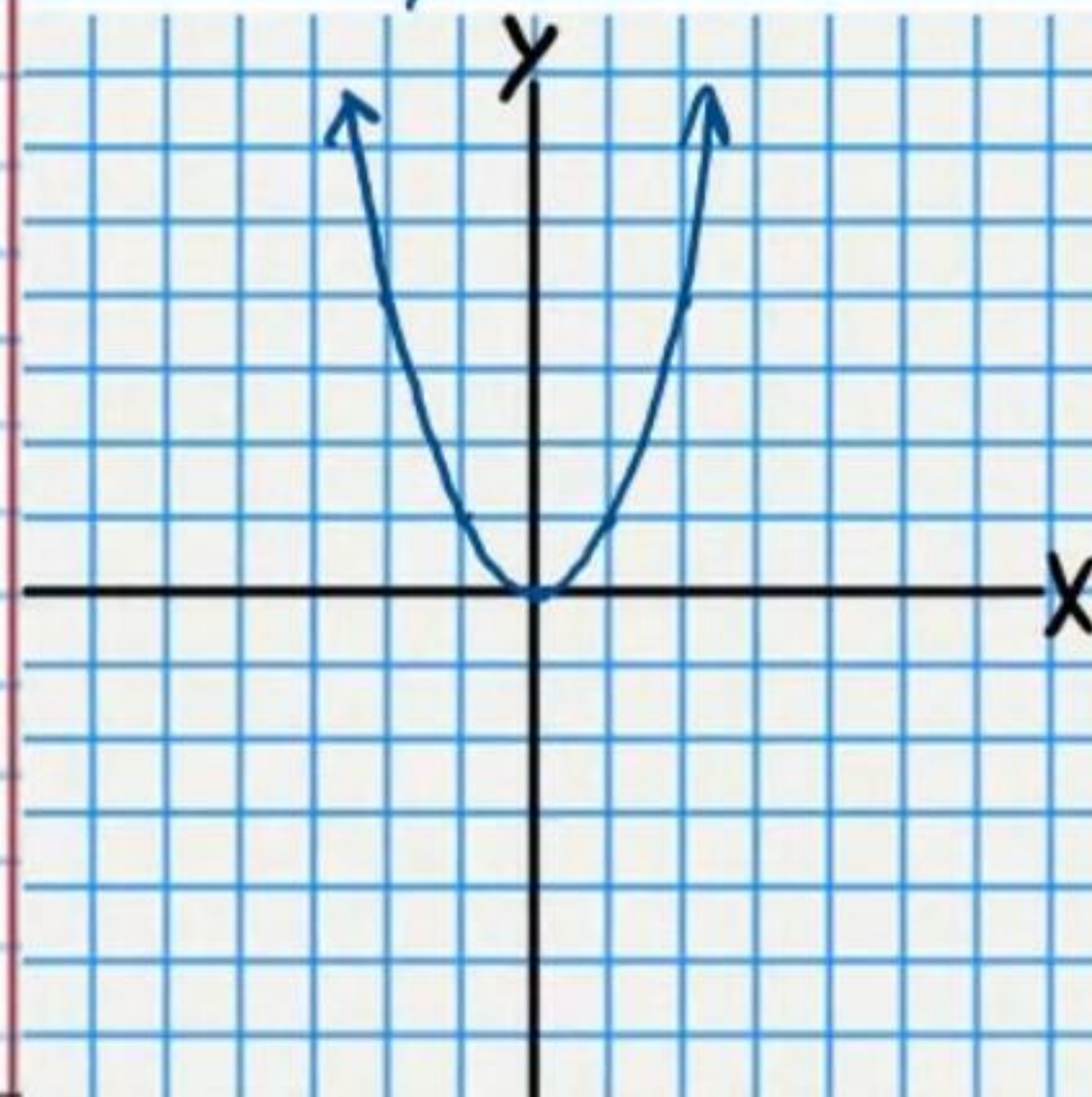
$-y = |x|$



⑫ $y = -x^2$

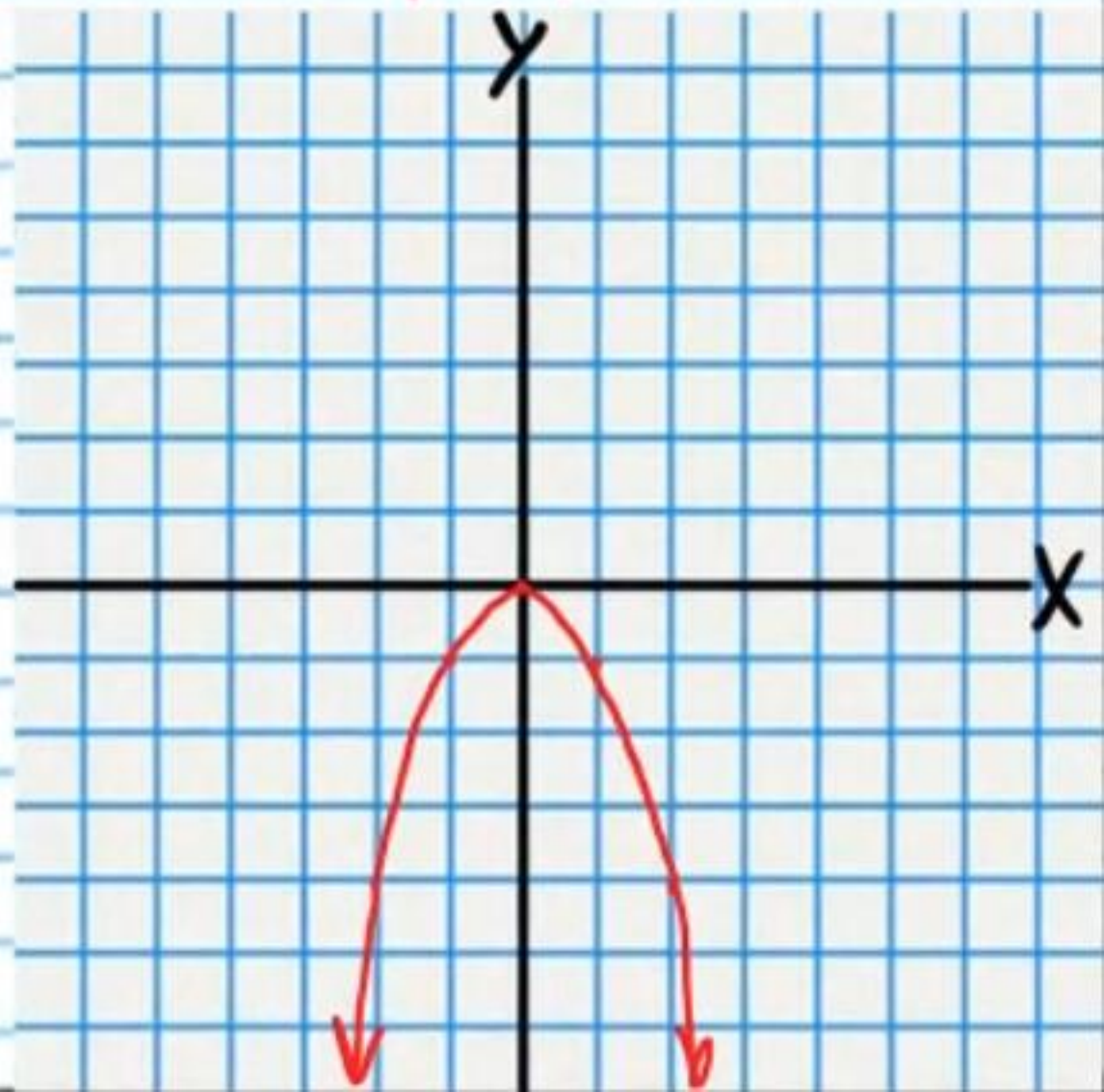
Parent Relation

$y = x^2$



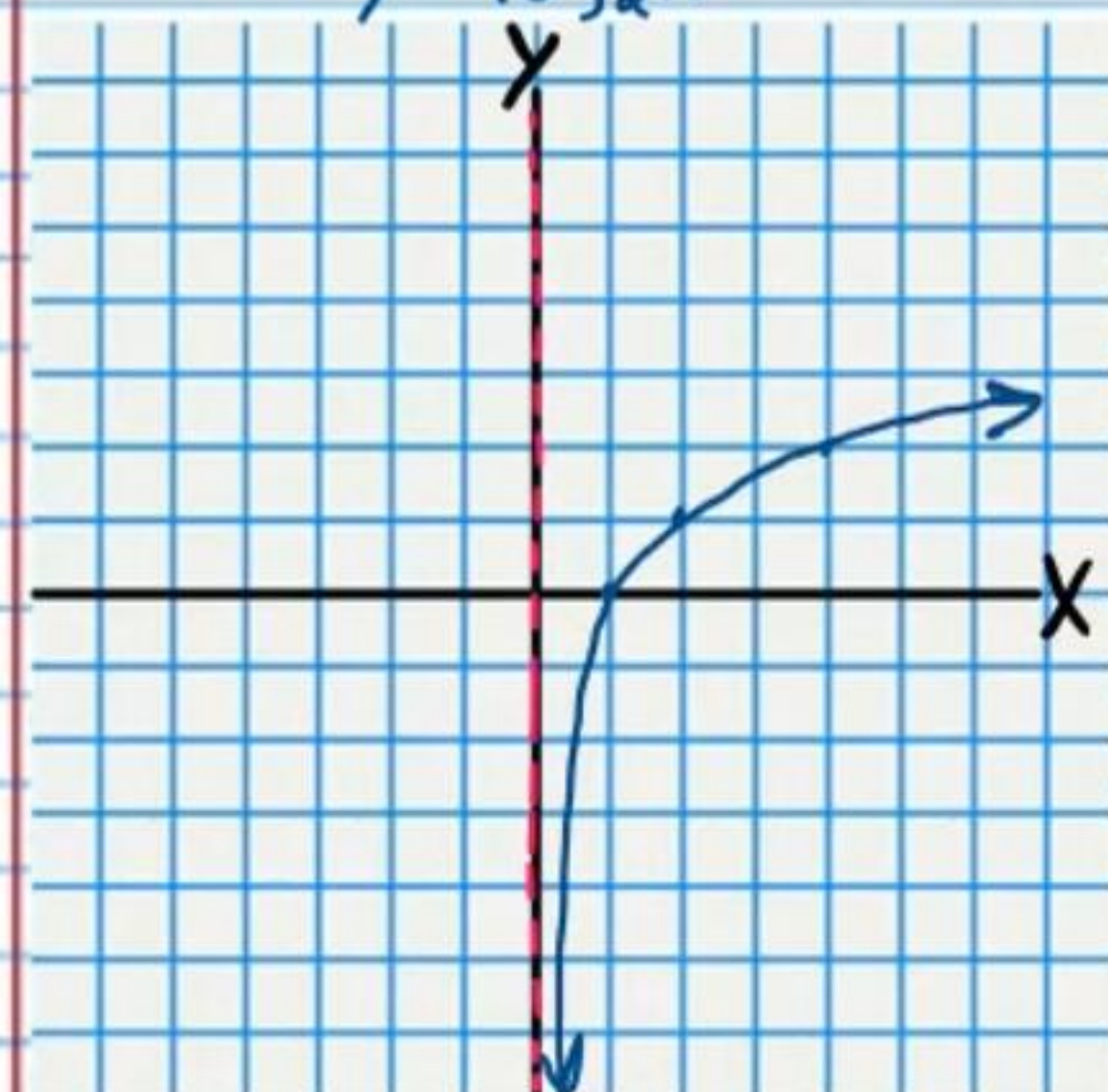
Transformation

$-y = x^2$

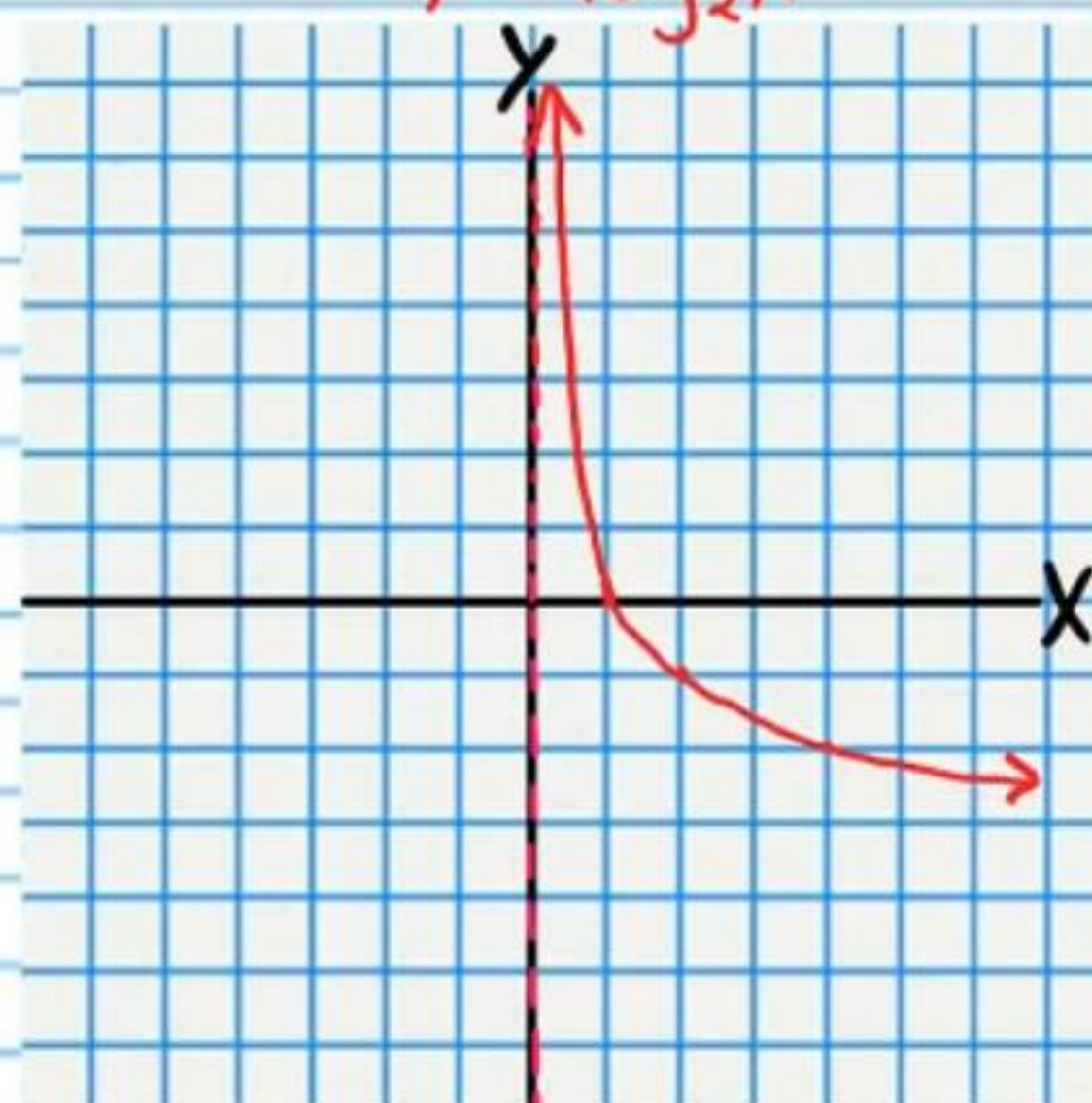


⑬ $y = -\log_2 X$

Parent Relation
 $y = \log_2 X$

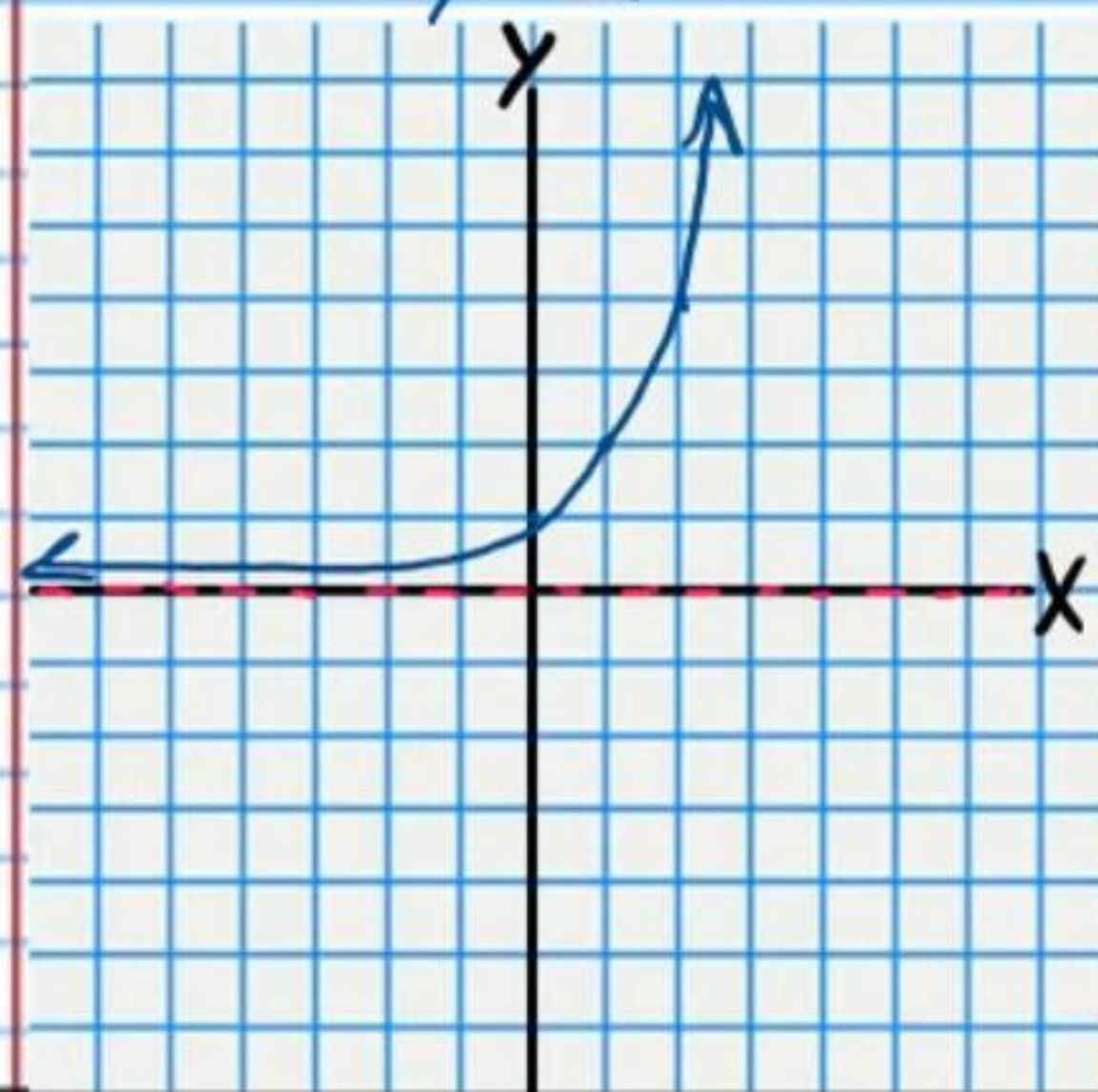


Transformation
 $-y = \log_2 X$

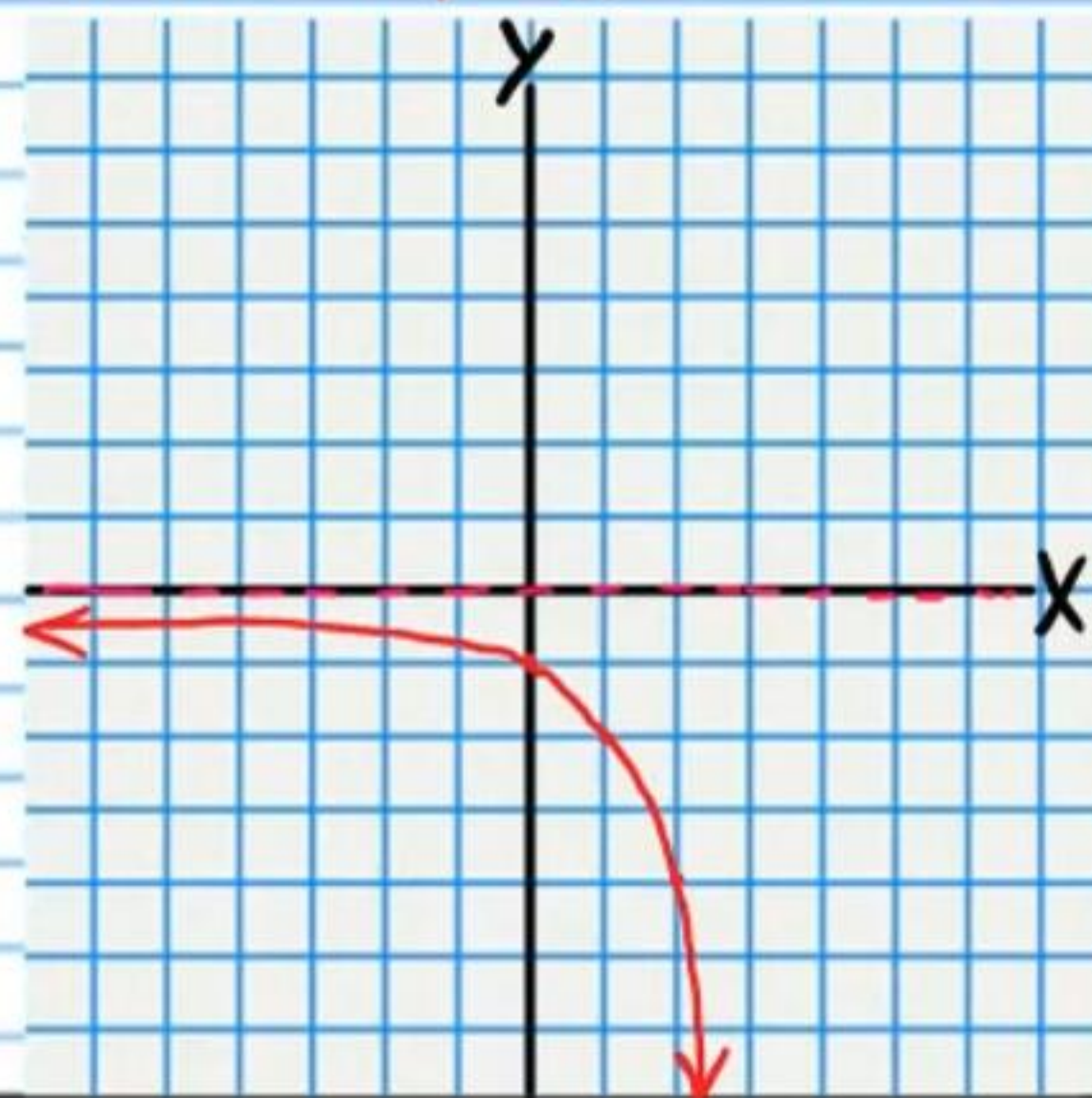


⑭ $y = -2^x$

Parent Relation
 $y = 2^x$

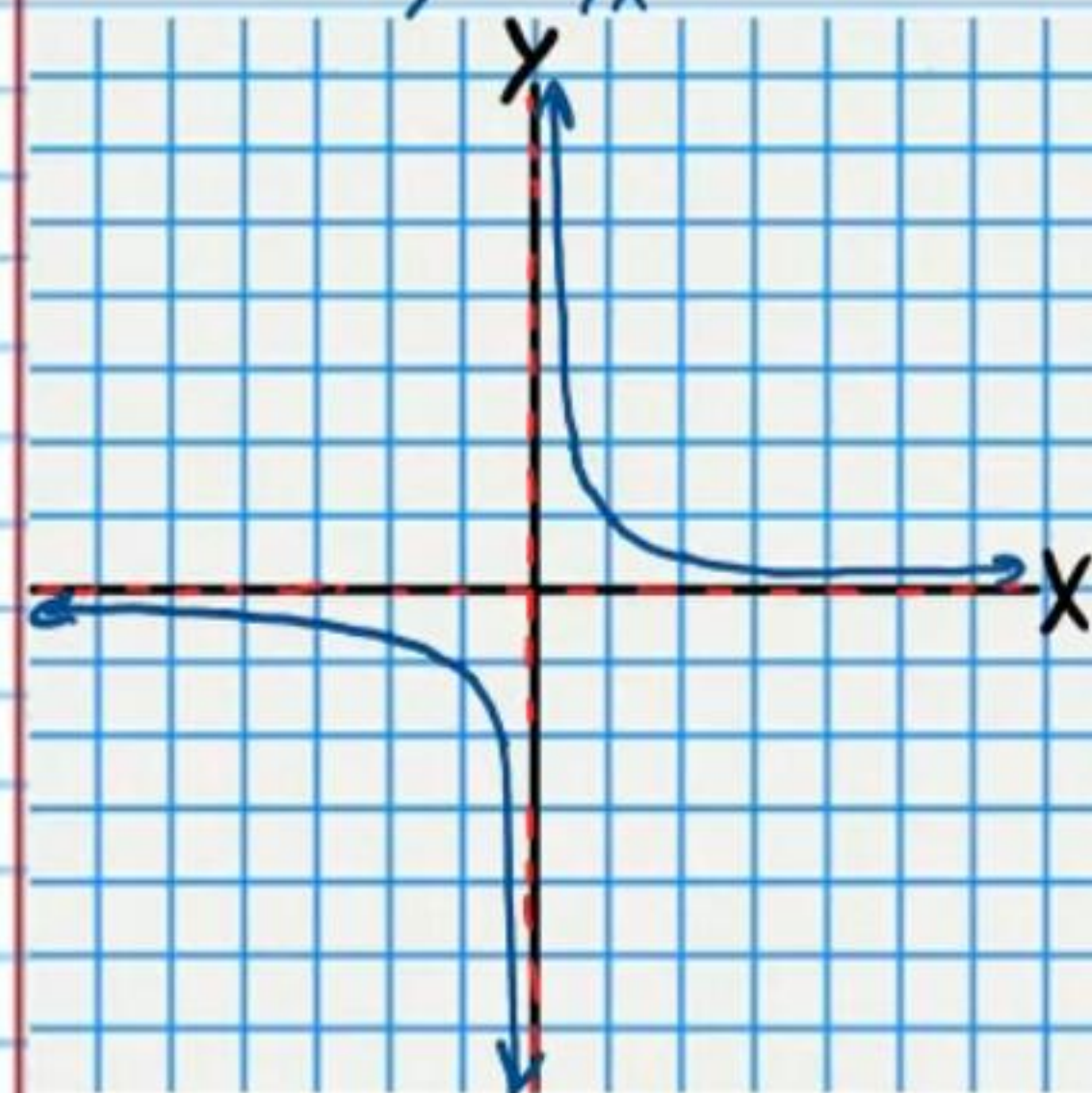


Transformation
 $-y = 2^x$

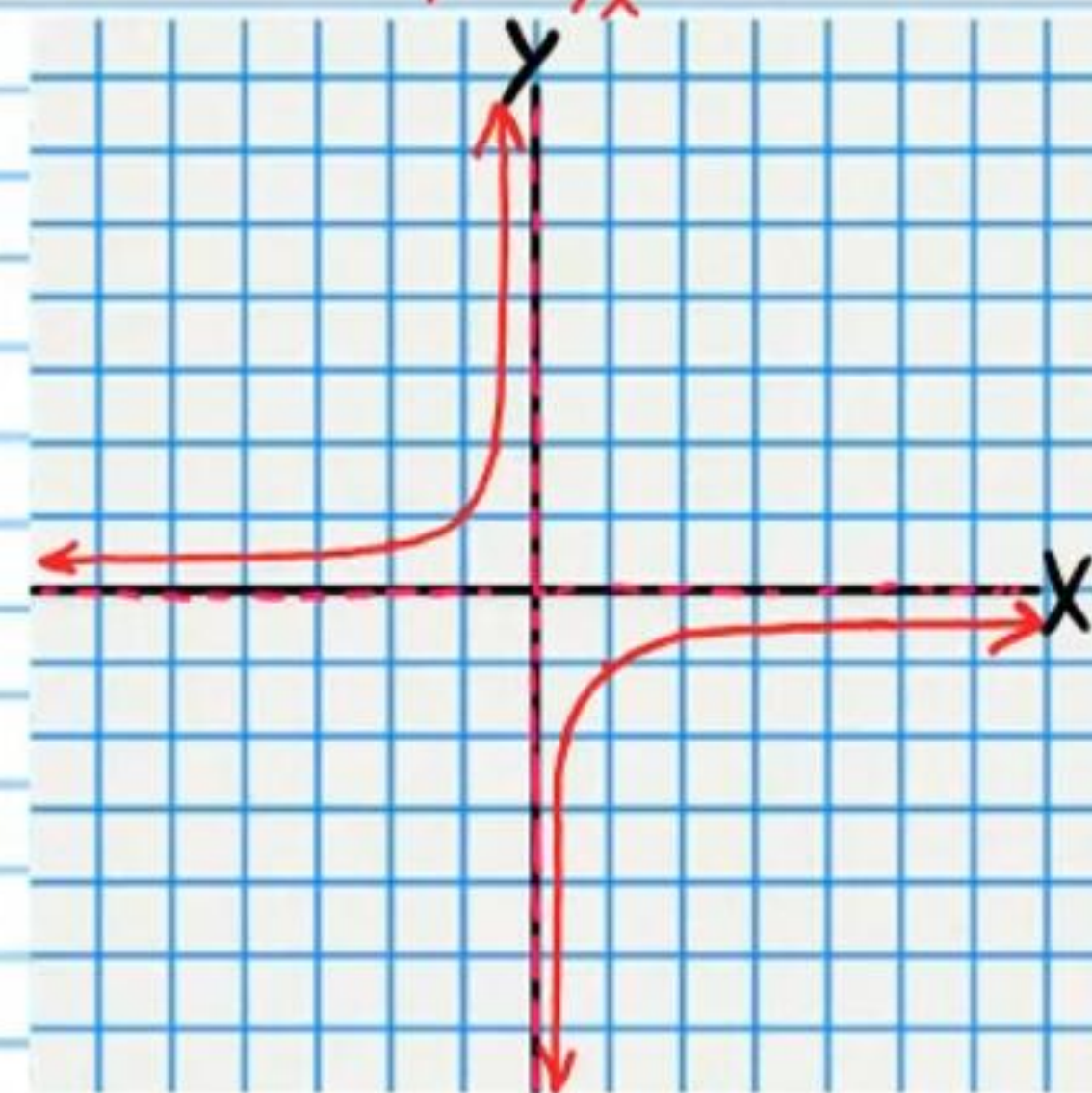


⑮ $y = -\frac{1}{x}$

Parent Relation
 $y = \frac{1}{x}$

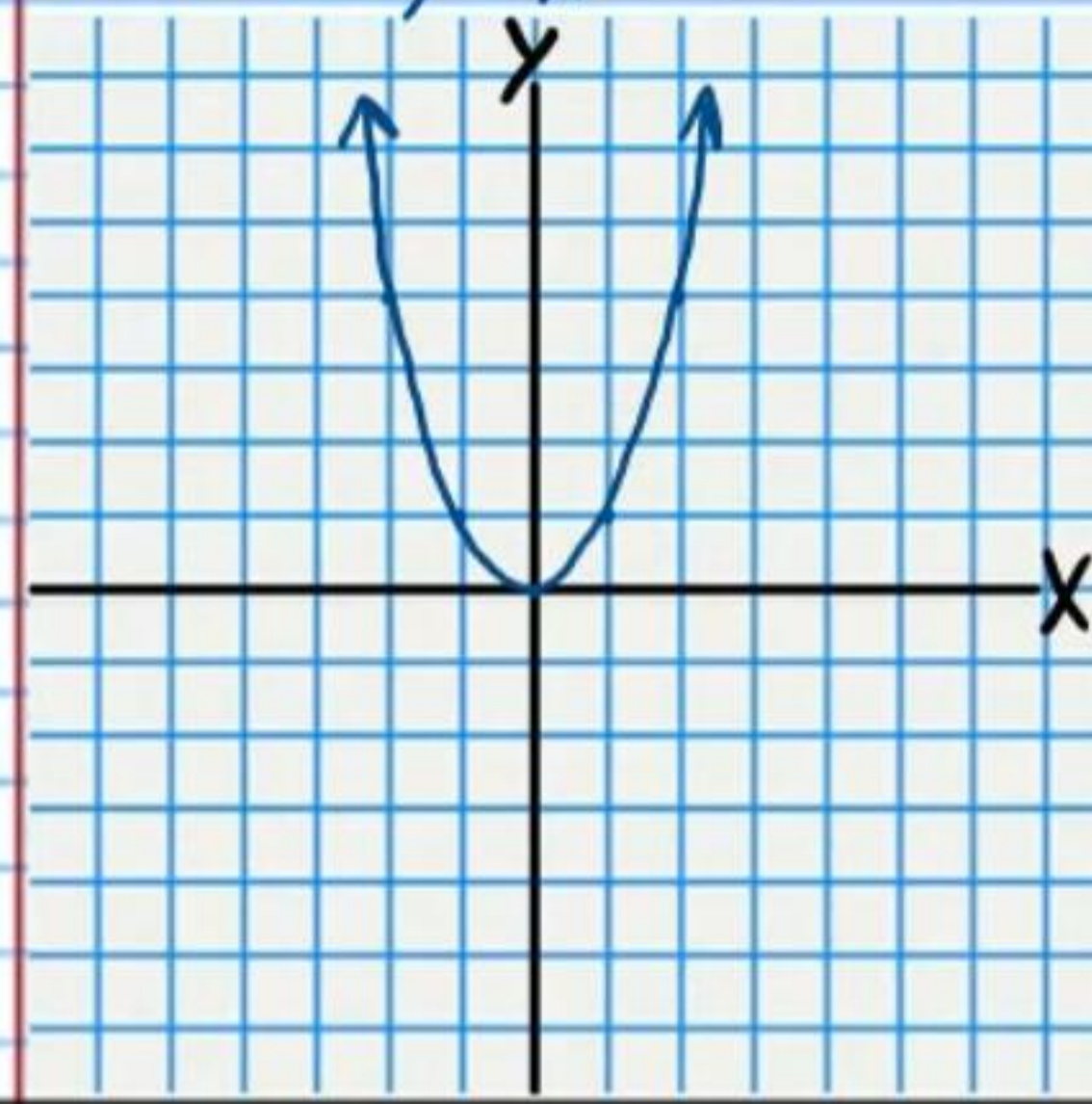


Transformation
 $-y = \frac{1}{x}$

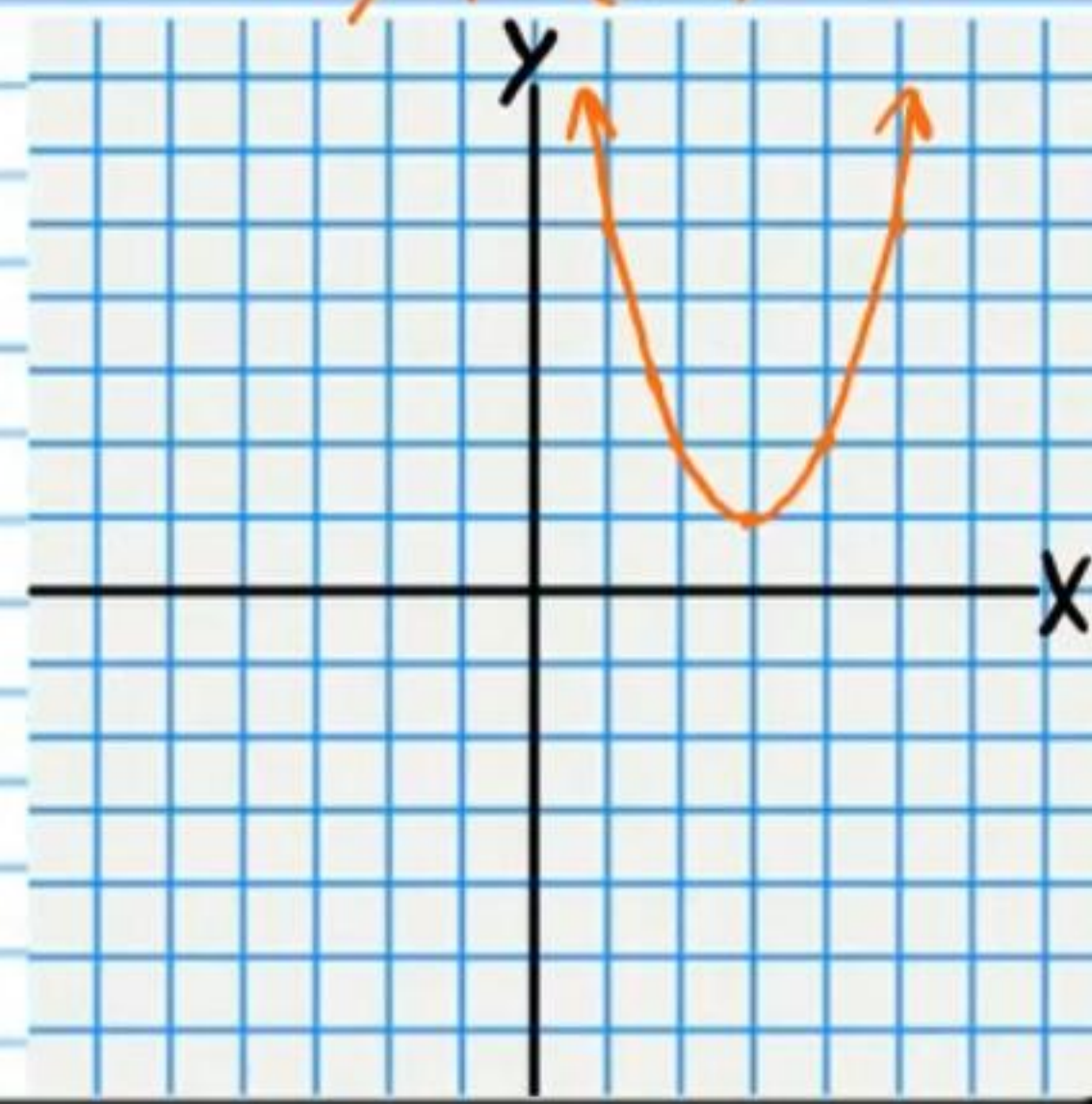


⑯ $y = (x-3)^2 + 1$

Parent Relation
 $y = x^2$

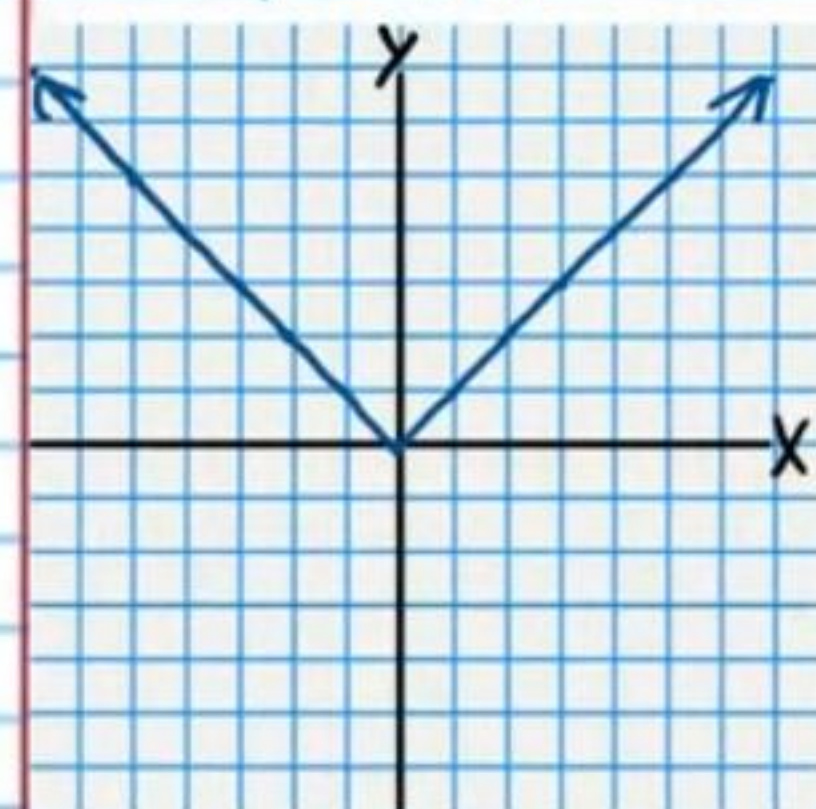


Transformation
 $y - 1 = (x - 3)^2$

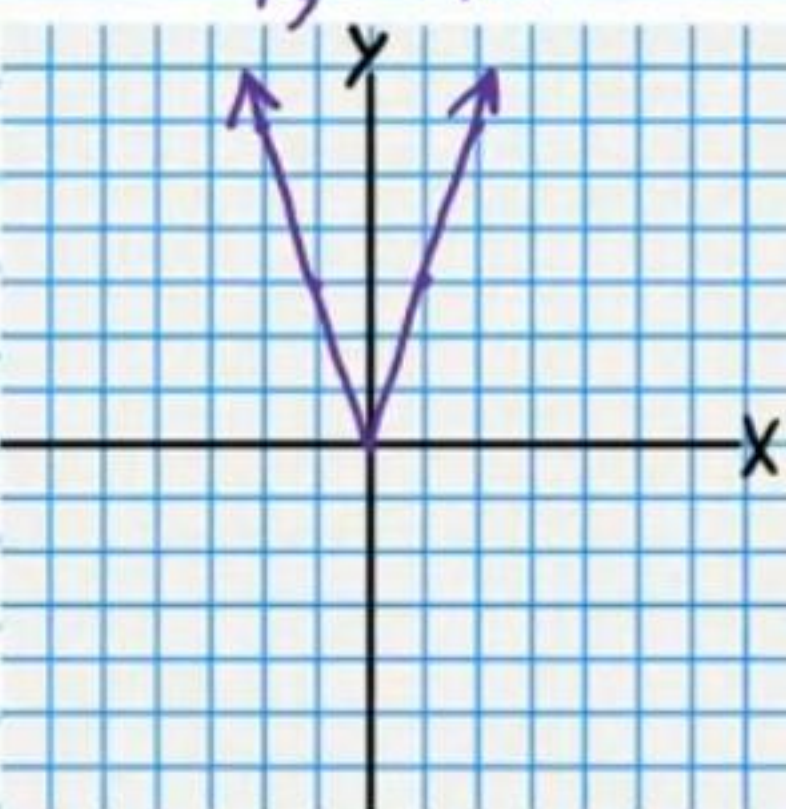


⑪ $y = 3|x+2|$

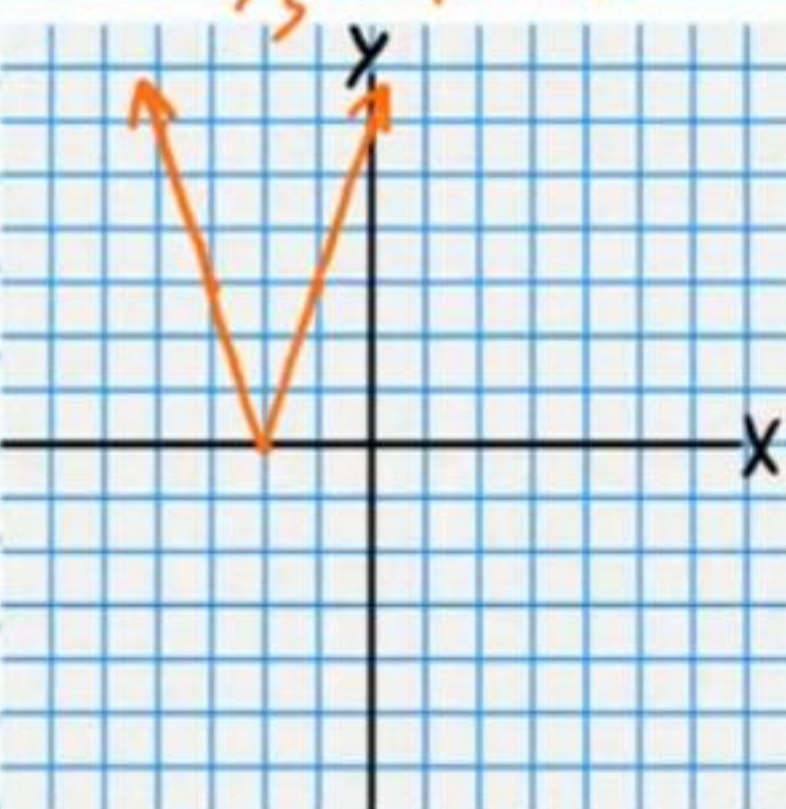
Parent Relation
 $y = |x|$



Transformation #1
 $y/3 = |x|$



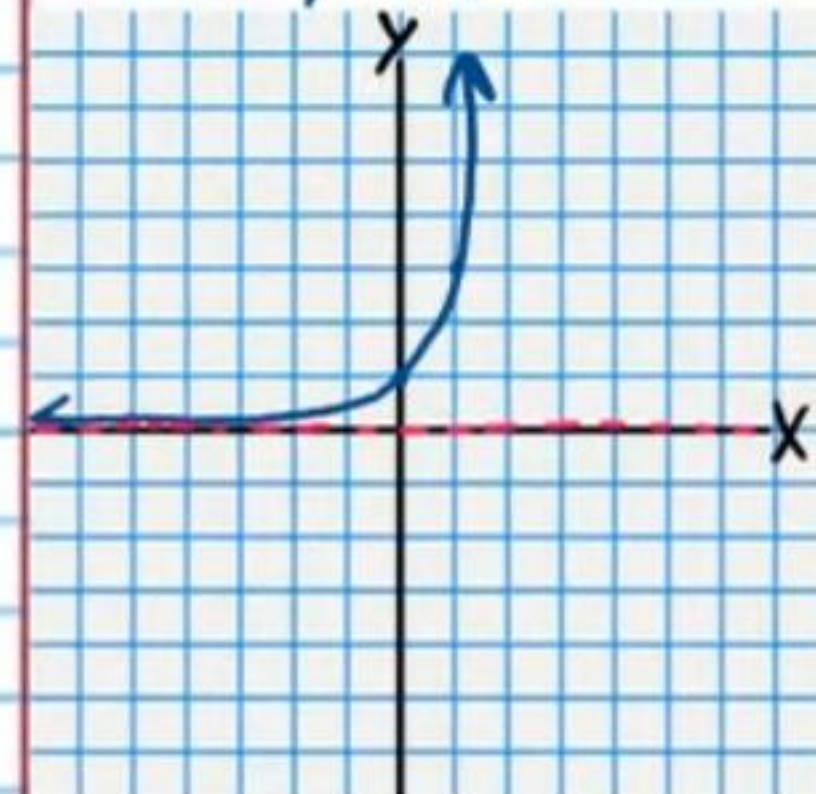
Transformation #2
 $y/3 = |x+2|$



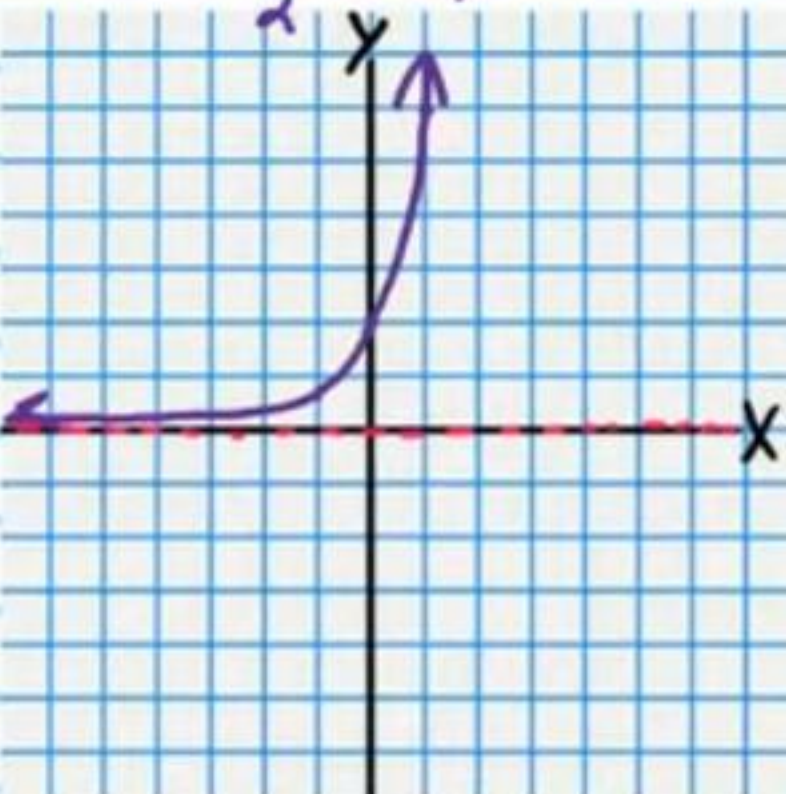
You may choose to translate first. That is okay. Your final graph should be the same.

⑫ $y = 2(3^{-x})$

Parent Relation
 $y = 3^x$



Transformation #1
 $y/2 = 3^x$



Transformation #2
 $y/2 = 3^{-x}$

